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# Probing Strange Metal Signatures in an Interaction-Disordered Hubbard model via Monte Carlo Mean Field Approximation

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# חיפוש אחר התנהגות "מתכת מוזרה" במודל הבארד עם אי-סדר באינטרקרציה באמצעות קירוב מונטה-קרלו-שדה-ממוצע

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# תקציר


ההתנהגות הלינארית בטממפרטורה של ההתנגדות החשמלית הנצפית במגוון ``מתכות מוזרות'' מפרה את ניבוי תורת נוזל פרמי $\rho \propto T^2$. התנהגות זו ממשיכה עד טמפרטורות נמוכות בהרבה מכל סקאלה שבה פיזורי אלקטרון--פונון יכולו להסבירה. בין ההסברים שהוצעו, הועלתה הסברה כי ``שלוליות'' על-מוליכות לא-הומוגניות המשובצות במארח מתכתי יכולות לייצר $\rho(T)$ לינארית שכזו.

בעבודה זו אנו חוקרים תרחיש זה באמצעות מודל הבארד המושך (Attractive-Hubbard Model), ומרחיבים אותו כך שיכלול: (i) אי-הומוגניות מרחבית בעוצמת האינטראקציה ו-(ii) שדה מגנטי אורביטלי המופעל על המערכת האחידה. אנו נוקטים בשיטת מונטה קרלו שדה ממוצע (MCMF), שבה האינטראקציה מפורקת באמצעות שדה זיווג עזר $\Delta_i$. בעוד שמונטה קרלו קוונטי דטרמיננטי (DQMC) מתחשב הן בתנודות המרחביות (התרמיות) והן בתנודות הזמניות (הקוונטיות) של השדה $\Delta$, MCMF מזניח את האחרונות ושומר רק על התנודות התרמיות. אפקטים קוונטיים משולבים אך ורק דרך הצימוד לפרמיונים. היתרון המרכזי של שיטה זו הוא יכולתה לספק ישירות פונקציות קורלציה על ציר התדר הממשי, ובכך לעקוף את הקושי הטמון בהמשכה האנליטית הנדרשת על ידי DQMC .

אנו מתחילים בהשוואה של שיטת MCMF מול תוצאות DQMC עבור מודל הבארד המושך וההומוגני. אנו מעריכים את טמפרטורת המעבר העל-מוליך $T_c$ כפונקציה של עוצמת האינטראקציה U. תוצאותינו לוכדות איכותית את המעבר מ-BCS ל-BEC. עם זאת, שיטת MCMF מעריכה בחסר באופן שיטתי את $T_c$ בכ-15%. במצב הנורמלי ($T > T_c$), הן פונקציית קורלציית הזיווג $P_s$ והן הקשיחות זרימת-העל (Superfluid-Stiffness) $\rho_s$ משוחזרות במדויק, מה שמהווה שיפור משמעותי על פני תורת שדה ממוצע BCS הסטנדרטית. לעומת זאת, מתחת ל-$T_c$, MCMF נוטה להעריך יתר על המידה תצפיות אלו, אי-התאמה הנובעת מהזנחת התנודות הזמניות (הקוונטיות) בשדה העזר $\Delta_i$.

בהמשך אנו מחשבים את תכונות הולכה, ובהן משקל דרודה $D$ והחלק הרגולרי של המוליכות האופטית $\sigma^{\mathrm{reg}}(\omega)$, שממנה נגזרת התנגדות ה-DC $\rho_{xx}^{\mathrm{reg}}$. ההתנגדות מציגה תלות לא-מונוטונית בטמפרטורה עם מינימום ב-$T_c < T_{\min}$, המפריד בין תחום טמפרטורות נמוכות שבו ההתנגדות הרגולרית מקיימת $d\rho/dT < 0$ (לצד משקל דרודה סופי), לבין תחום מתכתי בטמפרטורות גבוהות שבו $\rho \sim T^{\alpha}$ עם $0 < \alpha < 2$. נראה כי הסקאלה $T_{\min}$ מתואמת עם טמפרטורת הפסאודוגאפ $T^*$, אותה אנו מזהים כטמפרטורה שבה סוספטביליות הספין האחידה מגיעה למקסימום שלה.

בשלב הבא אנו משלבים אי-הומוגניות מרחבית על-ידי פיזור אקראי של מרכזי אינטראקציה מושכת על פני $p$ מאתרי הסריג ($0 < p < 1$). כל מרכז תורם פרופיל אינטראקציה גאוסיאני במרחב, כך שהאינטראקציה הממוצעת היא $\bar{\mathrm{U}} = \mathrm{U} \cdot p$. בשינוי של $p$ ו- $\mathrm{U}$ בו-זמנית, אנו יכולים לזהות משפחות של מערכות בעלות $T_c$ משותף אך שונות במידת אי-הסדר שלהן. אנו מוצאים כי סקאלת הפסאודוגאפ $T^*$ עוקבת אחר האינטראקציה הממוצעת $\bar{\mathrm{U}}$. כתוצאה מכך, הגדלת אי-הסדר לאורך מסלול איזו-$T_c$ מקטינה את $\bar{\mathrm{U}}$ ובהתאם גם את $T^*$. ובכך, מתרחב התחום המתכתי $d\rho/dT > 0$ לטמפרטורות נמוכות יותר.

בדומה למערכות פיזיקליות, ההתנגדות בטמפרטורות נמוכות של המודל מוסתרת על-ידי הופעת העל-מוליכות. ניסיונית, תחום טמפרטורות זה נחקר לעיתים קרובות באמצעות הפעלת שדה מגנטי חזק $B$ כדי לדכא את המצב העל-מוליך. אנו נוקטים בגישה דומה ומוסיפים שדה מגנטי למודל האחיד. אנו מוצאים כי אפילו השדה הקטן ביותר שאנו מסוגלים להכניס למערכת כבר הורס את קוהרנטיות הפאזה ועמה את העל-מוליכות, על אף שהזיווג המקומי מושפע אך במעט. השדה המגנטי מרחיב את התחום $d\rho/dT > 0$ לטמפרטורות נמוכות יותר, כאשר $T^*$ ו-$T_{\min}$ מדוכאים יחד ובצורה דרמטית יותר ביותר בעוצמות אינטראקציה חלשות.

ממצאינו חשפו התנהגות הולכה מעניינת בואריאנטים של מודל הבארד המושך. נדרשת עבודה נוספת כדי לאמת היבטים של תוצאות MCMF באמצעות חישובי DQMC . בנוסף, שילוב של אפקטים פיזיקליים כגון אנרגיית הטעינה (charging-energy) של האזורים המושכים עשוי להיות הכרחי כדי להפוך את חתימות המתכת המוזרה לברורות יותר.

# Abstract

The linear-in-temperature resistivity observed in various "strange metals" violates the Fermi-liquid prediction $\rho \propto T^2$, even at temperatures far below any scale at which electron–phonon scattering can explain it. Among the proposed explanations, it has been suggested that inhomogeneous superconducting "puddles" embedded in a metallic host can produce such a linear $\rho(T)$.

In this work, we investigate this scenario using the attractive Hubbard model, extended to include: (i) spatial inhomogeneity in the interaction strength and (ii) an orbital magnetic field applied to the uniform system. We employ the Monte Carlo Mean-Field (MCMF) approach, where the interaction is decoupled via an auxiliary pairing field $\Delta_i$. While Determinant Quantum Monte Carlo (DQMC) accounts for both the spatial (thermal) and temporal (quantum) fluctuations of the $\Delta$ field, MCMF neglects the latter, retaining only thermal fluctuations while incorporating quantum effects only through the coupling to fermions. The primary advantage of this method is its ability to provide correlation functions directly on the real-frequency axis, thereby bypassing the ill-posed analytic continuation inherent to DQMC.

We begin by benchmarking the MCMF approach against established DQMC results for the uniform attractive Hubbard model. Specifically, we evaluate the superconducting transition temperature $T_c$ as a function of the interaction strength $\mathsf{U}$. Our results qualitatively capture the BCS-to-BEC crossover. However, the MCMF method systematically underestimates $T_c$ by approximately 15%. In the normal state ($T > T_c$), both the pairing correlation function $P_s$ and the superfluid stiffness $\rho_s$ are accurately reproduced, representing a significant improvement over standard BCS mean-field theory. Conversely, below $T_c$, MCMF tends to overestimate these observables, a discrepancy attributable to the neglect of temporal (quantum) fluctuations in the auxiliary field $\Delta_i$.

We subsequently calculate transport properties, including the Drude weight $D$ and the regular part of the optical conductivity $\sigma^{\mathrm{reg}}(\omega)$, from which the

DC resistivity $\rho_{xx}^{\mathrm{reg}}$ is derived. The resistivity exhibits non-monotonic temperature dependence with a minimum at $T_{\mathrm{min}} > T_c$, separating a low-$T$ regime where the regular resistivity exhibit $d\rho/dT < 0$ (along side a finite Drude weight), and a high-$T$ metallic regime where $\rho \sim T^{\alpha}$ with $0 < \alpha < 2$. The scale $T_{\mathrm{min}}$ appears to correlate with the pseudogap temperature $T^*$, which we identify as the temperature at which the uniform spin susceptibility reaches its maximum.

Next, we incorporate spatial inhomogeneity by randomly distributing attractive interaction centers on a fraction $p$ of the lattice sites ($0 < p < 1$). Each center contributes a Gaussian interaction profile in real space, yielding a mean interaction $\bar{\mathrm{U}} = \mathrm{U} \cdot p$. By varying $p$ and $\mathrm{U}$ simultaneously, we can identify a family of systems that share a common $T_c$ while differing in their degree of disorder. We find that the pseudogap scale $T^*$ tracks the mean interaction $\bar{\mathrm{U}}$. Consequently, increasing the disorder along an iso-$T_c$ trajectory reduces $\bar{\mathrm{U}}$ and, by extension $T^*$. This, in turn, has the effect of extending the metallic $d\rho/dT > 0$ regime to lower temperatures.

As in physical the systems, the low-temperature resistivity of the model is masked by the onset of superconductivity. Experimentally, this regime is often explored by applying a strong magnetic field $B$ to quench the superconducting state. We use a similar approach and add a magnetic field to the uniform model. We find that even the smallest field we can resolve already destroys phase coherence and with it superconductivity, despite the fact that local pairing is only mildly affected. The magnetic field extends the $d\rho/dT > 0$ regime to lower temperatures, with $T^*$ and $T_{\mathrm{min}}$ suppressed together and most strongly at weak coupling.

Our findings have revealed interesting transport behavior in variants of the attractive Hubbard model. More work is needed to validate aspects of the MCMF results by DQMC calculations. Additionally, incorporating physical effects such as the charging energy of the attractive regions may be necessary to make the strange-metallic signatures more robust.

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I would like to thank my amazing wife Lotan, that supported me throughout all of this work.

May our lives be filled with love and curiosity.

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# Chapter 1

# Introduction

A persistent challenge to our understanding of metals is the family of anomalous "strange metals", which clearly violate the well-established Fermi-liquid (FL) theory. The most prominent of these is a resistivity that depends linearly on temperature, $\rho \propto T$, persisting down to very low temperatures [1–3]. In a conventional metal, the low-temperature resistivity takes the form

$$\rho = \rho_0 + \alpha_2 T^2 + \alpha_5 T^5, \tag{1.1}$$

where $\rho_0$ is the residual resistivity from impurity scattering, the $\alpha_2 T^2$ term arises from electron-electron scattering within FL theory, and the $\alpha_5 T^5$ term originates from electron-phonon scattering at low temperatures [4]. A linear phonon contribution does appear above the Debye temperature, but in strange metals the linearity persists far below any phonon scale—where phonons are frozen out—ruling them out as its origin.

The semiclassical picture of charge transport is captured by the Drude formula for the conductivity,

$$\sigma = \frac{ne^2\tau}{m}, \tag{1.2}$$

where $n$ is the density of conducting electrons, $-e$ is the electron charge, $m$ is their effective mass, and $\tau$ is the average time between scattering events. Within this picture, the resistivity $\rho = m/(ne^2\tau)$ grows as the scattering rate $1/\tau$ increases with temperature, and the associated mean free path $l = v_F\tau$ correspondingly shrinks. Once $l$ approaches the lattice constant $a$, the notion of a freely propagating particle between collisions loses meaning. This is the Mott–Ioffe–Regel (MIR) limit, and in conventional metals it manifests as a saturation of the resistivity at high temperature [5]. Strange metals are anomalous because their $\rho(T) \propto T$ continues through and beyond the MIR

value with no saturation, implying that charge is carried by something other than long-lived quasiparticles [6].

A clue to what sets the resistivity comes from the $T$-linear regime itself. Writing the scattering rate as $\hbar/\tau = \alpha k_B T$, one finds across many strange metals—from overdoped cuprates to heavy-fermion compounds—that the coefficient is of order unity, $\alpha \sim 1$ [2, 3]. The relaxation time then follows the single timescale that can be formed from temperature alone, $\tau \sim \hbar/k_B T$, sometimes referred to as "Planckian" dissipation. Such Planckian dissipation is often regarded as the fastest possible relaxation rate allowed by quantum mechanics. The fact that this timescale is set only by fundamental constants and temperature, independent of material-specific parameters such as the Fermi energy, suggests that the strange-metal behavior is governed by a universal mechanism.

Strange-metal behavior is not confined to a single class of materials: it appears in the high-temperature cuprate superconductors, in heavy-fermion compounds, and in a growing list of other correlated systems [6–8]. Within a given material the strange metal is typically reached by tuning an external parameter as doping, pressure, or magnetic field through a particular value. This recurrence across chemically and structurally dissimilar systems, reached by such different tuning knobs, is what motivates the search for a single common mechanism.

A leading idea links strange metallicity to quantum criticality. A quantum critical point (QCP) is a continuous phase transition at zero temperature, reached by tuning a non-thermal parameter; above it at finite $T$, quantum fluctuations are expected to dominate within a widening wedge of the phase diagram called the "quantum critical fan" or quantum critical region (QCR). In this view, the strange-metal phase is the QCR associated with a QCP at $T = 0$. In the cuprates, however, the strange-metal regime occupies a broad region of the temperature–doping plane rather than a narrow fan emanating from a single QCP, and the $T$-linear resistivity persists over an extended range of doping [9]. A further open challenge is constructing a unified theory that can explain the strange-metal behavior consistently across the different materials and tuning parameters.

Disorder appears to be intrinsic to the cuprates rather than incidental. Scanning tunneling spectra of a high-temperature cuprate superconductor are not uniform, varying vastly between regions a few nanometers apart [10]. Because superconductivity is highly sensitive to disorder in two dimensions [11], this nanoscale inhomogeneity has itself been suggested as a possible origin of the strange-metal behavior.

A specific proposal in this spirit is a model of superconducting puddles embedded in a metallic host [12]. Such a system is macroscopically metallic but hosts local superconducting domains, and their interplay together with the charging energy of the puddles is argued to produce an extended $T$-linear resistivity like that of the overdoped cuprates. Another model that incorporates inhomogeneous interactions to explain strange metallicity was put forward in Ref. [13], where electrons are coupled to quantum critical bosonic fluctuations via random couplings. Such granularity or inhomogeneity can emerge spontaneously in a disordered superconductor whose order parameter fragments into puddles the size of the coherence length [14].

In this work we explore two extensions of the attractive Hubbard model, in an effort to understand strange-metal behavior. In the first, we introduce disorder into the interaction by making its strength nonuniform. In the second, we apply a uniform magnetic field to suppress superconductivity to expose the underlying normal state. The same strategy was used experimentally to uncover the low-temperature linear-in-$T$ resistivity of the cuprates [1]. While the microscopics of the high-temperature superconductors can not be mapped onto the attractive Hubbard model (the belief is that it does map on the repulsive version), we take it as a paradigmatic model for probing the role that interaction–inhomogeneity play in superconductivity. As a computational tool we use the Monte Carlo mean-field (MCMF) approximation, which gives the optical conductivity and resistivity directly on the real-frequency axis. This is unlike determinant quantum Monte Carlo (DQMC), where real-frequency transport requires an ill-posed analytic continuation of imaginary-time data. This is what makes the transport study at the heart of this work feasible, though it comes at the cost of neglecting quantum (temporal) fluctuations of the pairing field, whose consequences and the regime where the approximation is reliable we examine in § 5.

Our main results concern the temperature dependence of the regular resistivity $\rho_{xx}^{\mathrm{reg}}$. For most systems we study it has a minimum at a temperature $T_{\mathrm{min}} > T_c$. Below the minimum the resistivity rises and eventually diverges as the system approaches the superconducting state; above it lies a $d\rho/dT > 0$ regime in which the resistivity follows a power law $\rho \sim T^{\alpha}$ with an exponent $\alpha < 2$, a departure from the Fermi-liquid value $\alpha = 2$. We find that adding either interaction disorder or a magnetic field extends this $d\rho/dT > 0$ regime to lower temperatures. Across all of these systems, the temperature of the resistivity minimum $T_{\mathrm{min}}$ tracks the pseudogap scale $T^*$, which we identify from the maximum of the spin susceptibility. For some magnetic field systems we did not find $T_{min}$ within the temperature regime we checked, further research is needed to conclude if there is a finite $T_{min}$ for

all magnetic fields.

The remainder of this thesis is organized as follows. § 2 introduces the attractive Hubbard model, its limits and the known effects of disorder introduced to it. § 3 develops the analytical framework, deriving the Monte Carlo mean-field (MCMF) approximation from the path-integral formulation through a Hubbard–Stratonovich decoupling in the pairing channel, and explains why it is well suited to transport calculations. § 4 details the numerical implementation: the Monte Carlo procedure, the traveling-cluster approximation, the treatment of the chemical potential, and the evaluation of observables from the Bogoliubov–de Gennes spectrum. § 5 benchmarks the method on the uniform system against mean-field and quantum Monte Carlo results, explore the limitations of the MCMF and gives transport results in the clean system. § 6 presents the interaction-disordered system, focusing on systems that maintain the same $T_c$ with changing disorder strength. § 7 extends the study to a uniform magnetic field, examining how it destroys superconductivity and extends the normal state to lower $T$. § 8 summarizes the main findings and outlines directions for future work.

# Chapter 2

# The attractive Hubbard model

## 2.1 Model and phase diagram

The attractive Hubbard model in 2D is defined by the Hamiltonian

$$H - \mu N = -t \sum_{\langle i,j\rangle,\sigma} \left[ c^{\dagger}_{i\sigma} c_{j\sigma} + c^{\dagger}_{j\sigma} c_{i\sigma} \right] - \sum_{i} \mathrm{U}_i n_{i\uparrow} n_{i\downarrow} - \mu \sum_{i,\sigma} n_{i\sigma} \qquad (2.1)$$

where $c_{i\sigma}$ ($c^{\dagger}_{i\sigma}$) are annihilation (creation) operators of an electron at site $i = (i_x, i_y)$, $i_x, i_y = 1, \ldots, L$ on a square lattice (so the lattice has $N_s = L^2$ sites), with spin $\sigma = \uparrow / \downarrow$, and $n_{i\sigma} = c^{\dagger}_{i\sigma} c_{i\sigma}$ is the number operator. The hopping term runs over nearest neighboring sites $\langle i, j\rangle$, and we set its amplitude $t = 1$ as the basic energy scale. The $\mathrm{U}_i$ term describes an on-site s-wave attractive interaction; we will describe first the uniform interaction system $\mathrm{U}_i = \mathrm{U}$. The chemical potential $\mu$ is chosen self-consistently to fix the filling factor $\langle n\rangle = \frac{1}{N_s} \sum_{i,\sigma} n_{i,\sigma}$, which ranges from 0 (empty lattice) to 2 (fully occupied system). Unlike the repulsive Hubbard model, the attractive version is free of the fermionic sign problem, and so can be simulated using quantum Monte-Carlo methods.

For the hole-doped model away from half filling, $0 < n < 1$ (The particle doped case is behaves similarly due to particle-hole symmetry), one finds below a critical temperature $T_c(n, \mathrm{U})$ a superconducting phase. Due to the Mermin-Wagner theorem, this 2D system can not exhibit true long-range order at finite temperatures, and the transition is of the Berezinskii-Kosterlitz-Thouless (BKT) type, associated with the onset of quasi-long-range order characterized by algebraically decaying correlations. This transition is accompanied by a jump in the superfluid stiffness $\rho_s$ at $T_c$ [15]. In a finite

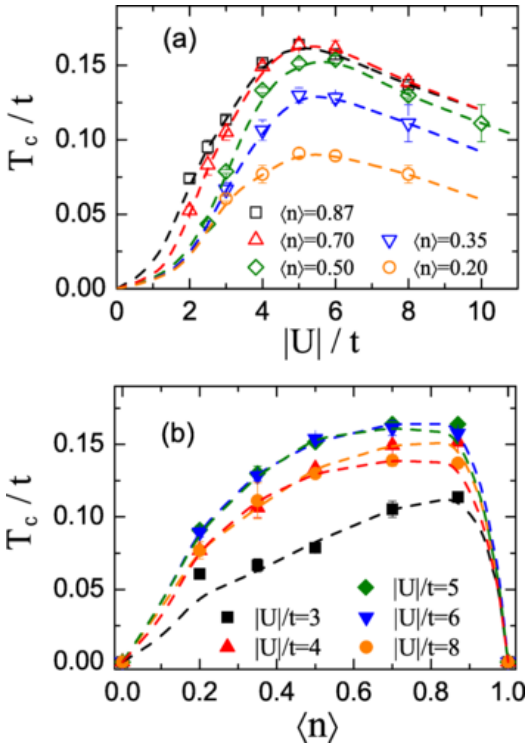



Figure 2.1: The critical temperature $T_c$ of the attractive Hubbard model as a function of (a) the interaction strength $\mathsf{U}$ and (b) the filling $n$. Taken from Ref.[17].

system the transition is rounded, but one can still extract the BKT transition temperature using the Halperin-Nelson criterion $\rho_s(T_c) = 2T_c/\pi$ [16].

Fig. 2.1 depicts the critical temperature $T_c(\mathsf{U}, n)$, as taken from Ref.[17] using a DQMC method. The critical temperature $T_c$ is non-monotonic as a function of $\mathsf{U}$, with a maximum at intermediate coupling and is suppressed at both weak and strong coupling. This is a manifestation of the BCS–BEC crossover, which we discuss below. As a function of doping $n$, $T_c$ is also non-monotonic, with a maximum at intermediate doping. At half filling ($n = 1$), the model acquires an additional pseudospin $SU(2)$ symmetry that rotates the s-wave superconducting order parameter into the staggered charge-density-wave (CDW) order [18]. The two are therefore exactly degenerate, and the order parameter combines into a single three-component vector with continuous $O(3)$ symmetry. Unlike the abelian $O(2)$ case away from half filling—where the Mermin-Wagner theorem forbids long-range order but a BKT transition still gives $T_c > 0$ — the non-abelian $O(3)$ order admits no finite-temperature transition in 2D, forcing $T_c(n = 1) = 0$.

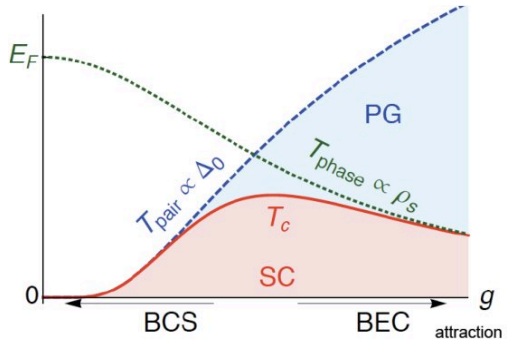



Figure 2.2: A sketch of the BCS-BEC crossover taken from [14]. The critical temerature $T_c$ is given by the minimum of the pair-formation scale $T^* = T_{pair}$ and the phase-coherence scale $T_{phase}$.

A central feature of the attractive Hubbard model is the BCS–BEC crossover, controlled by the ratio $\mathrm{U}/W$ of the interaction strength $\mathrm{U}$ and the bandwidth $W$.

In the weak-coupling (BCS) limit, $\mathrm{U} \ll W$, the gap is exponentially small, $\Delta \sim W \exp(-1/\rho_F \mathrm{U})$, and the coherence length $\xi_{\mathrm{BCS}} \sim v_F/\pi\Delta$ is exponentially large. Pair formation and phase coherence set in together at $T_c$, and standard BCS mean-field theory is accurate. Because $\xi_{\mathrm{BCS}}$ is so large at weak coupling, this limit is also hard to simulate: once $\xi_{\mathrm{BCS}} \gg L$ pairs do not fit in the system and finite-size artifacts contaminate the data.

In the strong-coupling (BEC) limit, $\mathrm{U} \gg W$, the physics is qualitatively different: electrons form tightly bound pairs whose size is comparable to a lattice spacing , with binding energy $\sim \mathrm{U}$. These local pairs behave as hard-core bosons that move only by virtual ionization, with a small effective hopping $t_B \sim t^2/|\mathrm{U}|$ and a correspondingly heavy mass. They form well above $T_c$, at a pseudogap temperature $T^*$ that scales with the binding energy $\mathrm{U}$ [18]. However, they only condense at the lower scale $T_c \sim t^2/\mathrm{U}$ set by their phase stiffness. In the pseudogap regime $T_c < T < T^*$, the amplitude of the superconducting order parameter is well established and the physics is governed by phase fluctuations that are effectively described by an XY model.

The two limits are joined by a smooth crossover with no sharp transition as sketched in Fig. 2.2.

The pairing temperature $T^*$ is usually estimated by the temperature where a gap is developed in the density of states, or where the spin susceptibility $\chi_s$ starts to drop. These two criteria capture the same physics, since the spin

susceptibility is suppressed by the formation of singlet pairs.

## 2.2 Disorder in the Hubbard model

As discussed in Chapter 1, disorder is believed to play a central role in the physics of overdoped cuprates. It is therefore natural to ask how disorder modifies the behavior of the attractive Hubbard model, a question addressed by a substantial body of numerical work. These studies fall into two broad categories, distinguished by the which term of the Hamiltonian includes the randomness.

The first and more studied category is potential disorder, in which a space dependent on-site energy is added, $\mu \to \mu + V_i$, with $V_i$ drawn from some distribution. Quantum Monte Carlo studies established the disorder-driven superconductor–insulator (SI) transition in this model [19], and real-space approaches later showed that with increasing disorder the superconductor breaks into spatially inhomogeneous "puddles" of pairing amplitude embedded in an insulating background, with a pseudogap that survives across the transition [14]. The same form of disorder has also been studied within the Monte Carlo mean-field (MCMF) framework we use here [20].

The second category, and the one relevant to this work, is interaction disorder, in which the attractive coupling itself is made spatially nonuniform, $\mathrm{U} \to \mathrm{U}_i$. In the simplest case the interaction is removed on a fraction of the sites, studied both with quantum Monte Carlo [21] and within MCMF [22]. Other works use a bimodal coupling within MCMF [23] or by a spatially correlated interaction using MF and time dependent Ginzburg–Landau (TDGL) [24]. Across these works, interaction inhomogeneity makes the pairing itself spatially inhomogeneous, with strongly paired regions coexisting with weakly paired ones, and its effect on $T_c$ ranges from enhancement at weak coupling to a complete loss of superconductivity under strong dilution. The effects of interaction disorder rely heavily on the way the disorder is implemented. To the best of our knowledge, however, none of these interaction-disorder studies report the temperature dependence of the regular resistivity, which is the focus of the present work.

# Chapter 3

# Analytical framework

This chapter sets up the analytical framework used throughout the thesis. We cast the attractive Hubbard model as a path integral (§3.1) and decouple the quartic interaction using a complex bosonic pairing field via a Hubbard–Stratonovich transformation (§3.2). The Monte Carlo mean-field (MCMF) approximation is then introduced: treating the pairing field as static in imaginary time (§3.3). We identify the BCS mean-field solution as its saddle point (§3.4). We close with the methodological reason for our choice, a direct access to transport observables (§3.5).

## 3.1 Path-integral formulation

To study the attractive Hubbard model, Eq. (2.1), we express the partition function $Z = \mathrm{Tr} e^{-\beta(H-\mu N)}$ as a functional integral over Grassmann fields $\psi_{i\sigma}(\tau)$ and $\bar{\psi}_{i\sigma}(\tau)$. Following the standard procedure of Trotter decomposition and the insertion of the resolution of identity in the Grassmann coherent state basis, the partition function takes the form:

$$Z = \int \mathcal{D}\bar{\psi}\mathcal{D}\psi \exp\left(-S[\bar{\psi},\psi]\right), \tag{3.1}$$

where the fields satisfy the anti-periodic boundary conditions $\psi_{i\sigma}(\beta) = -\psi_{i\sigma}(0)$. For the Hubbard Hamiltonian, the Euclidean action $S$ is:

$$S = \int_0^\beta d\tau \left[ \sum_{i,\sigma} \bar{\psi}_{i\sigma}(\partial_\tau - \mu)\psi_{i\sigma} - t \sum_{\langle i,j\rangle,\sigma} (\bar{\psi}_{i\sigma}\psi_{j\sigma} + \mathrm{h.c.}) - U \sum_i \bar{\psi}_{i\uparrow}\bar{\psi}_{i\downarrow}\psi_{i\downarrow}\psi_{i\uparrow} \right]. \tag{3.2}$$

The quartic interaction term arises from the normal ordering of the number operators, $n_{i\uparrow}n_{i\downarrow} = c^\dagger_{i\uparrow}c^\dagger_{i\downarrow}c_{i\downarrow}c_{i\uparrow}$. This formulation serves as the starting point for the Hubbard-Stratonovich transformation, which allows for the decoupling of the many-body interaction into a system of non-interacting fermions moving in a fluctuating auxiliary field.

## 3.2 Hubbard–Stratonovich decoupling in the pairing channel

To evaluate the partition function, we decouple the quartic interaction in Eq. (3.2) via a Hubbard–Stratonovich (HS) transformation. This can be done in various ways including the density, spin or pairing channels. While all these choices are equivalent on the exact level, they may generally lead to different results if one employs approximations. We choose the pairing channel because its saddle point corresponds to BCS mean-field theory, making the superconducting physics manifest. Technically, this decoupling scheme introduces a complex auxiliary bosonic field $\Delta_i(\tau)$ through the identity:

$$\begin{aligned} &\exp\left[\mathrm{U}\int d\tau \sum_i \bar\psi_{i\uparrow}\bar\psi_{i\downarrow}\psi_{i\downarrow}\psi_{i\uparrow}\right] = \\ &= \int \mathcal{D}\Delta^*\mathcal{D}\Delta \exp\left[-\int d\tau \sum_i \Big(\tfrac{1}{\mathrm{U}}|\Delta_i|^2 - \Delta_i^*\psi_{i\downarrow}\psi_{i\uparrow} - \Delta_i\bar\psi_{i\uparrow}\bar\psi_{i\downarrow}\Big)\right]. \end{aligned} \tag{3.3}$$

Substituting this into the partition function yields

$$Z = \int \mathcal{D}\Delta^*\mathcal{D}\Delta e^{-S_\Delta} \int \mathcal{D}\bar\psi\mathcal{D}\psi e^{-S_{\Delta,\psi}}, \tag{3.4}$$

where the original interaction is traded for a quadratic coupling to the bosonic field:

$$S_\Delta = \int d\tau \sum_i \frac{1}{\mathrm{U}}|\Delta_i|^2, \tag{3.5}$$

$$\begin{aligned} S_{\Delta,\psi} = \int d\tau \Bigg[ &\sum_{i,\sigma} \bar\psi_{i\sigma}(\partial_\tau - \mu)\psi_{i\sigma} - t\sum_{\langle i,j\rangle,\sigma}\left[\bar\psi_{i\sigma}\psi_{j\sigma} + \bar\psi_{j\sigma}\psi_{i\sigma}\right] \\ &- \sum_i \Delta_i^*\psi_{i\downarrow}\psi_{i\uparrow} - \sum_i \Delta_i\bar\psi_{i\uparrow}\bar\psi_{i\downarrow}\Bigg], \end{aligned} \tag{3.6}$$

The action $S_{\Delta,\psi}$ is now quadratic in the Grassmann fields, allowing the fermions to be integrated out.

## 3.3 The Monte Carlo mean field approximation

Many methods compute observables from this partition function, trading accuracy against computational cost. Exact diagonalization (ED) is exact but restricted to small systems, since the Hilbert space grows exponentially. The quantum Monte Carlo (QMC) family [25–27] samples the HS field — $\Delta_i(\tau)$ in the pairing channel, or any other — over the lattice and imaginary time, thus maintaining both quantum and thermal fluctuations. For the attractive Hubbard model QMC is free of the sign problem [28] and becomes numerically exact in the $\Delta\tau \to 0$ limit, which is computationally expensive. At the cheap end, mean-field theory replaces the field by its single saddle-point value (§3.4). This is computationally light, but the field can no longer fluctuate neither thermally nor quantum-mechanically, so all fluctuation physics is lost.

The Monte Carlo mean-field (MCMF) approximation resides between these extremes. It treats the HS field as static in imaginary time, $\Delta_i(\tau) = \Delta_i$, while still sampling its spatial configuration over the lattice by Monte Carlo. The field can therefore fluctuate thermally, from site to site, but not quantum-mechanically in $\tau$: MCMF keeps the thermal fluctuations and discards the quantum ones.

Discarding the quantum fluctuations has a price. In the exact theory the choice of decoupling channel is a matter of convenience. However, once the quantum fluctuations are gone the choice has physical consequences for the quality of the approximation. Decoupling in the pairing channel biases the model toward $s$-wave superconducting order and gives falsely $T_c > 0$ at half filling, where the enhanced $O(3)$ symmetry of the attractive Hubbard model forbids any finite-temperature transition. Some have attempted to correct this by decoupling in the pairing and charge channels simultaneously [29], bringing the results closer to QMC but still not recovering $T_c = 0$ at half filling. We therefore work only away from half filling.

Two features make MCMF well-suited to our problem. First, it retains the thermal fluctuations of the pairing field in both its amplitude $|\Delta_i|$ and its phase $\arg(\Delta_i)$. The phase fluctuations are particularly important in the strong interaction limit of the 2D Hubbard model, where they drive the BKT transition to the superconducting phase [16, 27, 30, 31]. Because MCMF keeps these thermal fluctuations but discards the quantum ones, the approximation is expected to hold for $T \gtrsim T_c$ — the regime we are after. Second, since the $\Delta$ field is static the electronic dynamics is governed by an effective Bogoliubo–de Gennes Hamiltonian. As we show below, the calculation of its

full quasi-particle spectrum $\varepsilon_\lambda$ at every MC step, makes transport observables directly accessible.

Specifically, in the presence of the static field the partition function becomes

$$Z = \int D\Delta\, D\Delta^*\, e^{-\beta \sum_i \frac{1}{U}|\Delta_i|^2}\, Z_e(\{\Delta_i\}), \tag{3.7}$$

where $Z_e(\{\Delta_i\})$ is the electronic partition function for a fixed configuration $\{\Delta_i\}$, expressed as

$$Z_e(\{\Delta_i\}) = \mathrm{Tr}\left[e^{-\beta K[\{\Delta_i\}]}\right], \tag{3.8}$$

where the effective Hamiltonian is

$$K[\{\Delta_i\}] = -\mu \sum_{i,\sigma} c^\dagger_{i\sigma} c_{i\sigma} - t \sum_{\langle ij\rangle \sigma} \left(c^\dagger_{i\sigma} c_{j\sigma} + \text{h.c.}\right) - \sum_i \left(\Delta^*_i c_{i\downarrow} c_{i\uparrow} + \Delta_i c^\dagger_{i\uparrow} c^\dagger_{i\downarrow}\right). \tag{3.9}$$

We can define a Nambu spinor $\Psi_{i\sigma}$:

$$\mathbf{\Psi} = \begin{pmatrix} c_{1\uparrow} \\ \vdots \\ c_{N_s\uparrow} \\ c^\dagger_{1\downarrow} \\ \vdots \\ c^\dagger_{N_s\downarrow} \end{pmatrix}, \qquad \Psi_{i\uparrow} = c_{i\uparrow}, \qquad \Psi_{i\downarrow} = c^\dagger_{i\downarrow}, \tag{3.10}$$

and with it the effective Hamiltonian $K$:

$$K = \mathbf{\Psi}^\dagger \hat{K} \mathbf{\Psi} - \mu N_s \tag{3.11}$$

where $\hat{K}$ is a $2N_s \times 2N_s$ block matrix:

$$\hat{K} = \begin{pmatrix} h & \Delta \\ \Delta^\dagger & -h^* \end{pmatrix} \tag{3.12}$$

where each block is the size of $N_s \times N_s$ and:

$$\begin{aligned} h_{ij} &= -\mu\delta_{ij} - t\delta_{\langle ij\rangle} \\ \Delta_{ij} &= -\Delta_i \delta_{ij} \end{aligned} \tag{3.13}$$

We can diagonalize it by writing $\hat{K} = U\, D\, U^\dagger$, where $U$ is a unitary matrix and $D$ is diagonal with the quasi-particle energies $\varepsilon_\lambda$ on its diagonal. We

introduce quasi-particle operators $\gamma_\lambda$:

$$\begin{aligned}\gamma_\lambda &= U^\dagger \Psi = \sum_{i,\sigma} U^*_{i\sigma,\lambda} \Psi_{i\sigma} \,, \\ \gamma^\dagger_\lambda &= \Psi^\dagger U = \sum_{i,\sigma} U_{i\sigma,\lambda} \Psi^\dagger_{i\sigma} \,,\end{aligned} \tag{3.14}$$

in terms of which the effective Hamiltonian is diagonal:

$$K = \sum_\lambda \varepsilon_\lambda \gamma^\dagger_\lambda \gamma_\lambda - \mu N_s. \tag{3.15}$$

The electronic partition function is then a product over the single-particle states:

$$Z_e[\{\Delta_i\}] = e^{\beta\mu N_s} \prod_\lambda \Big(1 + e^{-\beta\varepsilon_\lambda(\{\Delta_i\})}\Big) \,. \tag{3.16}$$

Combining these and dropping the constant $\beta\mu N_s$ term gives the MCMF action:

$$\boxed{S_{\mathrm{MCMF}}[\{\Delta_i\}] = \frac{\beta}{U} \sum_i |\Delta_i|^2 \;-\; \sum_\lambda \ln\Big(1 + e^{-\beta\varepsilon_\lambda(\{\Delta_i\})}\Big)} \tag{3.17}$$

and the partition function is now a classical $2N_s$-dimensional integral over a real-valued action:

$$Z_{\mathrm{MCMF}} = \int D\Delta \, D\Delta^* \, e^{-S_{\mathrm{MCMF}}(\{\Delta_i\})}. \tag{3.18}$$

We sample $\{\Delta_i\}$ by Monte Carlo with Boltzmann weight $e^{-S_{\mathrm{MCMF}}}$. At each step we update the configuration and recompute $S_{\mathrm{MCMF}}$ by diagonalizing $\hat{K}[\{\Delta_i\}]$. The same diagonalization yields the quasi-particle spectrum $\{\varepsilon_\lambda\}$ and eigenvectors, so observables can be measured at every Monte Carlo step. We detail the Monte Carlo procedure in §4.1, the observables we compute in §4.5, and the statistical analysis of the measurements in Appendix A.

## 3.4 Relation to BCS theory

The BCS mean-field solution is recovered as the saddle point of the MCMF action. The configuration $\{\Delta_i\}$ that extremizes $S_{\mathrm{MCMF}}$:

$$0 = \frac{\partial S_{\mathrm{MCMF}}}{\partial \Delta^*_i} = \frac{\beta}{U_i} \Delta_i - \frac{\partial}{\partial \Delta^*_i} \ln Z_e, \tag{3.19}$$

$$\frac{\partial}{\partial \Delta_i^*} \ln Z_e = \frac{1}{Z_e} \frac{\partial}{\partial \Delta_i^*} \mathrm{Tr}\left[e^{-\beta K}\right] = \beta \frac{1}{Z_e} \mathrm{Tr}\left[c_{i\downarrow} c_{i\uparrow} e^{-\beta K}\right] = \beta \langle c_{i\downarrow} c_{i\uparrow} \rangle. \quad (3.20)$$

Hence, we obtain the BCS gap equation:

$$\Delta_i = \mathrm{U}_i \langle c_{i\downarrow} c_{i\uparrow} \rangle. \quad (3.21)$$

Equation (3.21) is in general a set of $N_s$ coupled equations. It collapses to a single equation for $\Delta_{\mathrm{MF}}$ if the system is translationally invariant. When disorder in the interaction or in the on-site potential is present the full set must be solved numerically [23, 24].

The MCMF integral over $\{\Delta_i\}$ dresses the BCS mean field with the thermal fluctuations of the amplitude and phase of the pairing field. As $T \to 0$ the thermal fluctuations freeze out, and the MCMF integral is dominated by its saddle point alone, reducing to the plain BCS mean field. But that is exactly the regime where the discarded quantum fluctuations matter most, so MCMF is least reliable well below $T_c$, where it tends to overestimates observables such as the pair correlation function.

## 3.5 Why MCMF is well-suited for transport

Transport observables such as the optical conductivity $\sigma_{xx}^{\mathrm{reg}}(\omega)$ are all built from the current–current correlator $\Lambda_{xx}(q, i\omega_n)$, defined in §4.5.5. QMC samples this correlator on a finite imaginary-time grid, and the real-frequency $\Lambda_{xx}(q, \omega)$ must then be obtained by analytic continuation — a notoriously ill-posed step that strongly amplifies the statistical noise of the simulation [32]. In practice QMC can only deliver proxies for the regular conductivity in the DC limit, and not the quantity itself.

MCMF gives it directly. At every Monte Carlo step the electrons are non-interacting and only coupled to a static $\Delta_i$ configuration. The effective Hamiltonian can be diagonalized so we have the full quasi-particle spectrum in hand. The imaginary-time integral in $\Lambda_{xx}(q, i\omega_n)$ can then be carried out analytically and continued to real frequency by $i\omega_n \to \omega + i\delta$, yielding $\Lambda_{xx}(q, \omega)$ for all real frequencies. The only remaining obstacle is the $\delta \to 0$ extrapolation to every $\omega$ computed. The full derivation is given in Appendix B. The regular conductivity, and hence the DC resistivity, is therefore directly accessible in MCMF whereas it is largely out of reach of QMC. This is the central methodological reason for our choice of MCMF.

# Chapter 4

# Numerical implementation

This chapter lays out the numerical implementation of the MCMF approximation introduced in Ch. 3. The simulation pipeline samples the static pairing field $\{\Delta_i\}$ by Metropolis Monte Carlo (§4.1), accelerated by the Traveling Cluster Approximation (§4.2), with the filling factor fixed self-consistently by the chemical potential (§4.3). At every Monte Carlo step we diagonalize the BdG matrix to obtain the quasi-particle spectrum $\{\varepsilon_\lambda, U\}$. In §4.4 we explain how this spectrum is used to calculate expectation values of observables, and then in §4.5 discuss the details of calculating the various observables that are reported throughout Chapters 5–7.

## 4.1 Monte Carlo method

The MCMF action of Eq. (3.17) defines a classical statistical-mechanics problem for the field $\{\Delta_i\}$ with Boltzmann weights $e^{-S_{\mathrm{MCMF}}(\{\Delta_i\})}$. We sample this distribution by Metropolis Monte Carlo, which preserves detailed balance.

We initialize $\{\Delta_i\}$ with the solution of the self-consistent BCS gap equation (3.21) plus small random noise in amplitude and phase, and compute the initial action $S^0_{\mathrm{MCMF}}$ by diagonalizing $\hat{K}(\{\Delta_i\})$ to obtain the spectrum $\{\varepsilon^0_\lambda\}$. At each MC step, the configuration is updated locally on a random site $j$, by adding a small random perturbation $\Delta_j \to \Delta_j + \delta\Delta_j$, where the real and imaginary parts of $\delta\Delta_j$ are independent random variables drawn from a Gaussian distribution $\mathcal{N}(0, (0.05\,\mathrm{U})^2)$. With the new configuration $\{\Delta_i\}'$, the action is recomputed. The update is accepted with probability determined by the Metropolis algorithm $p = \min(1, e^{-\Delta S})$, where

$$\Delta S = S_{\mathrm{MCMF}}(\{\Delta_i\} + \delta\Delta_j) - S_{\mathrm{MCMF}}(\{\Delta_i\}). \tag{4.1}$$

One MC sweep consists of $N_s$ such attempted updates, so that on average each site is offered an update.

We discard the first $\sim 5 \times 10^3$ sweeps as thermalization, and then record observables once every 20–100 sweeps. Because the field changes only slightly per sweep, consecutive measurements remain correlated. We use binning analysis to determine the real error bars and the effective number of independent samples, as detailed in Appendix A.

## 4.2 Traveling Cluster Approximation

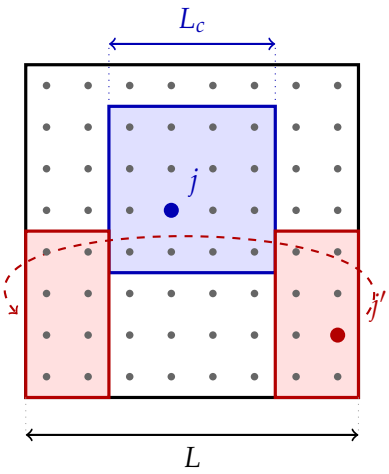
The Monte Carlo sweep of §4.1 is computationally expensive: each of the $N_s$ attempted updates requires a full diagonalization of the $2N_s \times 2N_s$ BdG matrix at cost $\mathcal{O}(N_s^3) = \mathcal{O}(L^6)$ [33], giving $\mathcal{O}(L^8)$ per sweep. The Traveling Cluster Approximation (TCA) [34] reduces this cost by exploiting the locality of the update: changing $\Delta$ at a site is expected to affect predominantly its neighborhood, and the change in the action may be approximated from a local cluster around it.

Specifically, we consider an $L_c \times L_c$ cluster (with $L_c \leq L$) around the updated site $j$ with periodic boundary conditions, see Fig. 4.1a. We then diagonalize the BdG matrix $\hat{K}_c(\{\Delta_i\}; j)$ restricted to that cluster and use its eigenvalues $\{\varepsilon_\lambda^c\}$ to construct the TCA action

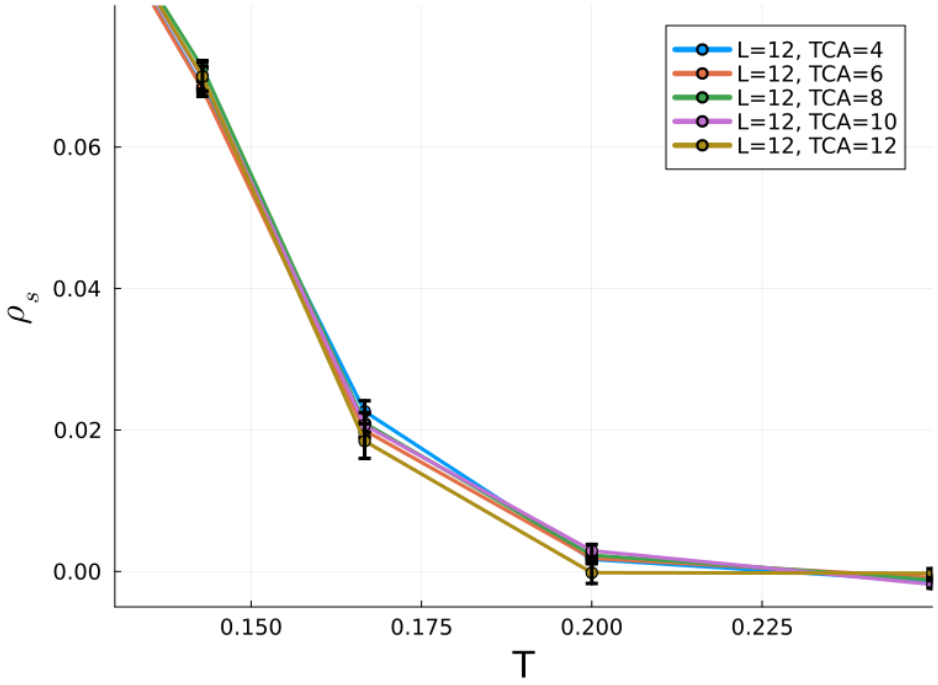
$$S_{\mathrm{TCA}}(\{\Delta_i\}; j) = \beta \sum_{i \in \mathrm{cluster}} \frac{|\Delta_i|^2}{\mathrm{U}_i} - \sum_{\lambda=1}^{2L_c^2} \ln\left(1 + e^{-\beta \varepsilon_\lambda^c}\right), \tag{4.2}$$

which enters the Metropolis test. The accept/reject step is otherwise identical to §4.1. The cluster diagonalization costs $\mathcal{O}(L_c^6)$, bringing the per-sweep cost to $\mathcal{O}(L^2 L_c^6)$ — a speedup of $(L/L_c)^6$ over the full-system MC. Crucially, the cluster spectrum is used only to sample $\{\Delta_i\}$; once a measurement is to be made, we diagonalize the full $2N_s \times 2N_s$ BdG matrix to obtain the exact spectrum and eigenvectors used for the computation of observables.

We note that the TCA approximation suffers from two problems. First, it misses correlations longer than $L_c$, as it can only captures correlations within the cluster. TCA is therefore unreliable in the deep-BCS limit ($\mathrm{U}/W \ll 1$, $\xi_{\mathrm{BCS}} \gg a$). The second problem is that detailed balance is not globally preserved. While detailed balance is maintained within the cluster, the global configuration update is not guaranteed to satisfy it. Given these issues, we have checked the validity of the TCA when we used it. Fig. 4.1b shows an example: $\rho_s$ of an $L = 12$ system has converged by $L_c \approx L/2$.




(a)




(b)

Figure 4.1: (a) Two $L_c = 4$ clusters on an $L = 8$ lattice, each built around its central site (large filled circle). The interior cluster (blue) fits entirely within the lattice; the edge cluster (red) wraps across the periodic boundary in $x$, the portion exiting the right edge re-entering at the left. (b) The superfluid stiffness for an $L = 12$, $\mathrm{U} = 4$, $n = 0.875$ system at several cluster sizes $L_c$; by $L_c \approx L/2$ the approximation is well converged.

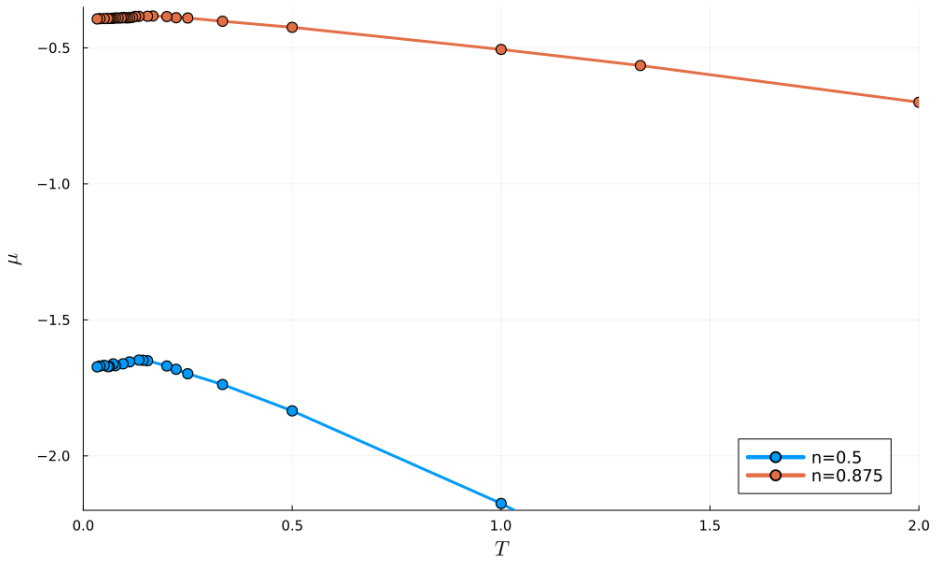



Figure 4.2: The chemical potential used to maintain constant filling $n = 0.5$ and $0.875$ at $L = 26$, $L_c = 10$, $\mathrm{U} = 4$.

## 4.3 The chemical potential

In the grand canonical formulation of §3.1 the chemical potential $\mu$ is the input and the filling $n$ is an observable. To study the system at a target filling we therefore tune $\mu$. The filling is defined by:

$$n = \langle \hat{n} \rangle = \frac{\langle \hat{N} \rangle}{N_s} = \frac{1}{N_s} \sum_{i,\sigma} \langle c^\dagger_{i\sigma} c_{i\sigma} \rangle. \tag{4.3}$$

For each simulation $(\mathrm{U}, T, n)$ we determine $\mu(T; \mathrm{U}, n)$ by a binary search: at each step we run a short simulation (thermalization plus $\sim 16$ uncorrelated measurements of $n$) and bracket $\mu$ until $|n - n_{\text{target}}| < 10^{-3}$. For disordered systems (Ch. 6) we repeat the search per disorder realization. Fig. 4.2 shows the resulting $\mu(T; \mathrm{U}, n)$.

Our $\mu$ values are offset from the conventional textbook form by $\mathrm{U}n/2$. The shift comes from the decoupling choice: had we decoupled the interaction in the density channel, we would have picked up a Hartree contribution $\mu \to \mu + \mathrm{U}n/2$ to the chemical potential. Decoupling in the pairing channel (§3.2) drops this term, so our half-filling sits at $\mu = 0$ rather than the conventional $\mu = -\mathrm{U}/2$.

## 4.4 Computing observables from the BdG spectrum

At every MC step we diagonalize the BdG $\hat{K}[\{\Delta_i\}]$ matrix to obtain the quasi particle spectrum and eigenvectors $\{\varepsilon_\lambda, U_{i\sigma,\lambda}\}$. In the following we outline how this information is used to calculate various correlation functions.

The effective Hamiltonian $K$ is diagonal in the $\gamma_\lambda$ operators, Eq. (3.14), Hence,

$$\langle \gamma_\lambda^\dagger \gamma_\mu \rangle = \delta_{\lambda\mu} f(\varepsilon_\lambda), \qquad f(\varepsilon) = \frac{1}{e^{\beta\varepsilon}+1}, \tag{4.4}$$

where $f(\varepsilon)$ is the Fermi–Dirac function. The time dependence of the correlation function is determined by the evolution of these operators, which in imaginary time takes the form

$$\gamma_\lambda(\tau) = e^{-\varepsilon_\lambda \tau}\gamma_\lambda, \qquad \gamma_\lambda^\dagger(\tau) = e^{+\varepsilon_\lambda \tau}\gamma_\lambda^\dagger. \tag{4.5}$$

To calculate the various electronic correlation functions that we discuss in the nest section, we express the electronic operators according to

$$c_{i\uparrow}(\tau) = \sum_\lambda U_{i\uparrow,\lambda}\, e^{-\varepsilon_\lambda \tau}\, \gamma_\lambda, \qquad c_{i\downarrow}(\tau) = \sum_\lambda U^*_{i\downarrow,\lambda}\, e^{+\varepsilon_\lambda \tau}\, \gamma_\lambda^\dagger. \tag{4.6}$$

Several of these correlation functions (like the pair-field structure factor $P_s$ and the current–current correlator $\Lambda_{xx}$) involve four fermionic operators. For any fixed configuration $\{\Delta_i\}$ the effective Hamiltonian is quadratic in the fermions, so by Wick's theorem:

$$\langle O_1 O_2 O_3 O_4 \rangle = \langle O_1 O_2 \rangle \langle O_3 O_4 \rangle - \langle O_1 O_3 \rangle \langle O_2 O_4 \rangle + \langle O_1 O_4 \rangle \langle O_2 O_3 \rangle. \tag{4.7}$$

Both normal ($\langle c^\dagger c \rangle$) and anomalous ($\langle cc \rangle$, $\langle c^\dagger c^\dagger \rangle$) pairings appear, since the pairing field $\{\Delta_i\}$ breaks particle-number conservation.

One can show that the BdG matrix obeys $\hat{C}^\dagger \hat{K} \hat{C} = -\hat{K}$ where $\hat{C} = \tau_2 \mathcal{K}$ with $\tau_2 = \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix}$ and $\mathcal{K}$ is the complex conjugation operator. Consequently, each state with eigenenergy $\varepsilon_\lambda$ and eigenvector $U_{i,\sigma,\lambda}$ is accompanied by a state with $\varepsilon_{\bar{\lambda}} = -\varepsilon_\lambda$ and eigenvector $U_{i,\sigma,\bar{\lambda}} = e^{i\theta_\lambda}\tau_2 U^*_{i,\sigma,\lambda}$ where $\theta_\lambda$ is a global phase. As a result,

$$U_{i\uparrow,\bar{\lambda}} = -e^{i\theta_\lambda} U^*_{i\downarrow,\lambda}, \qquad U_{i\downarrow,\bar{\lambda}} = e^{i\theta_\lambda} U^*_{i\uparrow,\lambda}. \tag{4.8}$$

The consequence $|U_{i\uparrow,\lambda}|^2 = |U_{i\downarrow,\bar{\lambda}}|^2$ simplifies several observable expressions in Appendix C.

## 4.5 Observables

This section lists the observables that we compute and gives their operator-level definitions, with the explicit BdG-eigenvector forms collected in Appendices C and B.

### 4.5.1 HS-field amplitudes

The simplest pairing diagnostic involves the HS field configuration $\{\Delta_i\}$ directly, with no BdG diagonalization needed. Because this is a complex field, we can define two amplitudes:

$$\Delta_p = \frac{1}{N_s}\sum_i |\Delta_i|, \qquad \Delta_c = \frac{1}{N_s}\Big|\sum_i \Delta_i\Big|. \tag{4.9}$$

$\Delta_p$ measures the local pair amplitude regardless of phase, whereas $\Delta_c$ is also a measure of the degree of phase coherence. The two coincide when phases are aligned and differ when they fluctuate. Hence $\Delta_c \leq \Delta_p$ always.

### 4.5.2 Pair-field structure factor

The $s$-wave pair-field structure factor is [17, 27]

$$P_s = \langle \hat{\Delta}^\dagger \hat{\Delta} + \hat{\Delta}\hat{\Delta}^\dagger \rangle, \qquad \hat{\Delta}^\dagger = \frac{1}{\sqrt{N_s}}\sum_i \hat{\Delta}_i^\dagger, \qquad \hat{\Delta}_i^\dagger = c_{i\uparrow}^\dagger c_{i\downarrow}^\dagger. \tag{4.10}$$

It obeys $P_s = 2\sum_l P_s(l) + \langle n \rangle - 1$ where [19]

$$P_s(l) = \frac{1}{N_s}\sum_i \langle \hat{\Delta}_i \hat{\Delta}_{i+l}^\dagger \rangle \tag{4.11}$$

is the real-space pair correlator. We expect the spatial decay of $P_s(l)$ to control the scaling of $P_s$, as we discuss in Chapter 5.

### 4.5.3 Density of states

The particle density of states is

$$\rho(\omega) = -\frac{1}{\pi}\sum_{i,\sigma} \mathrm{Im}\, G_{i\sigma,i\sigma}(i\omega_n \to \omega + i\delta), \tag{4.12}$$

with $G_{i\sigma,j\sigma'}(i\omega_n) = -\int_0^\beta d\tau\, e^{-i\omega_n\tau}\langle c_{i\sigma}(\tau)c^\dagger_{j\sigma'}(0)\rangle$ the Matsubara one-particle Green's function. The analytic continuation $i\omega_n \to \omega + i\delta$ is obtained in practice by choosing a width $\delta = 1\,\mathrm{MLS} = 4t/N_s$, where MLS denotes the mean level spacing of the BdG spectrum.

### 4.5.4 Spin susceptibility

The uniform static spin susceptibility is [35]

$$\chi_s = 3\int_0^\beta d\tau\, \langle \hat{S}_z(\tau)\hat{S}_z(0)\rangle, \qquad \hat{S}_z(\tau) = \frac{1}{2}\sum_i (n_{i\uparrow}(\tau) - n_{i\downarrow}(\tau)). \tag{4.13}$$

The spin susceptibility is suppressed when the system forms singlet pairs. Thus, $\chi_s$ can be used to track the onset of local pairing correlations.

### 4.5.5 The superfluid stiffness, Drude weight and conductivity

The Kubo formula for the linear response of a system to a vector potential $A_x(q,\omega)$ is [26]

$$\langle j_x(q,\omega)\rangle = -e^2[\langle -K_x\rangle - \Lambda_{xx}(q,\omega)]\, A_x(q,\omega). \tag{4.14}$$

Here, $K_x = -\frac{t}{N_s}\sum_{i,\sigma}\left(c^\dagger_{i+\hat{x},\sigma}c_{i,\sigma} + c^\dagger_{i,\sigma}c_{i+\hat{x},\sigma}\right)$ is the kinetic energy in the $x$ direction and $\Lambda_{xx}(q,\omega)$ is the current–current correlator

$$\Lambda_{xx}(q,i\omega_n) = \frac{1}{N_s}\int_0^\beta d\tau\, e^{i\omega_n\tau}\,\langle j_x(q,\tau)\, j_x(-q,0)\rangle, \tag{4.15}$$

with $j_x(q,\tau) = it\sum_{i,\sigma} e^{-iq\cdot r_i}\, e^{H\tau}\left(c^\dagger_{i+\hat{x},\sigma}c_{i,\sigma} - c^\dagger_{i,\sigma}c_{i+\hat{x},\sigma}\right)e^{-H\tau}$, the current-density operator. The behavior for real frequency $\omega$ follows by taking $i\omega_n \to \omega + i\delta$. The explicit BdG-eigenvector form of $\Lambda_{xx}$ is given in Appendix B.5. The current–current correlator encodes both thermodynamic properties and the transport response of the system, depending on the order of limits $q \to 0$ and $\omega \to 0$.

The static ($\omega = 0$) long-wavelength ($q \to 0$) limit of $\Lambda_{xx}$ can be taken in two different ways, longitudinal and transverse, depending on the direction of $q$:

$$\begin{aligned} \Lambda^L &\equiv \Lambda_{xx}(q_x \to 0,\, q_y = 0,\, \omega = 0), \\ \Lambda^T &\equiv \Lambda_{xx}(q_x = 0,\, q_y \to 0,\, \omega = 0). \end{aligned} \tag{4.16}$$

Gauge invariance fixes the longitudinal limit to be $\Lambda^L = -\langle K_x \rangle$ (see derivation in Appendix B.2). We use this identity as a check for our simulations. The transverse limit $\Lambda^T$ is not constrained by gauge invariance. Its deviation from $\Lambda^L$ defines the superfluid stiffness $\rho_s$ [17, 19, 25–27],

$$\rho_s = \tfrac{1}{4}\Big[\langle -K_x \rangle - \Lambda^T\Big]. \tag{4.17}$$

In practice, the limit $q \to 0$ is taken by calculating $\Lambda_{xx}$ for the lowest available momentum $q_y = \frac{2\pi}{L_y}$. The sensitivity to the order of limits is a signature of long-range phase coherence in the superconducting state.

The uniform ($q = 0$) finite $\omega$ limit of $\Lambda_{xx}$ yields a real-frequency conductivity whose real part decomposes as

$$\mathrm{Re}[\sigma_{xx}(\omega)] = D\,\delta(\omega) + \sigma_{xx}^{\mathrm{reg}}(\omega). \tag{4.18}$$

The weight $D$ of the zero-frequency $\delta$-function is the Drude weight, while $\sigma_{xx}^{\mathrm{reg}}(\omega)$ is the regular part of the real conductivity. They are given by

$$D = -e^2\pi[\langle K_x \rangle + \mathrm{Re}\,\Lambda_{xx}(q = 0,\, \omega \to 0)]\,, \tag{4.19}$$

and

$$\sigma_{xx}^{\mathrm{reg}}(\omega) = \frac{e^2}{\omega}\,\mathrm{Im}[\Lambda_{xx}(q = 0, \omega)]. \tag{4.20}$$

To compute these observables numerically, we extrapolate $\delta \to 0$ for every $\omega$ and we detail the procedure in Appendix B.6. In practice, the limit of $\omega \to 0$ for $D$ and $\sigma_{xx}^{\mathrm{reg}}$ is taken at frequencies smaller than the mean level spacing (MLS)$= 4t/N_s$, where the observables saturate as a function of $\omega$.

The regular resistivity follows by inverting the conductivity tensor, $\rho_{\mu\nu}^{\mathrm{reg}}(\omega) = \big(\sigma_{\mu\nu}^{\mathrm{reg}}(\omega)\big)^{-1}$, which reduces to $\rho_{xx}^{\mathrm{reg}} = 1/\sigma_{xx}^{\mathrm{reg}}$ in zero magnetic field.

# Chapter 5

# The uniform system

In this chapter we study the model with a uniform (site-independent) attractive interaction, which serves as the baseline for the disordered and magnetic-field systems of the following chapters. We open by benchmarking the MCMF scheme against QMC and the simpler MF approximation (§5.1): we compare the pairing amplitudes to the MF gap, analyze the high-temperature statistics of the Hubbard–Stratonovich pairing field. We extract the critical temperature $T_c$ from the $s$-wave pair-field structure factor $P_s$ and the superfluid stiffness $\rho_s$. Equipped with $T_c$, we map the BCS–BEC crossover via the dome in $T_c(U)$ (§5.2), and then probe the normal state above it: we identify the pseudogap regime $T_c < T < T^*$ from the density of states and the spin susceptibility, the latter fixing the pseudogap temperature $T^*$ (§5.3). Finally, we compute the temperature dependence of the Drude weight and the DC resistivity, concentrating on the regime above $T_c$ (§5.4).

Unless stated otherwise, all results in this chapter use the traveling-cluster approximation with cluster size $L_c = 10$ (or the full system when $L \leq 10$). All temperatures and energies are reported in units of the hopping $t$.

## 5.1 Benchmarks against MF and QMC

### 5.1.1 Pairing diagnostics

To build confidence in the MCMF scheme, we first benchmark it against the MF gap $\Delta_{\mathrm{MF}}$, the self-consistent solution of the BCS gap equation (3.21). Fig. 5.1 overlays the MCMF amplitudes $\Delta_p$ and $\Delta_c$ (Eq. (4.9)) on $\Delta_{\mathrm{MF}}$. The three agree as $T \to 0$ — where MCMF reduces to BCS-MF — but depart

on warming: $\Delta_c$ collapses at a lower temperature than $\Delta_{\mathrm{MF}}$, so $T_{c,\mathrm{MCMF}} < T_{c,\mathrm{MF}}$. In MF there are no phase fluctuations to destroy the long-range order, so it shows a genuine phase transition. Conversely, in MCMF these fluctuations survive and leave only a BKT transition. Their inclusion is what reduces $T_c$. Because $\Delta_c$ rises too gradually to locate $T_c$ precisely, later this chapter we turn to two different probes, the structure factor $P_s$ and the superfluid stiffness.

The two amplitudes differ in their system-size dependence: $\Delta_p$ is essentially $L$-independent, marking it as a local quantity, whereas $\Delta_c$ varies with $L$ at all temperatures, reflecting its long-range character. In App. D we show that the finite-size scaling (FSS) of $\Delta_c$ parallels that of $P_s$, consistent with the two being closely related measures of superconducting order.

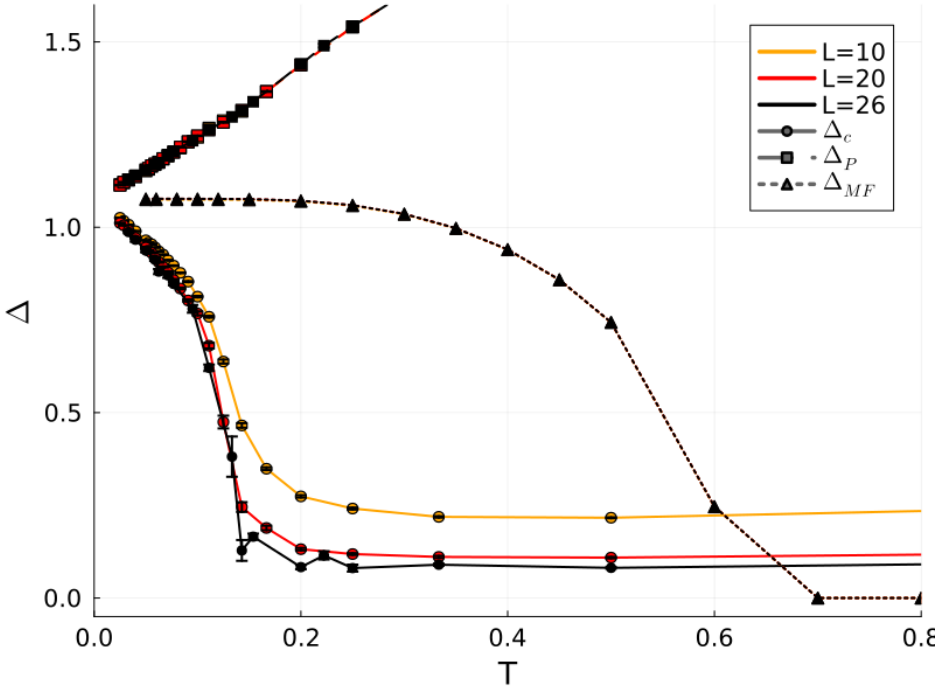



Figure 5.1: The condensate amplitude $\Delta_c$ (circles) and pairing amplitude $\Delta_p$ (squares) of the HS field, averaged over the lattice, vs. the MF energy gap $\Delta_{\mathrm{MF}}$ (triangles) obtained by solving the BCS gap equation (3.21). Different colors represent different system sizes. At these system sizes the MF solution is barely dependent on $L$.

Fig. 5.2 shows the high-temperature behavior of $\Delta_c$ and $\Delta_p$ for several interaction strengths U. Both grow with $T$ and, beyond a crossover scale that

increases with U, satisfy $\Delta_c = \Delta_p / L$. This relation is expected if the real and imaginary parts of $\Delta_i$ are independent, identically distributed (IID), and Gaussian.

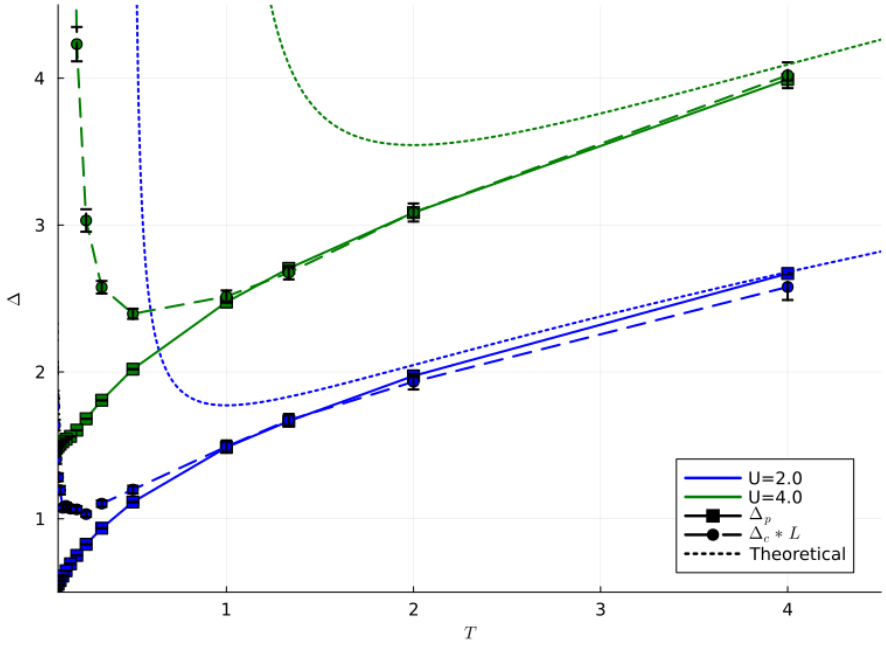



Figure 5.2: $\Delta_c L$ and $\Delta_p$ as a function of $T$ for different interaction strengths U for a system size $L = 16$. Also plotted is the theoretical value for $\Delta_p$ from the high-$T$ approximation to the action, Eq. (5.5).

Assuming the IID condition, the real and imaginary parts of $\Delta_i = x_i + iy_i$ have zero mean and variance $\sigma^2$. Define the complex lattice average

$$S = \frac{1}{N_s} \sum_i \Delta_i = A + iB, \tag{5.1}$$

so that $\Delta_c = \langle |S| \rangle$. By the central limit theorem (CLT), $A$ and $B$ are then normally distributed with variance $\sigma_S^2 = \sigma^2 / N_s$. Fig. 5.3 shows the distribution of $S$ over the measurement stream. At low temperatures it is non-Gaussian: the real and imaginary parts are correlated, forming a ring in the complex plane. At high temperatures it becomes Gaussian with zero mean — consistent with the $\Delta_i$ being IID, though not a proof of it. The change in the distribution of $S$ is a clear signature of the normal-to-superconducting transition, marking the onset of phase coherence at low $T$.

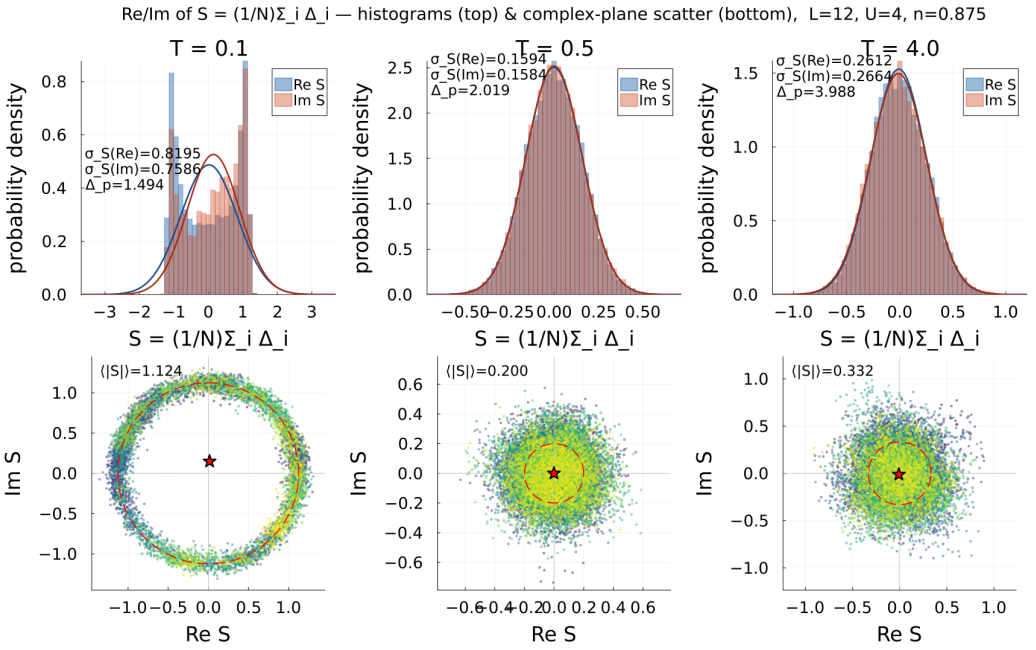



Figure 5.3: Distribution of the per-measurement complex value of $S = \frac{1}{N_s}\sum_i \Delta_i$ for $\mathrm{U} = 4$, $n = 0.875$, $L = 12$ at $T = 0.1$, $0.5$ and $4.0$. Top: histograms of $\Re[S]$ (blue) and $\Im[S]$ (orange) with zero-mean Gaussian fits (solid); Bottom: scatter of $(\Re[S], \Im[S])$ in the complex plane; the color represents "MC time". The complex average is given as a red star, and the red dashed circle marks $\langle|S|\rangle = \Delta_c$.

Since $S$ is a complex Gaussian at high $T$, its amplitude $|S|$ is Rayleigh distributed, with mean $\Delta_c = \langle|S|\rangle = \sqrt{\frac{\pi}{2}}\sigma_S$. If $x_i$ and $y_i$ are themselves IID and Gaussian, then $|\Delta|$ is likewise Rayleigh distributed, giving

$$\Delta_c = \sqrt{\frac{\pi}{2}}\frac{\sigma}{L} = \frac{\langle|\Delta|\rangle}{L} = \frac{\Delta_p}{L}. \tag{5.2}$$

Gaussianity is sufficient but not necessary here: other distributions can be fine-tuned to reproduce this relation.

We can derive this Gaussian behavior directly by expanding the MCMF action $S_{\mathrm{MCMF}}$ (Eq. (3.17)) at high $T$, writing $\ln(1+e^{-\beta\varepsilon_\lambda})$ as a power series in $x = \beta\varepsilon_\lambda$. To second order, the action reads:

$$S_{\mathrm{MCMF}} \approx \frac{\beta}{\mathrm{U}}\sum_i |\Delta_i|^2 \; + \frac{\beta}{2}\sum_\lambda \varepsilon_\lambda - \frac{1}{8}\beta^2\sum_\lambda \varepsilon_\lambda^2 + const. \tag{5.3}$$

Since the eigenvalues of the BdG matrix come in pairs of equal magnitude but opposite signs, the second term in Eq. (5.3) vanish. The third can be rewritten as $\sum_\lambda \varepsilon_\lambda^2 = \mathrm{Tr}\, H_{BdG}^2 = 2\sum_i |\Delta_i|^2 + 2N_s(4t^2+\mu^2)$, giving:

$$S_{\mathrm{MCMF}} \approx \frac{\beta}{\mathrm{U}}(1-\frac{1}{4}\beta\mathrm{U})\sum_i |\Delta_i|^2 + const. \tag{5.4}$$

The action is now quadratic in $|\Delta_i|$, so $\Re[\Delta_i]$ and $\Im[\Delta_i]$ are Gaussian with variance $\sigma^2 = \frac{1}{2}\frac{T\mathrm{U}}{1-\frac{\mathrm{U}}{4T}}$, and the pairing amplitude becomes

$$\Delta_P = \Delta_c L = \sqrt{\frac{\pi}{2}}\sigma = \frac{\sqrt{\pi}}{2}\sqrt{\frac{T\mathrm{U}}{1-\frac{\mathrm{U}}{4T}}}. \tag{5.5}$$

Fig. 5.2 shows that the approximation matches the data at high $T$, with the agreement extending to lower $T$ for smaller $\mathrm{U}$. This is expected: $\Delta_p$ grows with $T$, so at high $T$ the bandwidth is set by the pairing, $W \approx \Delta_p \gg \mu, t$, and the expansion parameter $x = \beta\varepsilon_\lambda \leq \beta W \approx \frac{\sqrt{\pi}}{2}\sqrt{\frac{\mathrm{U}}{T(1-\frac{\mathrm{U}}{4T})}}$ vanishes as $\beta \to 0$, making the expansion self-consistent. It is controlled for $T \gg \frac{\pi+1}{4}\mathrm{U} \approx \mathrm{U}$. This agrees with Fig. 5.2, where the approximation holds for $T \gtrsim \mathrm{U}$.

Carrying the expansion, Eq. (5.3), to the next order introduces an interacting term of the form $t^2\sum_{ij}|\Delta_i||\Delta_j|\cos(\theta_i-\theta_j)$, where $\theta_i$ are the phases of $\Delta_i$, reducing the model to an effective XY model that must be sampled by Monte Carlo. Simulating this XY model and comparing it to MCMF is an interesting direction, which we leave for future work.

Let us comment that Eq. (5.3) suggests that one can obtain $S_{\mathrm{MCMF}}$ by computing the trace of powers of the BdG matrix. This has the potential of accelerating the Monte Carlo simulations. However, the Taylor series of $\ln(1+e^{-x})$ converge only for $x < \pi$, and therefore can not be used at arbitrarily low temperatures.

### 5.1.2 The $s$-wave pairing structure factor $P_s$

Fig. 5.4 compares the $s$-wave pair-field structure factor $P_s(T)$ (Eq. (4.10)) across MF, MCMF, and the QMC benchmark of Ref. [27]. At high $T$ all three agree and are system-size independent. Below a characteristic temperature the curves for different $L$ split apart — this is the finite-size signature of the superconducting transition that we use to locate $T_c$. The critical temperatures are ordered as $T_{c,\mathrm{MCMF}} < T_{c,\mathrm{QMC}} < T_{c,\mathrm{MF}}$: MF overestimates $T_c$,

lying well above QMC (as $\Delta_c$ did in Fig. 5.1), whereas MCMF underestimates it, but by a smaller margin. Separately, the methods differ in the value of $P_s$ itself: at low $T$, MCMF overshoots the QMC result and approaches the MF value. This stems from the fluctuations each model retains — none in MF, thermal only in MCMF, and both thermal and quantum in QMC. As $T \to 0$ the thermal fluctuations freeze out, so MCMF approaches MF, while the quantum fluctuations captured only by QMC suppress the pairing, so that its $P_s$ saturates at a lower value.

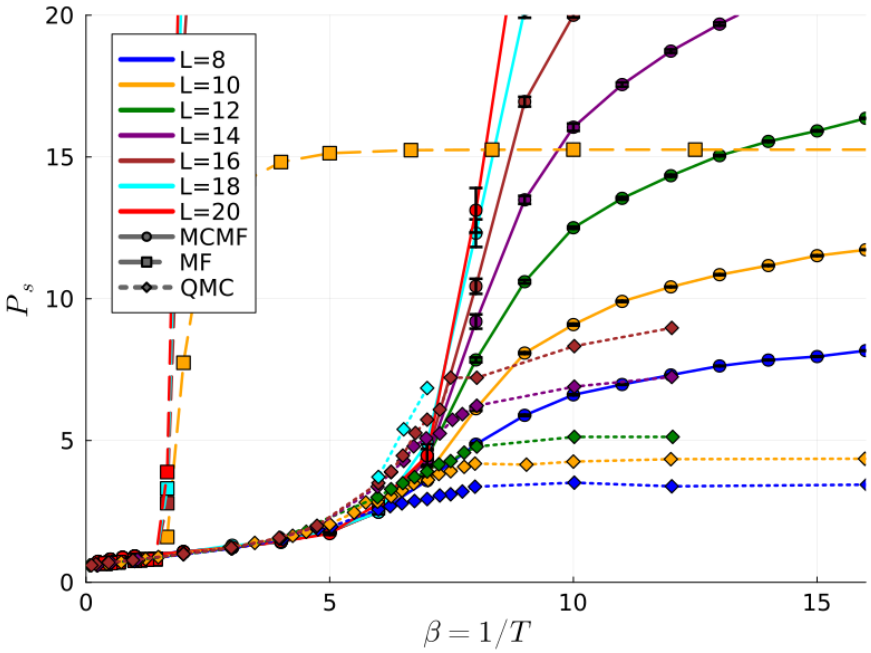



Figure 5.4: The s-wave pair-field structure factor $P_s$ for the MF (squares), MCMF (circles), and QMC (diamonds) benchmark (from Ref. [27]). The interaction strength is $\mathrm{U} = 4$ and the filling is $n = 0.5$. Different system sizes are denoted by different colors as given in the legend. The QMC benchmark has error bars, but we have stripped them to make it easier on the eye.

The real-space pair correlation $P_s(l) \propto \langle c^\dagger_{i\uparrow} c^\dagger_{i\downarrow} c_{i+l\downarrow} c_{i+l\uparrow} \rangle$ is governed by the BKT transition: above $T_c$ it decays exponentially with correlation length $\xi(T) = \exp[A/(T - T_c)^{1/2}]$, while below $T_c$ it decays algebraically with an exponent $\eta(T)$ varying from $\eta(T_c) = 1/4$ to $\eta(0) = 0$ [15]. Since $P_s$ scales as the integral of $P_s(l)$ over the 2D system, its finite-size scaling reflects this decay, and just above $T_c$ we expect [27]:

$$P_s(T) = L^{2-\eta(T_c)}\, f\Big(\frac{L}{\xi(T)}\Big)\,. \tag{5.6}$$

Where $f(x)$ is an unknown scaling function. We extract $T_c$ from finite size scaling (FSS) of $P_s$ by collapsing the data onto this form and fitting $A$ and $T_c$, with the full procedure given in App. D. For $\mathrm{U} = 4.0$, $n = 0.5$ this yields $T_c^{P_s} \approx 0.125 \pm 0.005$ (Fig. 5.5), consistent with the $T_c^{P_s,\mathrm{QMC}} = 0.13$ that Ref. [27] obtained the same way.

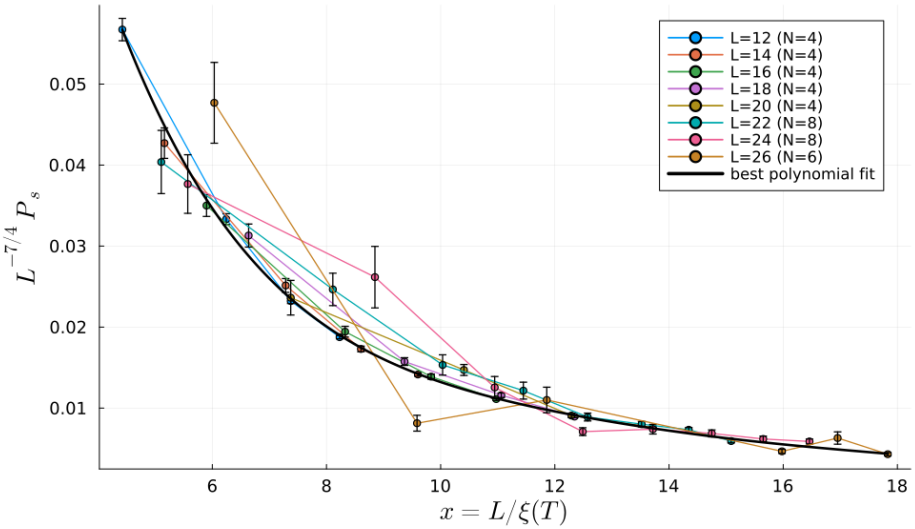



Figure 5.5: Finite-size scaling collapse of $L^{-7/4}P_s$ vs. $L/\xi(T)$ for $\mathrm{U} = 4$, $n = 0.5$, using system sizes $L = 12$–$26$. The fit procedure for $A$ and $T_c$ is detailed in App. D.

### 5.1.3 The superfluid stiffness

Extracting $T_c$ from $P_s$ is somewhat indirect, relying on the FSS form and the BKT correlation length. A more robust route is the crossing of the superfluid stiffness $\rho_s$ (Eq. (4.17)) with the BKT universal-jump line $\rho_s(T_c) = \frac{2}{\pi}T_c$ [16]. In computing $\rho_s$, we approximate the $q_y \to 0$ limit of $\Lambda_{xx}(q_x = 0, q_y, \omega = 0)$ by the smallest accessible momentum, $q_y = 2\pi/L$.

Fig. 5.6 compares the MCMF result to the MF and QMC benchmarks for $\rho_s$, together with the BKT line. As for $P_s$, MF overestimates $T_c$ and MCMF underestimates it by a smaller margin: the QMC data give $T_c^{\mathrm{QMC}} = 0.14 \pm 0.02$ and MCMF gives $T_c^{\mathrm{MCMF}} = 0.11 \pm 0.001$, an underestimation of about 20%. While the QMC estimates for $T_c$ based on $\rho_s$ and $P_s$ agree with each

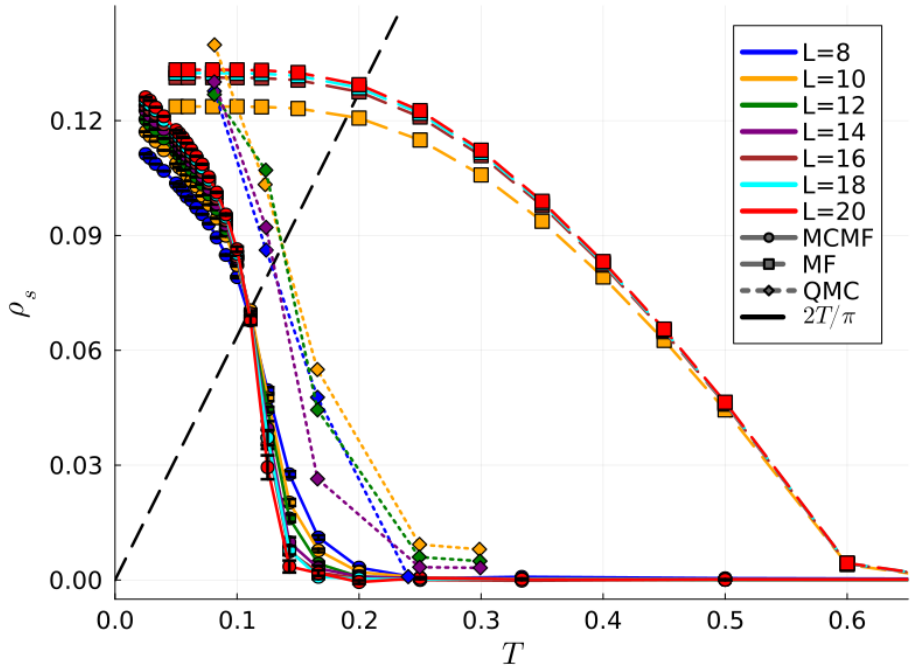



Figure 5.6: The superfluid density $\rho_s$ for the MCMF, MF, and QMC benchmark at $\mathrm{U} = 4$, $n = 0.5$, for various system sizes (different colors). The black dashed line of the BKT line $\frac{2T}{\pi}$; the critical temperature $T_c$ is given by the crossing point of the stiffness and this line.

other they differ for MCMF. We attribute this to the sparse temperature grid just above $T_c$ — precisely the region that dominates the FSS collapse of $P_s$. We therefore report critical temperatures from the BKT crossing of $\rho_s$ throughout the rest of this work.

In the MCMF data the $\rho_s(T; L)$ curves cross at a common point around $T_c$, where the ordering of the curves by size inverts: for $T > T_c$, $\rho_s$ of larger systems fall faster toward zero (the BKT jump in the thermodynamic limit), and for $T < T_c$ larger systems have higher stiffness, tending to $\rho_0^{L\to\infty}$. Such crossing is absent in the QMC data. We believe that the difference stems from the two possible ways of extracting the $q_y \to 0$ limit of $\Lambda_{xx}(q, \omega = 0)$. We use for this end the value of the correlation function at the smallest accessible momentum $q_y = 2\pi/L$, which itself vanishes as $L \to \infty$. Conversely, one may attempt to extrapolate the data to $q_y \to 0$ for each system size $L$. While the QMC study of Ref. [27] does not give details on which route was used, our group's QMC simulations reproduced their data by using the $q_y \to 0$ extrapolation.

## 5.2 BCS–BEC crossover

As discussed in Chapter 2, the attractive Hubbard model crosses over from BCS to BEC as U increases, producing the dome in $T_c(\mathrm{U})$. Fig. 5.7 shows that the MCMF approximation recovers this behavior. At $n = 0.875$ the dome peaks at $\mathrm{U} \approx 4$ with $T_c^{\max} \approx 0.135$ (the location and height of the maximum depend on filling, as mapped in earlier works [17, 27]). The same plot includes a DQMC benchmark from Ref. [17]. Both curves trace a BCS–BEC dome, agreeing well at weak coupling ($\mathrm{U} < 4$) but differing substantially at strong coupling. The position of the maxima also differ, with the MCMF dome peaking at a lower U than the QMC one.

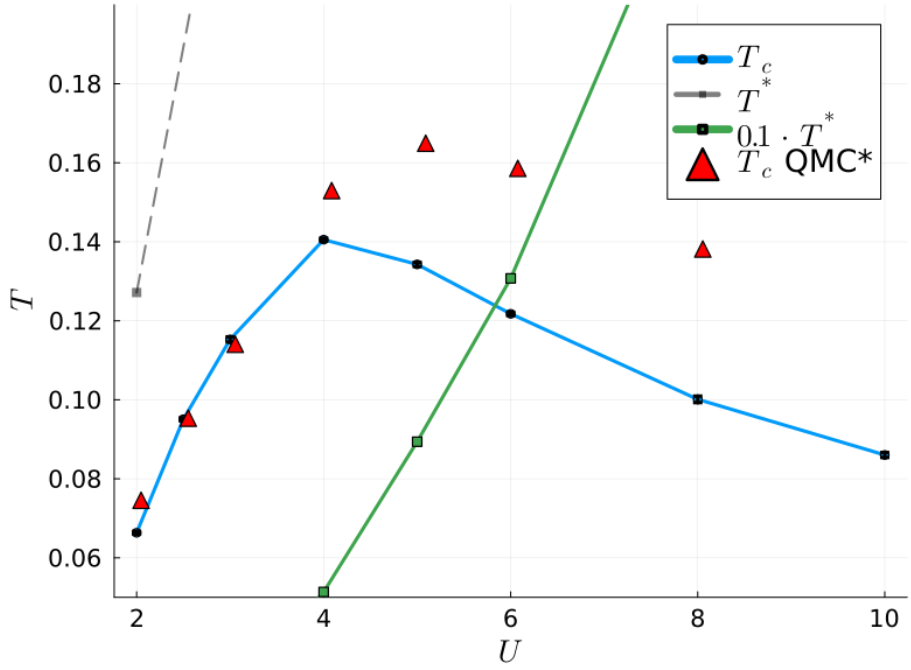



Figure 5.7: The MCMF critical temperature $T_c$ (blue line) and the pseudogap temperature $T^*$, scaled by a factor of 0.1 (green line) as a function of the interaction strength U for systems of size $L = 18$ and filling $n = 0.875$. The black dashed line gives the unscaled $T^*$. Red circles are a benchmark for $T_c$ from a DQMC simulation of the same parameters with linear size $L = 16$, from Ref. [17]. $T_c$ is obtained from the intercept of the superfluid density with the BKT line $\rho_s(T_c) = \frac{2}{\pi} T_c$, as in Fig. 5.6, and $T^*$ is read from the maximum of $\chi_s$.

## 5.3 Pseudogap regime

As discussed in Chapter 2, the pseudogap regime is the temperature window $T_c < T < T^*$ in which Cooper pairs have formed but their phases remain incoherent [17, 35], with $T^*$ marking the onset of pairing. Two routes are commonly used to extract $T^*$: the opening of a gap in the density of states (DOS), and the attainment of a maximum of the spin susceptibility $\chi_s$ with a subsequent decrease at lower temperatures. We apply both, but quote $T^*$ from the latter, since the DOS estimate is affected by a MCMF artifact discussed below.

Fig. 5.8 shows the DOS $\rho(\omega)$ at several temperatures for two interaction strengths. A true gap with symmetric coherence peaks is present for $T < T_c$. On warming into $T_c < T < T^*$ the gap partially fills and the peaks diminish but stay prominent, marking the pseudogap regime. The gap never fully closes — even at $T = 2$ the DOS remains suppressed at the Fermi level — so $T^*$ cannot be read reliably from the DOS.

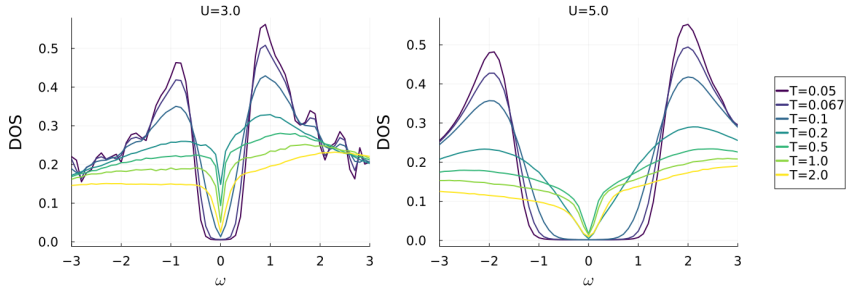



Figure 5.8: The density of states (DOS) against $\omega$ for an $L = 20$ system at $n = 0.875$, at interaction strengths $\mathrm{U} = 3$ (left) and $\mathrm{U} = 5$ (right); the shared legend on the right gives the temperatures. The relevant temperature scales are $T_c \approx 0.115$, $T^* \approx 0.25$ for $\mathrm{U} = 3$ and $T_c \approx 0.134$, $T^* \approx 0.89$ for $\mathrm{U} = 5$, where $T_c$ is computed from the BKT crossing of $\rho_s$ and $T^*$ from the maximum of $\chi_s$.

This persistent high-$T$ gap can be understood from the Gaussian approximation to the HS-field statistics introduced in §5.1.1. To test this picture directly, we sample $\Re[\Delta_i], \Im[\Delta_i] \sim \mathcal{N}(0, \sigma^2)$ independently on every site, averaging over 300 configurations, with $\sigma$ fixed so that $\Delta_p$ matches its measured MCMF value and $\mu$ set self-consistently to give $\langle n \rangle = 0.875$. As Fig. 5.9 shows by comparing this Gaussian-model DOS to the MCMF DOS at $\mathrm{U} = 4$, the model reproduces the MCMF structure, with a Fermi-level gap that widens with increasing $T$.

The DOS rises roughly linearly for small energies. This follows from the high-$T$ limit $|\Delta| \gg t, \mu$ where the spectrum tracks the Rayleigh distribution of $|\Delta_i|$, giving $\rho(\omega) \approx \frac{\omega}{\sigma^2} \exp[-\frac{\omega^2}{2\sigma^2}]$ with $\sigma = \sqrt{\frac{2}{\pi}}\Delta_p$. The slope slope $\sigma^{-2} = \frac{\pi}{2}\Delta_p^{-2}$ decreases with $\Delta_p$ and therefore with $T$, causing the gap widening.

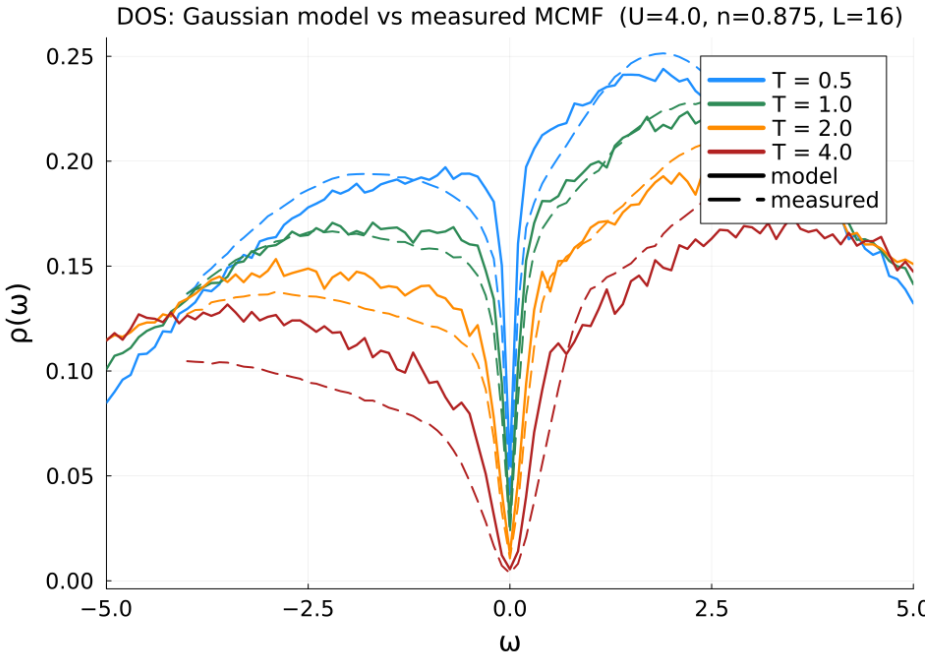



Figure 5.9: The Gaussian-model DOS (solid) and the measured MCMF DOS (dashed) at $\mathrm{U} = 4$, $n = 0.875$, $L = 16$, for $T = 0.5, 1, 2, 4$ (one color per temperature). The model samples the $\Re$ and $\Im$ parts of $\Delta_i$ from a Gaussian distribution with variance chosen such that $\Delta_p^{\mathrm{MCMF}} = \Delta_p^{\mathrm{Gaussian}}$. The chemical potential $\mu$ was chosen to keep the same filling.

We identify $T^*$ from the maximum of the spin susceptibility $\chi_s$ (Eq. (4.13)). Once Cooper pairs begin to form, flipping an electron's spin requires breaking a pair, which costs energy. $\chi_s$ therefore drops sharply below the pair-formation temperature $T^*$. Fig. 5.10 (inset) shows $\chi_s$ for MCMF alongside a DQMC benchmark from Ref. [17]. MCMF gives a lower $T^*$ but reproduces the abrupt drop. Above $T^*$ it systematically underestimates $\chi_s$, a consequence of the same high-$T$ artifact that overestimates the pairing. From the main panel of Fig. 5.10 we can see that the temperature of the maximum in $\chi_s$, identified with $T^*$, increases with U, consistent with the BCS-BEC crossover. Fig. 5.7 demonstrates that the pseudogap window between $T_c$ and $T^*$ widens with U.

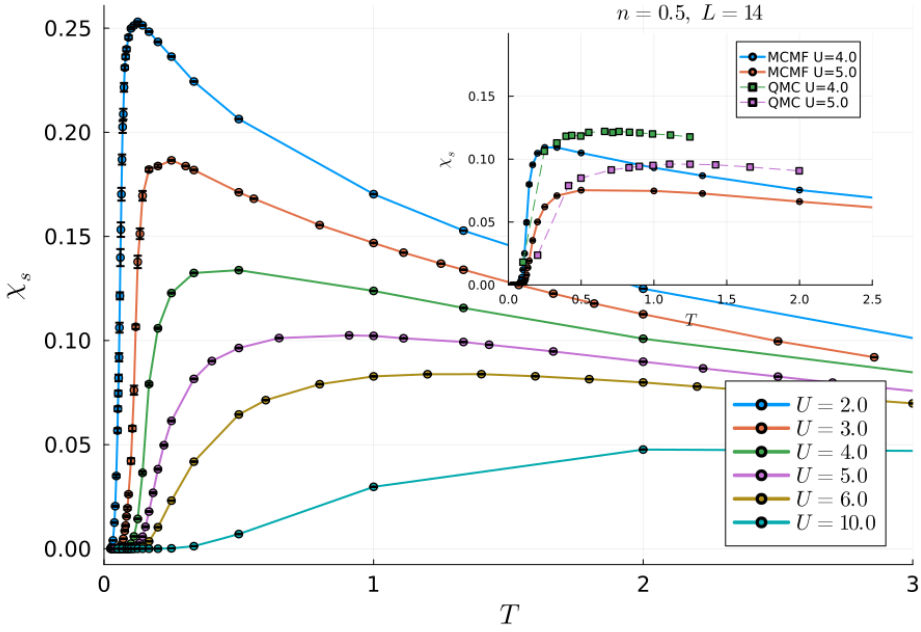



Figure 5.10: The spin susceptibility $\chi_s$ for various interaction strengths U. (Outer) A system size $L = 18$ with filling $n = 0.875$. (Inset) Comparison to [17] DQMC results for an $L = 14$, $n = 0.5$ system. The MCMF results predict systematically lower $T^*$ and underestimate $\chi_s$ for temperatures above it.

## 5.4 DC transport

In the following we present results for the two components of the real part of the conductivity $\Re\sigma_{xx}(\omega) = D\,\delta(\omega) + \sigma_{xx}^{\text{reg}}(\omega)$: a Drude weight $D$ carried by a delta function at $\omega = 0$, and the regular part $\sigma_{xx}^{\text{reg}}(\omega)$.

### 5.4.1 Drude weight and phase classification

Fig. 5.11 shows the MCMF Drude weight $D(T)$, which is barely affected by finite-size effects and stays finite at every temperature and system size. For large systems, $D/\pi$, $4\rho_s$, and $\langle -K_x \rangle$ converge to a common value As $T \to 0$. The behavior of the pair $(\rho_s, D)$ classifies the uniform system: If both $\rho_s > 0$ and $D > 0$, the system is a superconductor. A metal maintains a finite $D$ but has vanishing $\rho_s$. Both quantities vanish for an insulator [26].

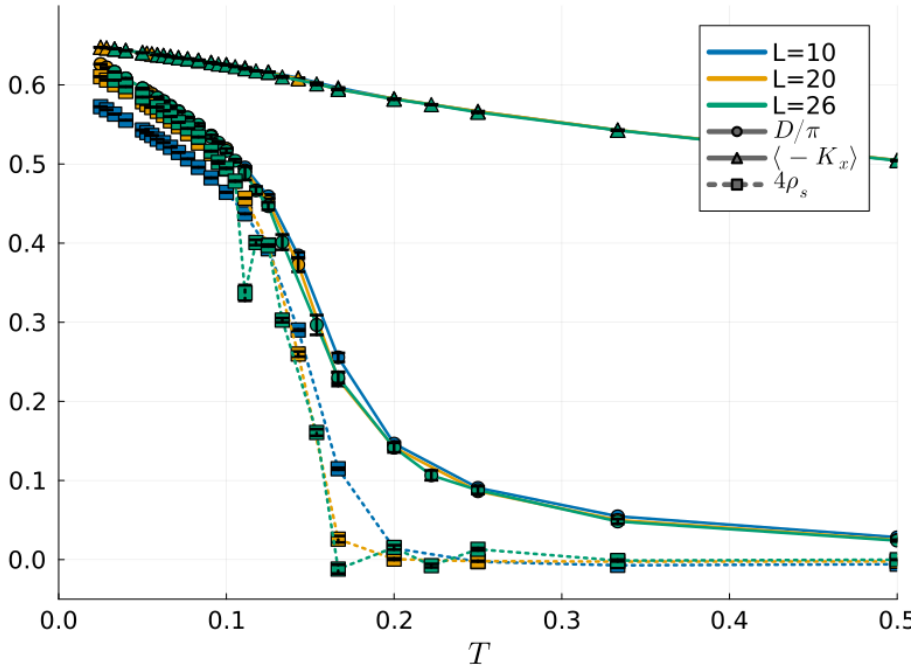



Figure 5.11: The Drude weight $D/\pi$ (circles), kinetic energy $\langle -K_x \rangle$ (triangles) and superfluid stiffness $4\rho_s$ (squares) at $\mathrm{U} = 4$, $n = 0.875$, for system sizes $L = 10, 20, 26$ (colors).

### 5.4.2 DC regular resistivity

The DC regular resistivity $\rho_{xx}^{\mathrm{reg}} \equiv \left(\sigma_{xx}^{\mathrm{reg}}(\omega \to 0)\right)^{-1}$ is the natural diagnostic for the dissipative response above $T_c$. Although our model is well suited to computing the finite-frequency conductivity $\sigma_{xx}^{\mathrm{reg}}(\omega > 0)$, we use it here only to reach the DC limit $\omega \to 0$ and do not present finite-frequency data; the full scheme is given in Appendix B.

Fig. 5.12 shows the DC regular resistivity $\rho_{xx}^{\mathrm{reg}}$ for several interaction strengths. At high $T$, $\rho_{xx}^{\mathrm{reg}}$ rises with $T$ as in a metal. Upon cooling it decreases up to a minimum at $T_{\mathrm{min}} > T_c$, below which it turns up. The minimum scale $T_{\mathrm{min}}$ grows with U and scales similarly to the pseudogap temperature $T^*$, though it always lies below it. This is expected if pair formation depletes the density of states at the Fermi level, lowering the carrier number and raising the resistivity. We return to this correspondence between $T_{\mathrm{min}}$ and $T^*$ when we add interaction disorder and magnetic fields to the discussion in the following chapters. At low $T$ the regular resistivity $\rho_{xx}^{\mathrm{reg}}$ diverges, as the

DC current response is taken over by the Drude-weight channel, which grows as $T$ decreases. $\rho_{xx}^{\text{reg}}$ increases with U for all temperatures: approaching the BEC limit, the effective pair-hopping energy falls as $\sim t^2/\mathrm{U}$, reducing the conductivity.

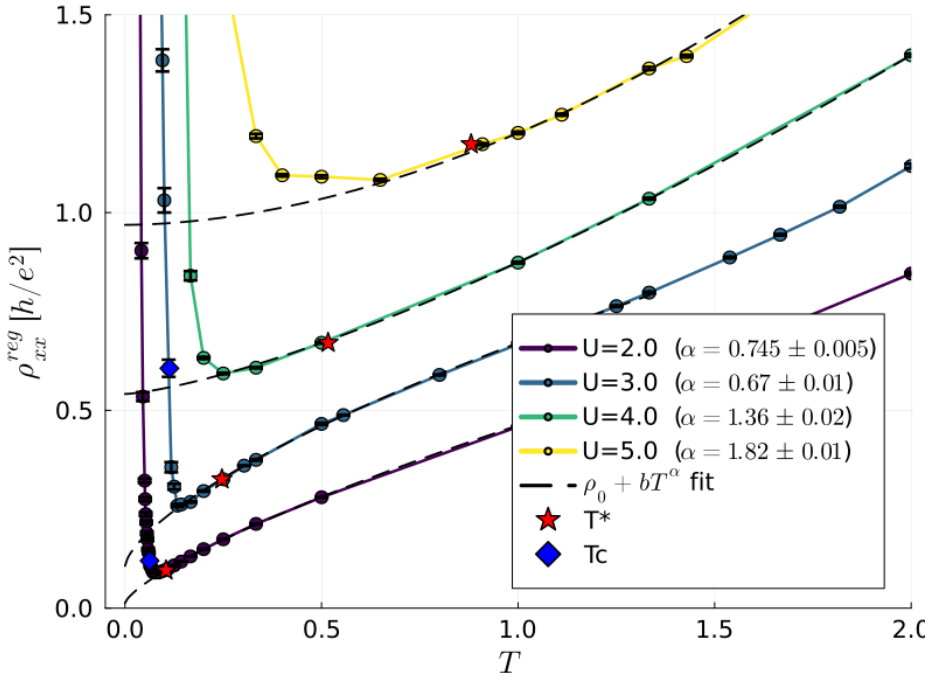



Figure 5.12: The DC regular resistivity $\rho_{xx}^{\text{reg}}$ for $L = 20$, $n = 0.875$ systems with different interaction strengths U (different colors). The dashed lines are a fit to $\rho = \rho_0 + bT^\alpha$ for a temperature range above $T^*$. The values for $\alpha$ are given in the legend. $T^*$ and $T_c$ are denoted as a red star and a blue diamond respectively. The $y$-values of these temperatures are chosen only to align them with the correct system. $T_c$ is not visible in the plot for all U values.

To characterize the curvature of $\rho_{xx}^{\text{reg}}(T)$ above $T^*$, we fit it to a power law $\rho = \rho_0 + bT^\alpha$. The fitted exponent varies with U, from $\alpha = 1.82 \pm 0.01$ at $\mathrm{U} = 5$ to $\alpha = 0.67 \pm 0.01$ at $\mathrm{U} = 3$, with an apparently non-monotonic trend that may simply reflect the narrow temperature window of the fit.

The following chapters track the regular resistivity for $T \gtrsim T^*$ as we add disorder in the interaction (Chapter 6) and magnetic fields (Chapter 7). Both push the power-law regime down to lower temperatures, in step with the suppression of $T^*$. we return to this correlation in those chapters.

# Chapter 6

# Interaction-disordered system

## 6.1 The interaction-disordered system

As discussed in Chapter 1, a metallic host containing superconducting "puddles" has been proposed to sustain an extended regime of linear-in-$T$ resistivity [12]. To realize such an inhomogeneous landscape within the attractive Hubbard model, we make the interaction spatially disordered: a fraction $0 < p < 1$ of the sites act as "interaction centers," each contributing a Gaussian profile to the local coupling. Consonantly, the interaction landscape becomes

$$\mathrm{U}_i = \frac{\mathrm{U}}{\mathcal{N}} \sum_{j=1}^{N_d} \exp\left( -\frac{\|r_j - r_i\|^2}{2\sigma_d^2} \right). \tag{6.1}$$

The number of interaction centers is $N_d = p\,N_s$. We take $\sigma_d = 0.8$ (in lattice units), and the normalization $\mathcal{N}$ is chosen such that the clean limit $p = 1$ recovers the uniform coupling $\mathrm{U}_i = \mathrm{U}$. The spatial mean is $\bar{\mathrm{U}} = \mathrm{U}p$. Lowering $p$ at fixed $\bar{\mathrm{U}}$ thus dilutes the uniform system into a sparse array of isolated "SC puddles," as Fig. 6.1 illustrates for several values of $p$.

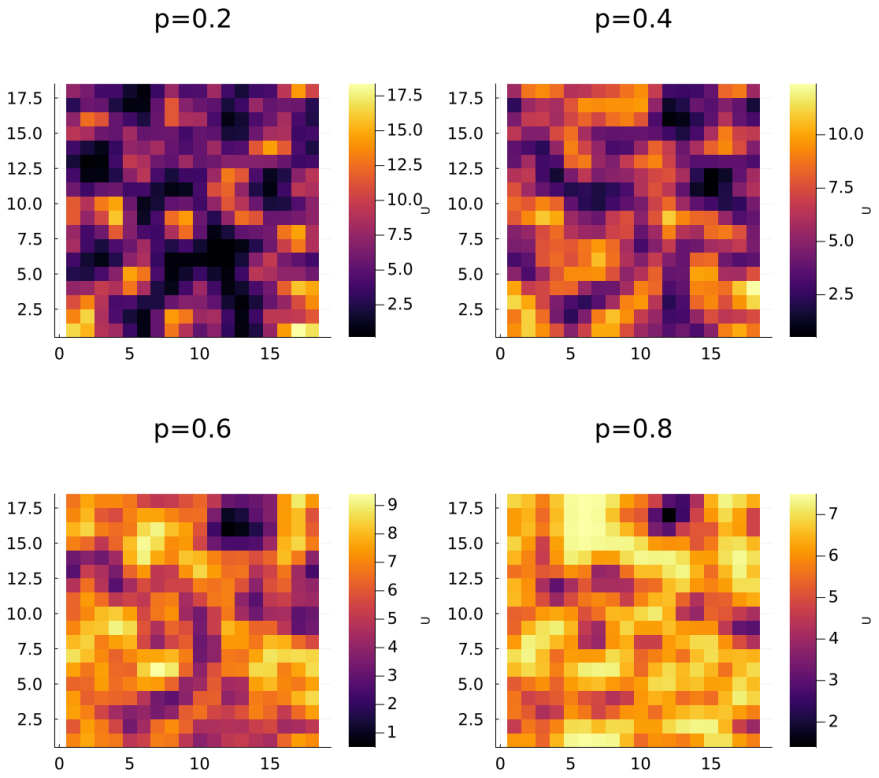



Figure 6.1: The disordered interaction landscape for an $L = 18$ system with $\bar{U} = 6.0$ at a few values of $p$.

## 6.2 The LDOS

To probe the effects of disorder, we examine the local density of states (LDOS), defined at site $i$ by

$$\rho_i(\omega) = -\frac{1}{\pi} \sum_{\sigma} \mathrm{Im}\, G_{i\sigma,i\sigma}(i\omega_n \to \omega + i\delta). \tag{6.2}$$

Following Ref. [24], we assign each site to a bin according to its spectral gap, i.e., the position $\omega > 0$ of its LDOS peak at the lowest accessible temperature. We then track how each bin's LDOS (avarged over disorder realization) changes with $T$. As Fig. 6.2 shows, the bins spread over a range of gap sizes at low $T$ and stay distinct on warming, with the most gapped sites still gapped well above $T_c$. Interaction disorder therefore produces spatially inhomogeneous pairing that survives to high temperatures.

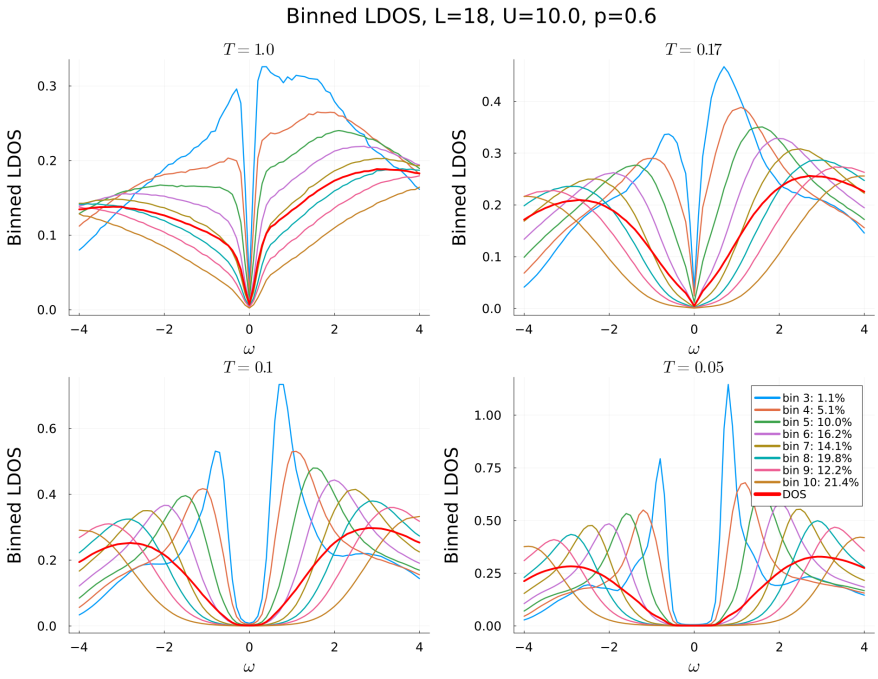



Figure 6.2: The binned LDOS for a $U = 10$, $p = 0.6$ system ($L = 18$, $\bar{U} = 6$), at different temperatures. Sites are grouped into bins by their local spectral gap at the lowest $T$ (bin 1 = least gapped, bin 10 = most gapped); each bin's share of the lattice sites (disorder-averaged) is given in the legend. The lowest two bins are empty, while the most gapped bin holds $\approx 21\%$ of the sites. For this system $T_c \approx 0.118$ and $T^* \approx 1.36$.

## 6.3 The critical temperature

Next, we determine $T_c$ for systems with different disorder levels $p$ and interaction strengths $U$. For every disorder level and interaction strength that we have checked, the system undergoes a superfluid transition at a finite $T_c$. This is expected: even in the limit of isolated interaction grains, some finite Josephson coupling survives between them, driving condensation at a finite, though possibly very low, temperature.

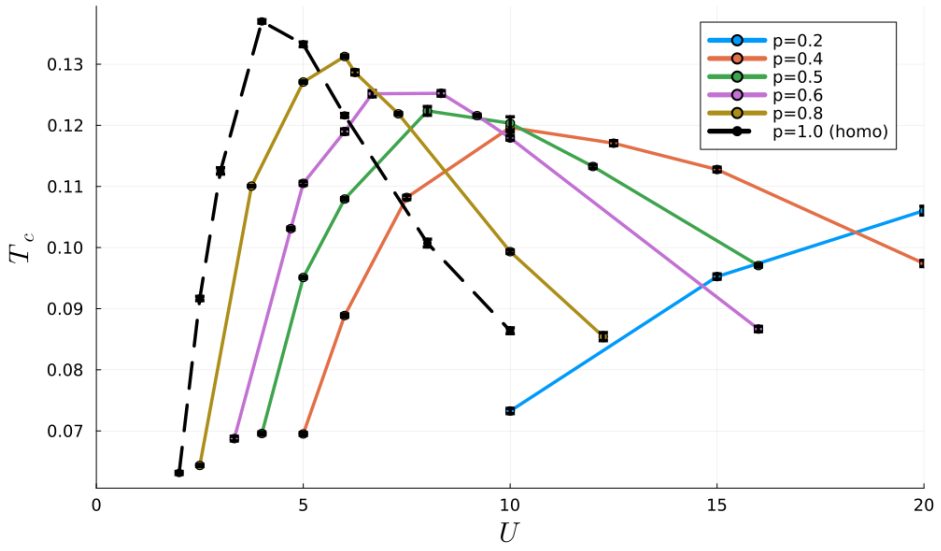
Fig. 6.3a shows that the effect of disorder depends on the interaction strength. At weak coupling ($U < 3$) disorder lowers $T_c$, whereas at larger $U$ it acts non-monotonically, raising $T_c$ before suppressing it. Each fixed-$p$ curve still forms a dome in $U$, so the BCS–BEC crossover survives across all disorder levels.

Plotting the same data against the mean interaction $\bar{U}$ clarifies the picture. As Fig. 6.3b shows, the $T_c(\bar{U}; p)$ domes share nearly the same peak, at $\bar{U} \approx 4$, for every $p$. On the BEC side ($\bar{U} \geq 3$) they are ordered by disorder — the more disordered the system, the lower its $T_c$; the curves cross near $\bar{U} \approx 3$, and on the BCS side ($\bar{U} < 3$) the ordering inverts, with disorder now enhancing $T_c$. The non-monotonicity at fixed $U$ then follows from the accompanying change in $\bar{U}$ and the resulting movement along the BCS–BEC crossover.

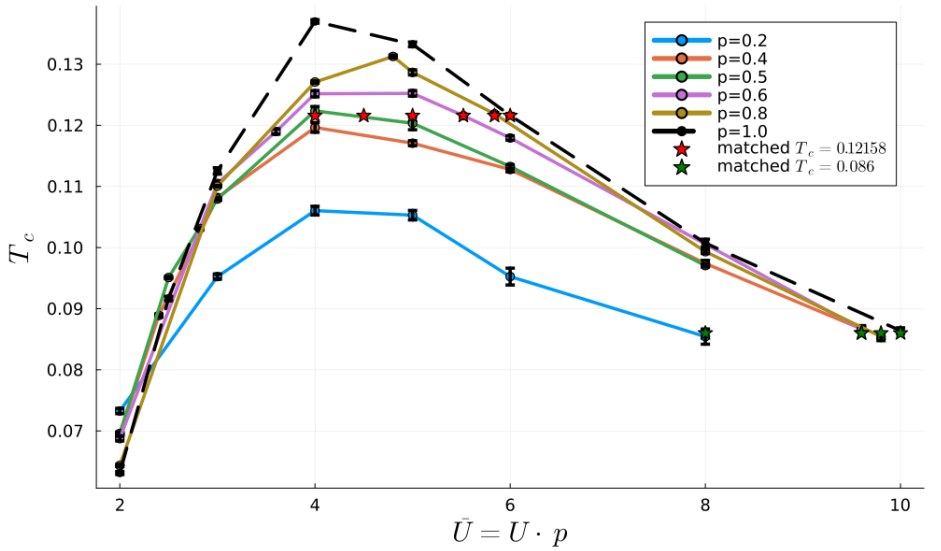
A simple physical picture explains why the $T_c(\bar{U}, p)$ domes keep a similar shape across disorder levels, with the BCS and BEC limits contributing differently. In the BCS limit the coherence length $\xi$ is exponentially large, so each pair averages over the $U_i$ landscape and responds only to its mean $\bar{U}$, which therefore sets the scale. In the BEC limit the system maps onto an XY model with Josephson couplings $J \sim t^2/U$, and as disorder grows it fragments into grains that are weakly coupled internally (large local $U$) and even more weakly coupled to one another. Two sweeps then lower $T_c$: raising $\bar{U}$ at fixed $p$ raises $U$ and weakens the coupling both within and between grains; lowering $p$ at fixed $\bar{U}$ pushes the grains apart (weaker intergrain coupling) and raises the local interaction (weaker intragrain coupling). Notably, the dome maximum stays at the same $\bar{U} \approx 4$ for every $p$. This points to the BCS side controlling this maximum, where $T_c \sim \exp(-1/\rho_F \bar{U})$ depends only on $\bar{U}$; and on the contrary the BEC limit depends on $p$ more heavily.

## 6.4 Disorder at fixed $U$

Having mapped $T_c$, we turn to how transport and the pseudogap evolve under disorder. We first hold the bare interaction at $U = 6$ and lower the disorder parameter $p$, so the mean interaction $\bar{U} = Up$ falls as the system dilutes. Fig. 6.4 shows $\rho_s$, $\rho_{xx}^{\text{reg}}$, and $\chi_s$ for several values of $p$. The stiffness (Fig. 6.4(a)) yields a non-monotonic $T_c$ that rises with $p$ before falling, as explained above. The resistivity (Fig. 6.4(b)) develops a minimum at $T_{\text{min}}$ that moves to lower $T$ with increasing disorder, extending the $d\rho/dT > 0$ regime to lower temperatures, while its overall magnitude drops as $\bar{U}$ decreases. The susceptibility (Fig. 6.4(c)) peaks at the pseudogap temperature $T^*$, and with increasing disorder this peak grows taller and moves to lower $T$ — again tracking the falling $\bar{U}$. The two effects line up: the $\rho_{xx}^{\text{reg}}$ minimum occurs near the $\chi_s$ peak and both shift to lower $T$ together as disorder grows, reinforcing the uniform-system link between the resistivity upturn and $T^*$. Because $\bar{U}$ varies along this sweep, though, we cannot yet disentangle the effect of disorder from that of the mean interaction.




(a)




(b)

Figure 6.3: The critical temperature $T_c$ for systems with various disorder parameters $p$ (colors) and interaction strength U. (a) as a function of the bare interaction U. (b) as a function of the mean interaction $\bar{\mathrm{U}} = \mathrm{U} \cdot p$. The red and green stars are iso-$T_c$ families that are examined in the text. All systems are of size $L = 18$ with cluster size $L_c = 10$ at $n = 0.875$.

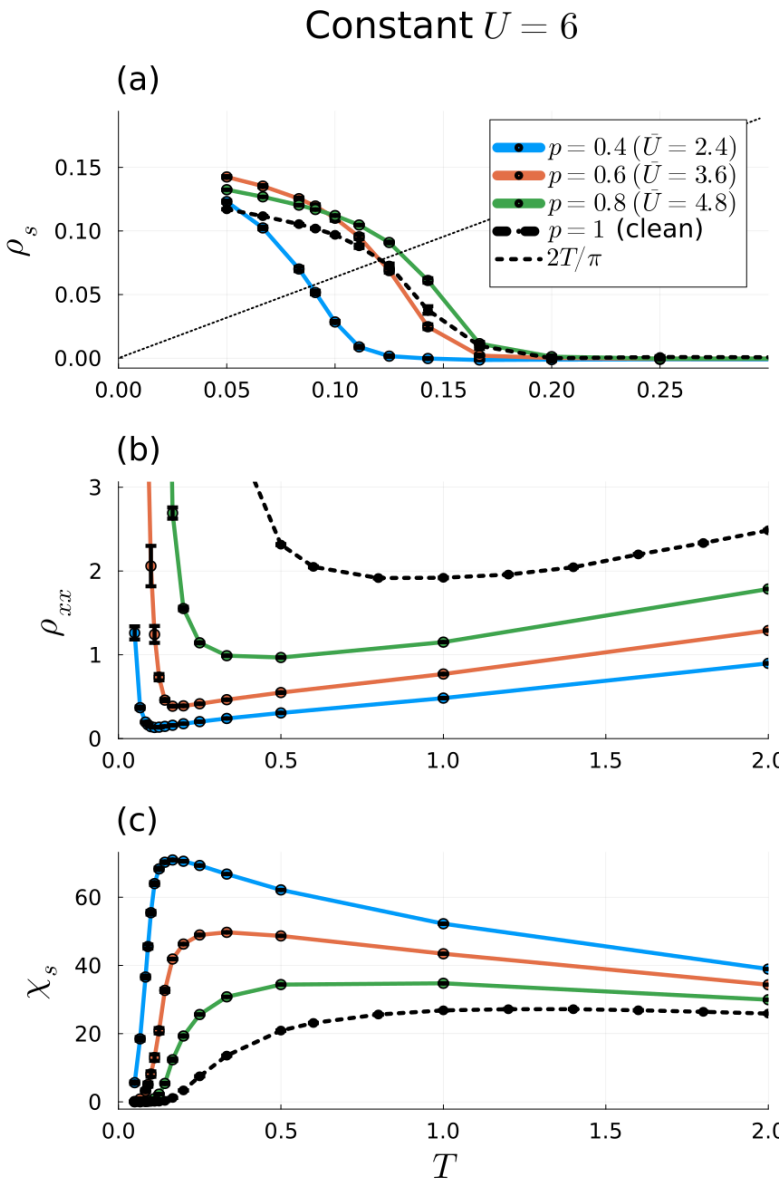



Figure 6.4: Temperature dependence along a constant-bare-interaction disorder sweep at $U = 6$, $n = 0.875$, $L = 18$, for disorder parameters $p$ (colors). (a) The superfluid stiffness $\rho_s$; the straight black dotted line through the origin is the BKT relation $\frac{2}{\pi}T$, whose crossing with each curve gives $T_c$. (b) The regular resistivity $\rho_{xx}^{\text{reg}}$. (c) The spin susceptibility $\chi_s$, whose maximum defines $T^*$.

## 6.5 The iso-$T_c$ family

We define an iso-$T_c$ family as a set of systems that span a range of the disorder parameter $p$ but share nearly the same $T_c$, like those marked in Fig. 6.3b. We build such a family starting from the uniform system ($p = 1$) on the BEC side ($\bar{U} > 3$) and lowering $p$ while keeping $T_c$ fixed. For a given $T_c$ this can be carried out only down to a minimum fraction $p_{\min}$, i.e. maximum disorder, at which the dome maximum equals $T_c$ itself.

Results for the family marked by red stars in Fig. 6.3b are shown in the panels of Fig. 6.5. It lies along $T_c \approx 0.1215$, with $p$ running from 1 at $U = 6$ (the homogeneous system) down to $p_{\min} = 0.45$ at $U = 10$.

As in the constant-U sweep, the iso-$T_c$ family's regular resistivity $\rho_{xx}^{\text{reg}}$ (Fig. 6.5(b)) keeps a $d\rho/dT > 0$ regime that reaches to lower $T$ as disorder grows. The temperature dependence becomes closer to linear in the more disordered systems. The pseudogap temperature $T^*$, deduced from the maximum of $\chi_s$, likewise falls with disorder (Fig. 6.5(c)).

With this model, then, we can extend the near-linear regime to lower $T$ while holding $T_c$ fixed — though only up to a maximum disorder level. A second family, at lower $T_c \approx 0.0865$ (green stars in Fig. 6.3b), reaches further into the SC-puddle regime; for it we stopped at $p = 0.2$ rather than pushing to the maximal disorder. It behaves similarly and is collected in Appendix E.

Because the onset of the metallic regime ($d\rho/dT > 0$) tracks $T^*$ even in the presence of disorder, its larger extent may stem more from the falling $T^*$ than from the disorder itself. To test this we plot $T^*$ against the mean interaction $\bar{U}$ (Fig. 6.6). $T^*$ rises with $\bar{U}$ at every disorder level, and adding disorder lowers it — slightly at small $\bar{U}$, more at large $\bar{U}$. The exception is the strongest disorder ($p = 0.2$), where $T^*$ appears to saturate as a function of $\bar{U}$. Confirming whether this is physical or a finite-size artifact will require larger systems.

Pairing here is a local process, governed by the interaction strength surrounding each site, so the local pairing — and with it the local pseudogap — varies across the lattice, as the binned LDOS of Fig. 6.2 showed. The pseudogap temperature $T^*$ then marks where a substantial fraction of the system has paired, a scale set by the mean interaction $\bar{U}$.

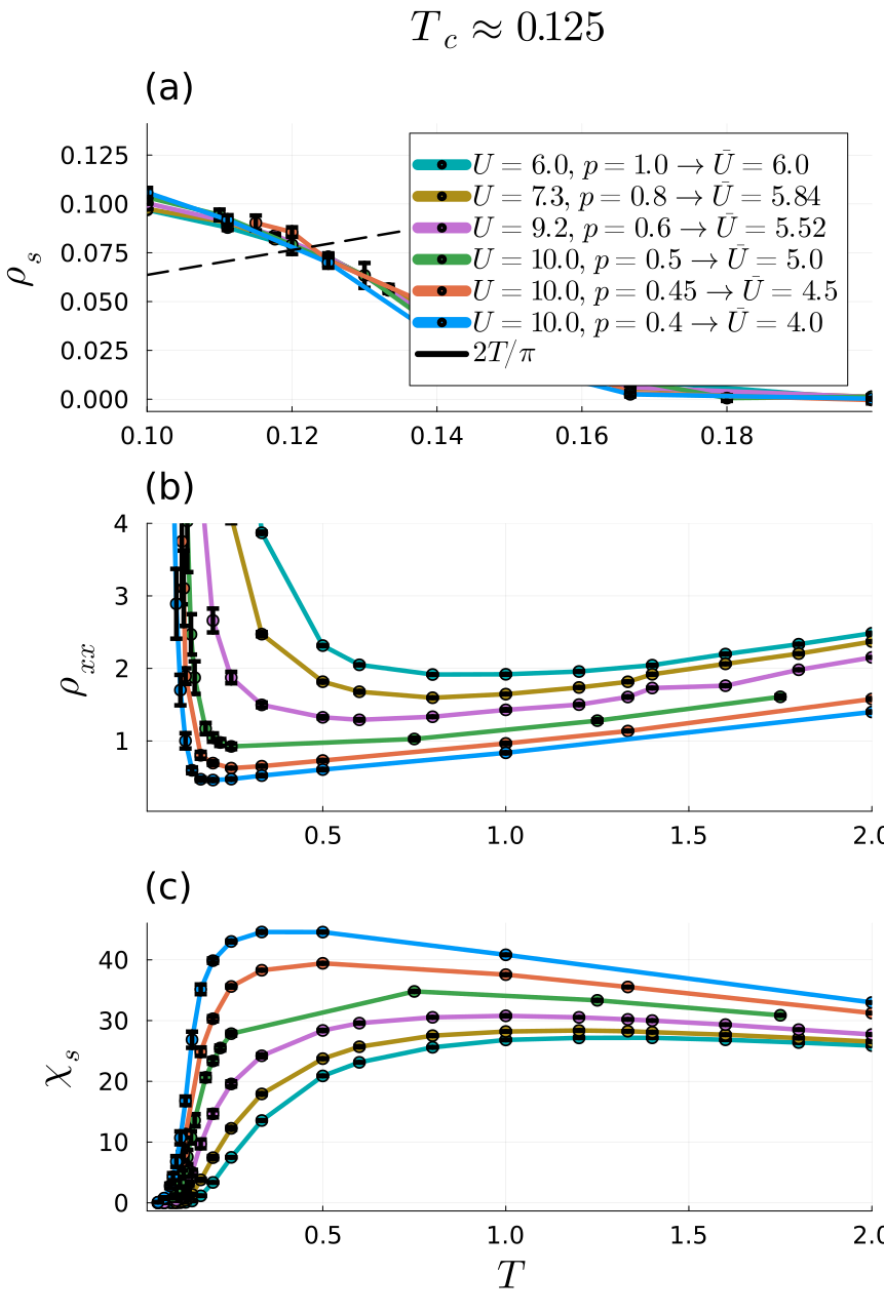



Figure 6.5: Temperature dependence for a family of systems with similar critical temperature $T_c \approx 0.1215$, as denoted in Fig. 6.3b by the green stars. (a) The superfluid stiffness $\rho_s$, accompanied by the BKT line $\frac{2}{\pi}T$. (b) The regular resistivity $\rho_{xx}^{\text{reg}}$. (c) The spin susceptibility $\chi_s$, which its maxima determines the pseudogap temperature $T^*$.

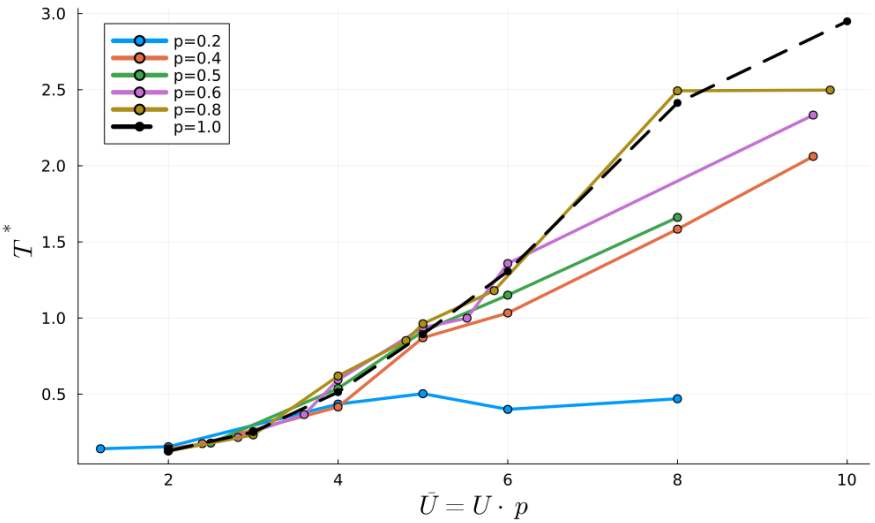



Figure 6.6: The pseudogap temperature $T^*$ for disordered systems with parameter $p$ at different mean interaction strengths $\bar{\mathrm{U}} = \mathrm{U} \cdot p$. $T^*$ is computed by finding the maximum of the spin susceptibility $\chi_s$.

In summary, adding disorder, whether at fixed U or along an iso-$T_c$ line, extends the $d\rho/dT > 0$ regime to lower temperatures and appears to make the temperature dependence closer to linear. We attribute this mainly to the drop in the mean interaction $\bar{\mathrm{U}}$, which lowers $T^*$ and, with it, the temperature range over which the regime persists.

This model invites further study. The Gaussian width $\sigma_d = 0.8$ of each interaction center may have an important influence, and varying it would clarify the role it plays. In particular, at $\sigma_d \approx 0.3$ the model becomes close to an "on-off" site-depletion model like that of Ref. [22], where a "critical disorder" drives $T_c$ to zero — something we do not observe. The difference may be that our model has no zero-interaction sites ($\mathrm{U}_i > 0$ everywhere), so the more strongly interacting medium keeps the intergrain coupling higher.

# Chapter 7

# The system in a magnetic field

## 7.1 Motivation

Cuprate superconductors are known to show strange-metal behavior once their superconductivity is stripped away by a magnetic field [1], which exposes the underlying normal state without superconductivity masking the resistivity. In this chapter we ask how a magnetic field affects the regular resistivity $\rho_{xx}^{\mathrm{reg}}(T)$ in the uniform-interaction attractive Hubbard model.

## 7.2 Adding the magnetic field

We add a magnetic field via the Peierls substitution, attaching a phase to each creation and annihilation operator:

$$c_{r\sigma} \;\longrightarrow\; \exp\!\left( i\,\frac{e}{\hbar}\int_0^r \mathbf{A}(\mathbf{r}')\cdot \mathbf{dr}' \right) c_{r\sigma}, \tag{7.1}$$

with $\mathbf{A}(\mathbf{r})$ the vector potential that generates the field through $\mathbf{B} = \nabla \times \mathbf{A}$. Equivalently, each hopping amplitude is replaced by

$$t_{i,i+a} \;\longrightarrow\; \exp\!\left( i\,\frac{e}{\hbar}\int_{r_i}^{r_{i+a}} A_a(r')\,dr' \right) t_{i,i+a} \;\approx\; \exp\!\Big( i\,\frac{e\,a}{\hbar}\,A_a(r_i) \Big)\, t_{i,i+a}, \tag{7.2}$$

where $A_a$ is the component of $\mathbf{A}$ along the bond $a$, and where we have assumed it changes little on the scale of a lattice spacing. The kinetic part of the Hamiltonian becomes

$$H_0 = -t \sum_{i,a,\sigma} \Big[ e^{iA_a(r_i)}\, c_{i\sigma}^{\dagger} c_{i+a\sigma} + e^{-iA_a(r_i)}\, c_{i+a\sigma}^{\dagger} c_{i\sigma} \Big], \tag{7.3}$$

with $a = \pm\hat{x}, \pm\hat{y}$ and in units where $|a| = e = \hbar = 1$. In the same way we modify every operator whose expectation value we compute.

We choose $\mathbf{A}(\mathbf{r})$ to produce a uniform magnetic field. A natural choice is the Landau gauge $\mathbf{A} = (0, Bx)$. Periodic boundary conditions then require $\exp[iA_a(r_i)] = \exp[iA_a(r_i + L_y\hat{y})]$, which quantizes the field to $B_n = \Phi_0 n / L_y$, where $\Phi_0 = h/e$ is the flux quantum.

To gain finer resolution in $B$, we instead use the nearly-symmetric gauge $\mathbf{A}(\mathbf{r}) = B\,(yL_x, xL_y)$ on near-symmetric lattices with $L_y = L_x + 1$. The boundary conditions then yield $B_n = \frac{\Phi_0}{L_x L_y}\, n$, a much finer spacing.

In the non-interacting limit (NIL, $\mathrm{U} = 0$), a magnetic field quantizes the spectrum into Landau levels, which become more degenerate as $B$ increases. On a lattice these levels form the Hofstadter butterfly spectrum, whose discreteness has a dramatic effect on the resistivity and produces the quantum Hall effect (QHE). We do not show these results here, but we verified both the butterfly spectrum and the QHE in our $\mathrm{U} = 0$ simulations, which validates our magnetic-field implementation.

We work at $n = 0.875$ on a small lattice, $L = (L_x, L_y) = (10, 11)$, showing results mainly for $\mathrm{U} = 3.0$. The small size is forced upon us. In a magnetic field the cluster size set the available $B$ values. Therefore, using a smaller cluster within the traveling-cluster approximation is unavailable and the full lattice must be diagonalized at every step.

## 7.3 $P_s$ at finite $B$

As Fig. 7.1 shows, even the lowest accessible field $B_{\mathrm{min}} = \Phi_0 / N_s \approx 0.01\,\Phi_0$ destroys the $s$-wave pairing correlation $P_s$ entirely, down to the lowest $T$. In other words, the smallest allowed field already quenches superconductivity. Zooming in on the $B > 0$ systems (inset), $P_s$ first falls with $B$ and then partially rises again at larger $B$, without ever returning to its $B = 0$ value. This non-monotonicity follows from the jumps in $\mu(B)$ required to hold the filling fixed as Landau levels cross the Fermi level.

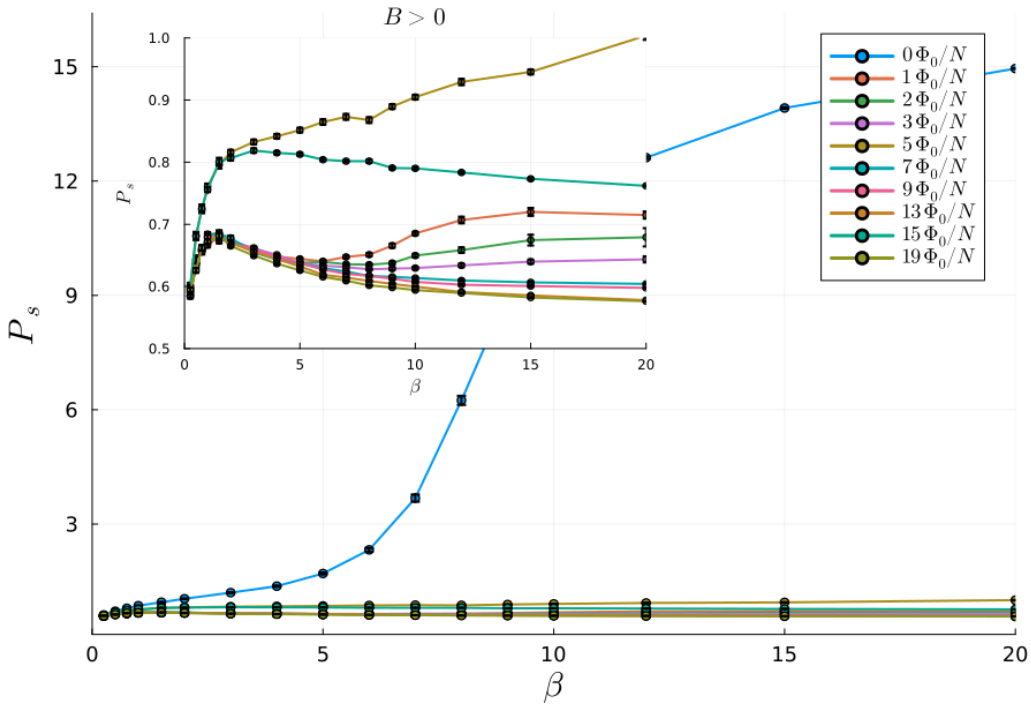



Figure 7.1: The $s$-wave pairing correlation $P_s$ at different magnetic fields as a function of $T$ for $\mathrm{U} = 3.0, n = 0.875$ . The inset presents the same data only for the $B > 0$ systems.

## 7.4 The DOS and pairing amplitudes

Fig. 7.2 shows the DOS as a function of $B$ at a low temperature, $T = 0.05$. At low fields a superconducting gap is visible and it shrinks with $B$, so pairing is suppressed as the system is placed in a magnetic field. In contrast to $P_s$, which vanished abruptly with $B$, the gap survives — coherence is destroyed by a magnetic field more easily than pairing is.

At higher fields, Landau levels form and distort the DOS. Holding the filling $n$ fixed forces $\mu(B)$ to move non-monotonically as those levels cross the Fermi level. Our interest is in the low-field regime that quenches superconductivity but barely shifts $\mu$, i.e., up to about $B = 0.18\Phi_0 \approx 20\Phi_0/N_s$.

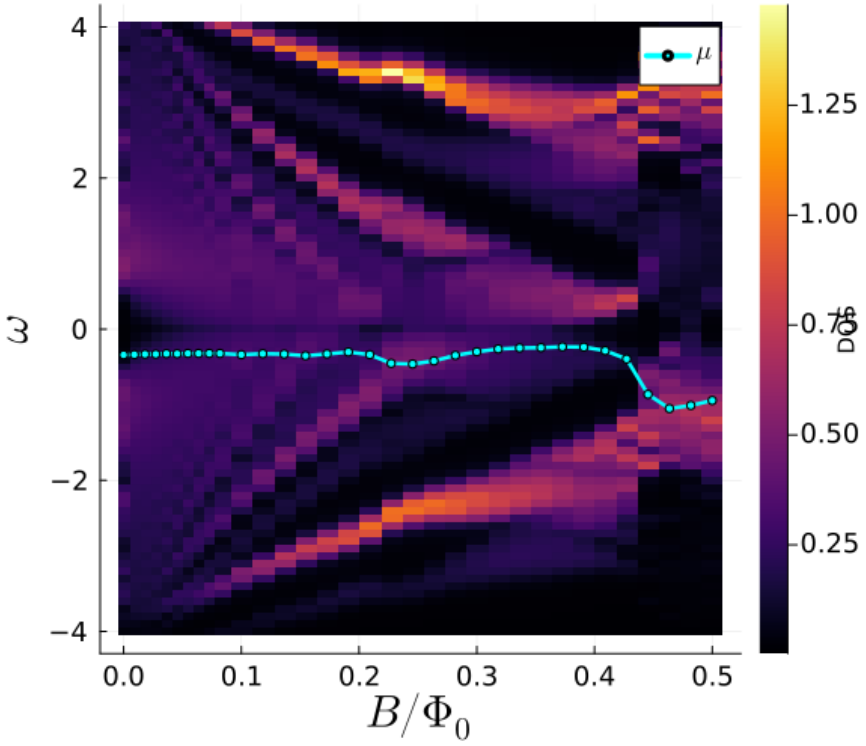
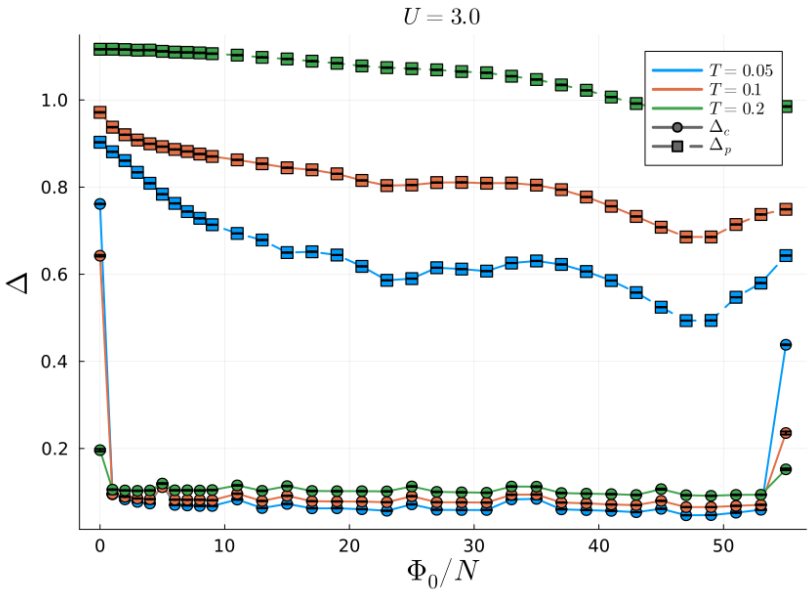


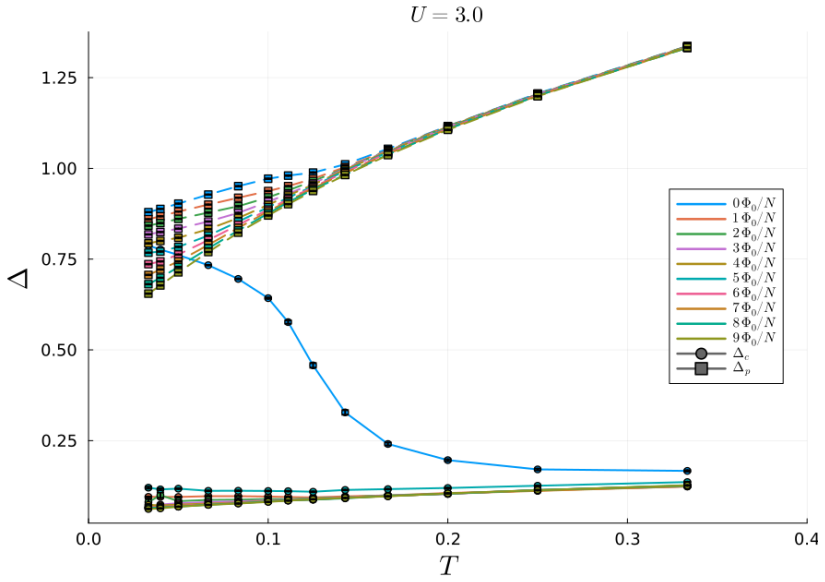

Figure 7.2: A heatmap of the DOS($\omega$) as a function of magnetic field $B$, for $\mathrm{U} = 3.0$ at $T = 0.05$ and $n = 0.875$. The magnetic field is given in units of the flux quantum $\Phi_0$. The cyan curve indicates the chemical potential $\mu(B)$ which is chosen to keeps the filling constant.

At no finite field does the gap close completely. There is always a narrow residual gap that reopens only at very high $B$, when Landau levels cross $\omega = 0$ and drive large jumps in $\mu$ to keep the filling fixed. We believe this residual gap is the same MCMF artifact discussed in Chapter 5, which leaves a finite $|\Delta_i|$ at every field. Fig. 7.3a tracks the pairing amplitudes $\Delta_c$ and $\Delta_p$ versus $B$ and $T$: $\Delta_c$ collapses at the smallest finite field, signaling the loss of phase coherence, while $\Delta_p$ decreases only gradually with $B$, so the local pairing amplitude $|\Delta_i|$ stays nonzero at every $T$ and $B$.

Fig. 7.3b shows the temperature dependence. At every field $\Delta_p$ grows with $T$ much like the $B = 0$ curve, joining it near $T \approx 0.15$. The condensate amplitude $\Delta_c$, by contrast, is far smaller at $B > 0$ than in the phase-coherent $B = 0$ system at low $T$; only at high $T$, in the phase-incoherent regime, do the curves come together.




(a)




(b)

Figure 7.3: The condensate amplitude $\Delta_c = \frac{1}{N_s} < |\sum_i \Delta_i| >$ (circles, solid) and pairing amplitude $\Delta_p = \frac{1}{N_s} \sum_i < |\Delta_i| >$ (squares, dashed) for a system with $\mathrm{U} = 3$, $n = 0.875$, $L = (10, 11)$. (a) As a function of magnetic field $B$ (in $\Phi_0/N_s$) at a few low temperatures (colors), over the full accessible flux range. (b) As a function of temperature $T$ for the ten smallest fields $B = 0$–$9\,\Phi_0/N_s$ (colors).

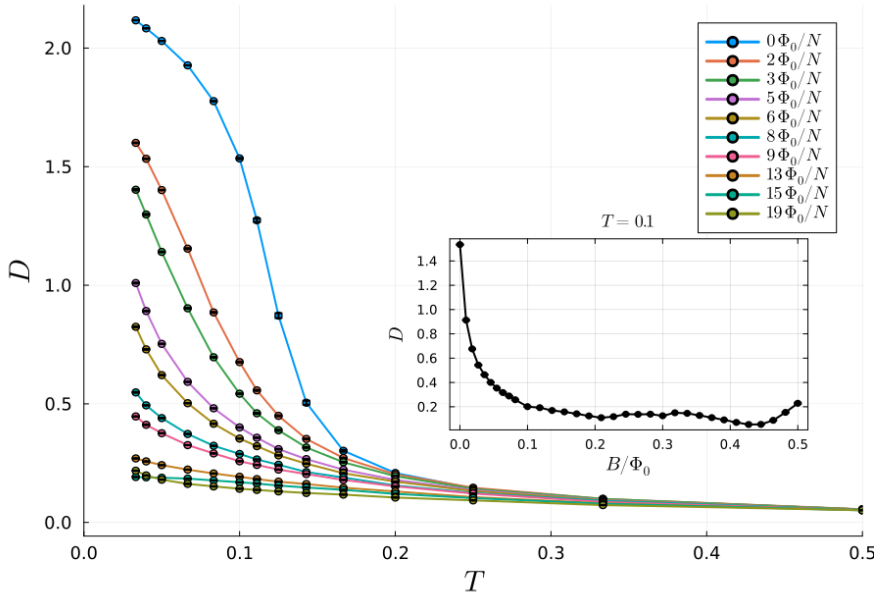



Figure 7.4: The Drude weight in magnetic-field systems. The main panel shows $D$ as a function of temperature $T$ for different magnetic-field strengths at $\mathrm{U} = 3$. The inset shows the Drude weight $D$ at $T = 0.1$ as a function of magnetic field in units of the flux quantum $\Phi_0 = h/e$.

## 7.5 The Drude weight and superfluid stiffness

The Drude weight $D$ falls as $B$ is applied (Fig. 7.4). At high temperatures the $B > 0$ curves merge with the $B = 0$ one, and $D$ stays finite at all $T$. As the inset of Fig. 7.4 shows, $D$ never vanishes at any finite field, so the system never turns insulating.

Classifying the system as superconductor, metal, or insulator also requires the superfluid stiffness $\rho_s$ (Eq. (4.17)). In the finite-field systems the transverse correlation $\Lambda_{xx}(q_x = 0, q_y, \omega = 0)$ is non-monotonic at low $T$ (Fig. 7.5). Whereas the longitudinal mode $\Lambda_{xx}(q_x, q_y = 0, \omega = 0)$ stays flat and tends to $-\langle K_x \rangle$ when $q_x \to 0$, as gauge invariance demands, this holds as a reassuring consistency check. Because the transverse mode fluctuates wildly, its $q_y \to 0$ limit, and with it $\rho_s$, is hard to pin down.

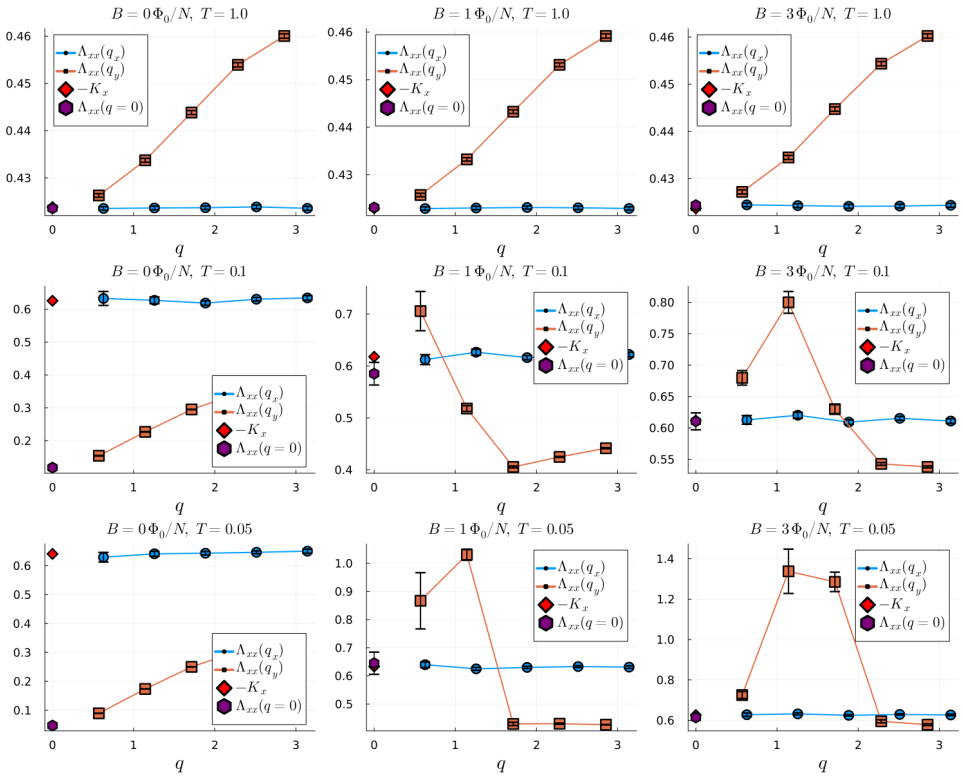



Figure 7.5: The current–current correlation $\Lambda_{xx}(q, \omega = 0)$ in different limits: $(q_x, q_y = 0)$, $(q_x = 0, q_y)$ and $(q = 0)$ denoted by blue, orange and purple symbols respectively. The red diamonds mark the kinetic energy in the $x$ direction, $\langle -K_x \rangle$. The data is for $\mathrm{U} = 4.0$, $n = 0.875$ and various values of $T$ and $B$.

The superfluid stiffness is the curvature of the free energy under a phase twist: $\rho_s = \frac{\partial^2 F}{\partial \Phi^2}\Big|_{\Phi=0}$, with $\Phi$ a magnetic flux. This definition assumes the system sits at a minimum of $F$, where the first derivative $\frac{\partial F}{\partial \Phi}\Big|_{\Phi=0}$ vanishes. Once a static field is applied the system is no longer at a minimum of $F$, and the curvature should be calculated near the minimum, which is determined by the part of the vector potential generating the uniform $B$. We argue instead that, on a 2D torus (periodic boundaries), $\rho_s$ can be read off from the $q = 0$ value of $\Lambda_{xx}(q, \omega = 0)$, this is the response to a constant uniform vector potential. Such a potential cannot be gauged away on the torus, and the system responds to it just as it would to a transverse mode $A_x(q_y \to 0)$ [36].Appendix F proves this for the XY model, which approximates our system (Section 5.1.1). Figure 7.5 also shows $\Lambda_{xx}(q = 0, \omega = 0)$ (taking

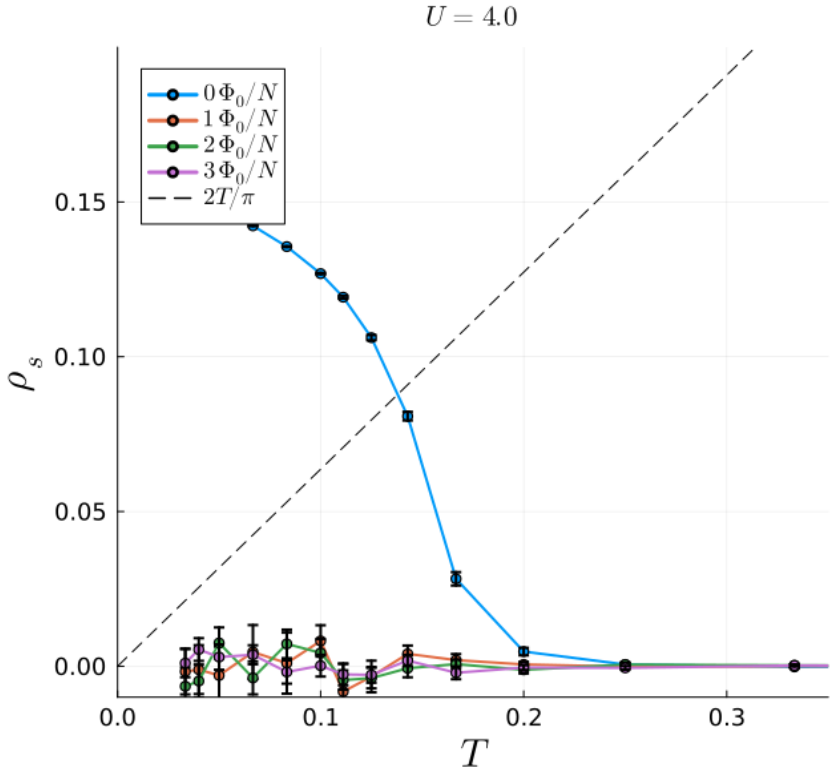



Figure 7.6: The superfluid stiffness $\rho_s$ in magnetic-field systems with $\mathrm{U} = 4$. Here the limit $q_y \to 0$ for $\Lambda_{xx}(q_x = 0, q_y, \omega = 0)$ is taken directly by the $q_y = 0$ value as explained in the text.

the $\omega = 0$ limit first). Wherever a clean $q_y \to 0$ extrapolation exists, as is the case for $B = 0$ and also at high $T$ for the $B > 0$ systems, the $q = 0$ value reproduces it. Only at low $T$ and finite $B$ does the transverse mode misbehave. There the $q = 0$ prescription still gives sensible values, whereas the lowest momentum $q_y = 2\pi/L_y$ and the direct $q_y \to 0$ extrapolation both yield negative or wildly fluctuating $\rho_s$.

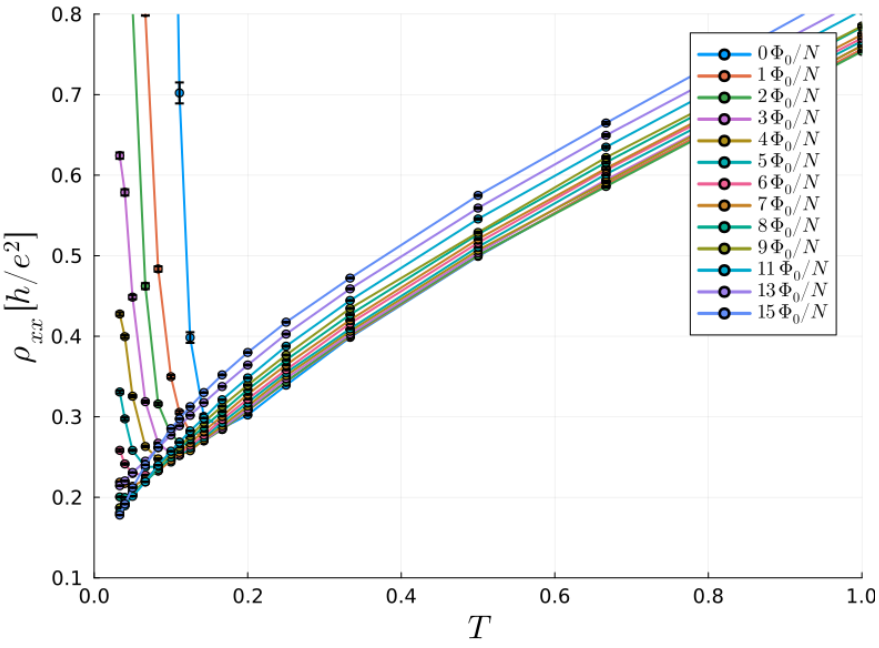
Figure 7.6 presents the stiffness obtained this way. Like $P_s$, $\rho_s$ is completely suppressed by any finite magnetic field, confirming that even our smallest accessible field exceeds the critical field $H_c$. With $\rho_s = 0$ and $D > 0$, every finite-field system is therefore metallic.

## 7.6 Regular resistivity

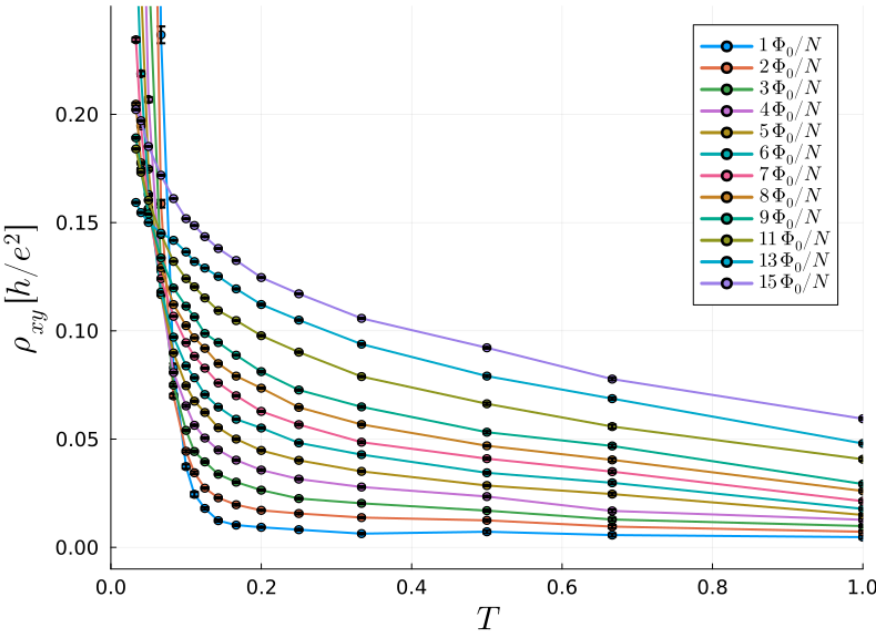
With this classification in hand, we turn to the regular resistivity of the magnetic-field systems (Fig. 7.7). The Hall resistivity $\rho_{xy}^{\mathrm{reg}}$ increases with $B$, as expected for the Hall response. The non-diverging regime of the longitudinal resistivity $\rho_{xx}^{\mathrm{reg}}$ extends to lower $T$ as the field grows, down to a minimum at $T_{\mathrm{min}}(B)$ below which $\rho_{xx}^{\mathrm{reg}}$ diverges. $T_{\mathrm{min}}(B)$ itself decreases with $B$. For some of the fields shown, no minimum is reached within the temperature range which we have checked. However, we can not exclude the possibility that it exits at even lower $T$.

Following the analysis of the previous chapters, we now ask whether and how the pseudogap temperature $T^*$ and the resistivity-minimum temperature $T_{\mathrm{min}}$ correlate in a magnetic field. As Fig. 7.8 shows, both scales are suppressed as $B$ increases, and they decrease together: the correlation between $T^*$ and $T_{\mathrm{min}}$ seen at $B = 0$ persists for finite field.

The suppression of $T^*$ reflects the expected pair-breaking effect of the magnetic field, which introduces a phase difference between the members of a superconducting pair. This effect is strongly U-dependent, governed by the evolution of the coherence length $\xi$. At weak coupling (low U), where $\xi$ spans many lattice sites, the magnetic-field-induced phase difference is significantly over the spatial extent of a pair, leading to efficient pair-breaking and a steep decline in $T^*$ with $B$. In contrast, at strong coupling (high U), the pairs are tightly bound within approximately a single lattice spacing ($\xi \sim a$). In this limit, the pairs respond to the field as composite neutral bosons rather than being internally dissociated, rendering $T^*$ significantly more robust against magnetic fields.




(a)




(b)

Figure 7.7: The regular resistivity of magnetic-field systems. (a) The longitudinal resistivity $\rho_{xx}^{\text{reg}}(\omega \to 0)$, (b) The Hall resistivity $\rho_{xy}^{\text{reg}}(\omega \to 0)$ (the $B = 0$ Hall resistivity is not shown, as it vanishes identically).

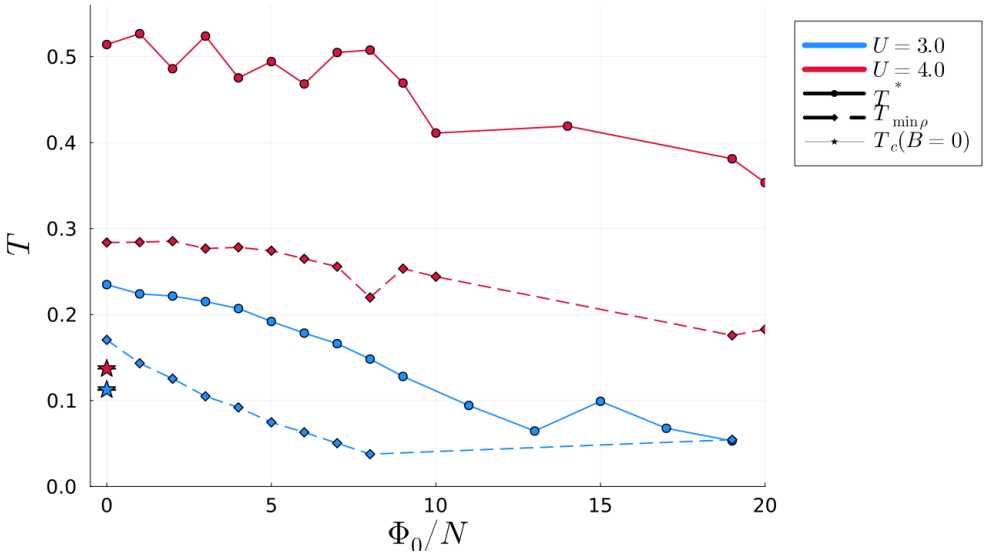



Figure 7.8: The pseudogap temperature $T^*$ (circles, solid line), the minimum of the regular longitudinal resistivity temperature $T_{\min}$ (diamonds, dashed line), and the zero-field superconducting transition temperature $T_c(B = 0)$ (stars), as a function of the magnetic field in units of $\Phi_0/N_s$, for $\mathrm{U} = 3,4$ (colors) at $n = 0.875$. Fields for which $\rho_{xx}(T)$ shows no interior minimum within the simulated temperature range are omitted.

Beyond the low-field regime the trends in $T^*$ and $T_{\min}$ become irregular. We attribute this to the non-monotonic motion of $\mu(B)$.

To conclude, introducing a magnetic field extends the $d\rho_{xx}/dT > 0$ regime to lower temperatures. Although in some cases we do not observe the upturn in the resistivity, this might be due to the fact that we have not reached low enough temperatures. Several questions remain open. As just alluded to — confirming that $T_{\min}$ stays finite at every $B$ and $\mathrm{U}$ requires reaching lower temperatures than we have accessed here. A finite-size scaling study is also needed, both to check that these results are not finite-size artifacts and, by improving the field resolution on larger lattices, to reach the weak fields below our current smallest $B$. This may allow to study the system in the presence of weak fields that do not fully quench superconductivity.

# Chapter 8

# Conclusion

This work was motivated by the experimental observation of strange metallic behavior in the cuprate superconductors, in particular the existence of a wide temperature range where these materials exhibit linear-in-$T$ resistivity. We set out to test whether such a behavior can be reproduced within the attractive Hubbard model with two extensions: a spatially inhomogeneous interaction and a uniform magnetic field.

Using the Monte Carlo mean-field (MCMF) approximation, we gained direct access to real-frequency transport observables, which let us study the resistivity without having to resort to use proxies or carry out analytic continuation, as required by determinant quantum Monte Carlo (DQMC). The price to pay is that MCMF neglects the quantum (temporal) fluctuations of the pairing field, retaining only its thermal fluctuations. Consequently, we have found that MCMF underestimates the critical temperature $T_c$ by about 15–20%. We also found a high-temperature artifact, where a residual gap in the density of states fails to close. We traced this to high-$T$ Gaussian fluctuations of the pairing field and derived a simple model that gives such a behavior.

In the interaction-disordered model, the BCS–BEC crossover induces $T_c$ domes for various disorder levels. They nearly collapse onto a single curve when plotted against the mean interaction strength, $\bar{\mathsf{U}} = \mathsf{U}p$; in all cases, the transition temperature peaks near $\bar{\mathsf{U}} \approx 4$. While the metallic regime ($d\rho/dT > 0$) extends to lower temperatures as disorder increases along an iso-$T_c$ contour, this expansion appears to be driven by the reduction in $\bar{\mathsf{U}}$ rather than by the spatial inhomogeneity itself. Specifically, a lower $\bar{\mathsf{U}}$ suppresses the pairing crossover temperature $T^*$, which is strongly correlated with the resistivity minimum $T_{\mathsf{min}}$, thereby shifting the onset of the metallic-

like behavior.

In the presence of a magnetic field, phase coherence is lost far more readily than pairing; even the minimal accessible field strength suppresses the pair-field structure factor $P_s$ and stiffness $\rho_s$, whereas the pseudogap scale $T^*$ is only gradually diminished. The magnetic field extends the metallic regime ($d\rho/dT > 0$) to lower temperatures, with $T^*$ and the resistivity minimum $T_{\mathsf{min}}$ being suppressed in tandem. This suppression is most pronounced at weak coupling, where the long coherence length $\xi$ renders the spatially extended pairs highly susceptible to the field-induced phase gradient, leading to efficient pair-breaking.

In conclusion, we could not find true $T$-linear resistivity that persist down to arbitrary low temperatures. Nevertheless, it appears that the addition of either interaction disorder or a magnetic field to the attractive Hubbard model expand the range where the model exhibits metallic transport. In particular, for some magnetic field strengths we have not found a temperature where the regular resistivity has a minimum, requiring further data at lower $T$ to asses whether there is a finite $T_{\mathsf{min}}$ for every $B$. In any case, the expansion of the metallic regime can be tied to the lowering of $T^*$.

The natural next step is to combine the two ingredients into an interaction-disordered system in a magnetic field. Larger system analysis would make the field resolution finer, potentially enabling the treatment of weak magnetic fields that do not fully quench superconductivity. For the disordered systems enlarging the system size would enable to reduce finite-size effects and further validate our findings, especially for the highly disordered systems.

Finally, the addition of charge fluctuation as an additional HS field can be important for future work, as the superconducting puddles scenario, which constituted part of the motivation for this work relies on the inclusion of an intra-puddle charging energy in the model [12].

# Appendix A

# Autocorrelation time analysis

In a Monte Carlo simulation the configuration is evolved by a small amount at each step, and therefore successive measurements of an observable are correlated. Treating them as statistically independent would underestimate the uncertainty on their mean. In this appendix we illustrate, for a representative observable, how we quantify this correlation and obtain a reliable error bar, using two complementary tools: a binning (blocking) analysis and the integrated autocorrelation time.

## A.1 Binning analysis

We vary the MC field by a small amount at each step and, once every 20–100 sweeps, record a measurement of the observables of interest. A representative stream is shown in Fig. A.1 for the pair observable $\langle \Delta^\dagger \Delta \rangle$ on an $L = 14$, $L_c = 8$ system at $T = 0.2$, $\mathrm{U} = 5.0$, $n = 0.875$. The signal drifts only slightly between consecutive measurements — the hallmark of strong autocorrelation.

To obtain an error bar that accounts for this correlation we use binning. Let $\{x_i\}_{i=1}^{M}$ be the stream of $M$ measurements. We group them into $N_b$ consecutive bins of size $M_b = M/N_b$ and average within each bin,

$$y_i = \frac{1}{M_b} \sum_{j=(i-1)M_b+1}^{i\,M_b} x_j. \tag{A.1}$$

Treating the bin averages $\{y_i\}$ as independent samples, the squared standard

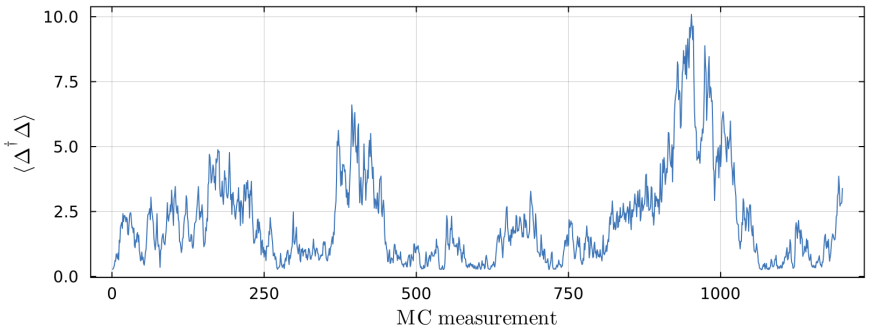



Figure A.1: A segment of the measurement stream of $\langle \Delta^\dagger \Delta \rangle$ for an $L = 14$, $L_c = 8$ system at $T = 0.2$, $\mathrm{U} = 5.0$, $n = 0.875$. Consecutive measurements drift smoothly over scales of order $\tau$, reflecting their correlation.

error of the mean is

$$\sigma_{\bar{x}}^2 = \frac{1}{N_b(N_b-1)} \sum_{i=1}^{N_b} (y_i - \bar{x})^2, \qquad \bar{x} = \frac{1}{M} \sum_i x_i, \tag{A.2}$$

where $\bar{x}$ is the global mean, equal to the mean of the bin averages for equal-sized bins. As $M_b$ grows past the autocorrelation time the bins become effectively uncorrelated and $\sigma_{\bar{x}}$ saturates at the true error.

Figure A.2 shows $\sigma_{\bar{x}}$ as a function of $M_b$. The error rises and then plateaus within a few autocorrelation times. At the plateau it is about seven times larger than the naive estimate from the raw stream ($M_b = 1$), consistent with the expected factor $\sqrt{\tau} \approx 6.7$ (with $\tau$ the autocorrelation time defined below) — an unambiguous signature that the raw data are correlated. We take the plateau value as the reported error bar; in practice we evaluate Eq. (A.2) at $M_b = 10\tau$, comfortably inside the plateau.

## A.2 Integrated autocorrelation time

The correlation can be characterised directly through the autocorrelation function

$$C(k) = \frac{1}{M-k} \sum_{i=1}^{M-k} (x_i - \bar{x})(x_{i+k} - \bar{x}), \tag{A.3}$$

which measures the correlation between measurements separated by $k$ measurements. As shown in Fig. A.3a, $C(k)$ falls rapidly and then fluctuates about zero. One may estimate the autocorrelation time either by fitting

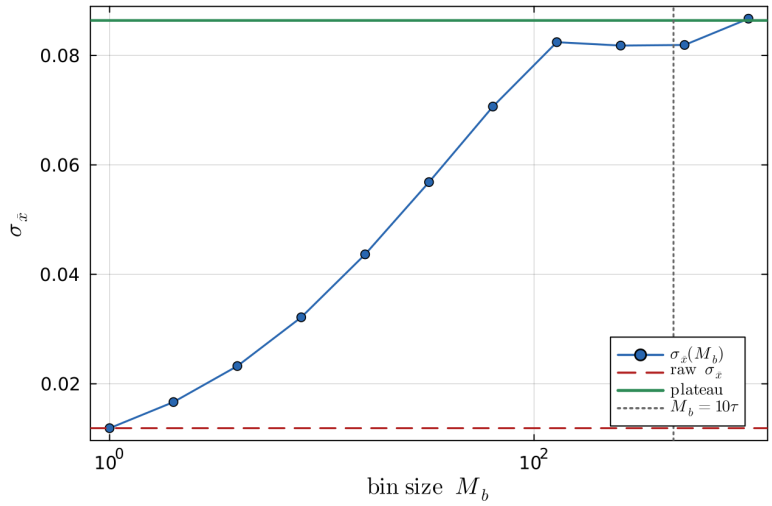


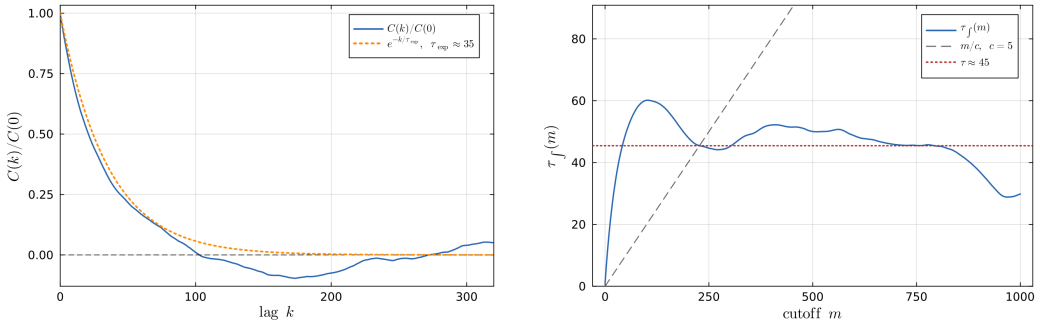

Figure A.2: Standard error of the mean, $\sigma_{\bar{x}}$, as a function of the bin size $M_b$. The error saturates once $M_b$ exceeds a few autocorrelation times; the black vertical dotted line marks the bin size $M_b = 10\tau$ used for the anlysis of this data, and the red horizontal dashed line is the naive estimate from the raw ($M_b = 1$) stream.

the initial decay to $C(k) \propto e^{-k/\tau_{\text{exp}}}$ (orange curve), or, more robustly, by integrating the normalised function up to a cutoff $m$,

$$\tau_{\int}(m) = 1 + \frac{2}{C(0)} \sum_{k=1}^{m} C(k). \tag{A.4}$$

Extending the sum to large $m$ accumulates noise from the fluctuating tail of $C(k)$, so we truncate the sum self-consistently following Sokal [37]. Specifically, we take $\tau$ to be $\tau_{\int}(m)$ evaluated at the smallest $m$ satisfying $\tau_{\int}(m) = m/c$, with $c = 5$. As Fig. A.3b shows, $\tau_{\int}(m)$ rises, settles onto a plateau, and the windowing condition selects $\tau \approx 45$.

As Fig. A.2 shows, the bin size needed for the binning error to saturate is a few times the autocorrelation time $\tau$ found above. We therefore use $M_b = 10\tau$, which lies safely within the saturated regime, and adopt the corresponding $\sigma_{\bar{x}}$ as the error bar throughout this work. The resulting bin averages are shown in Fig. A.4: in contrast to the raw stream of Fig. A.1, they scatter randomly about the mean with no residual drift, confirming that consecutive bins are effectively uncorrelated. With $N/\tau \approx 414 \gg 50$ the stream comprises many independent samples, so the estimate of $\tau$ is reliable [ForemanMackey2013].




(a) (b)

Figure A.3: (a) The autocorrelation function $C(k)/C(0)$ together with an exponential fit $e^{-k/\tau_{\text{exp}}}$ to its initial decay ($\tau_{\text{exp}} \approx 35$). (b) The integrated autocorrelation $\tau_{\int}(m)$ and the windowing line $m/c$ with $c = 5$; their intersection identifies the autocorrelation time $\tau \approx 45$, about which $\tau_{\int}(m)$ fluctuates.

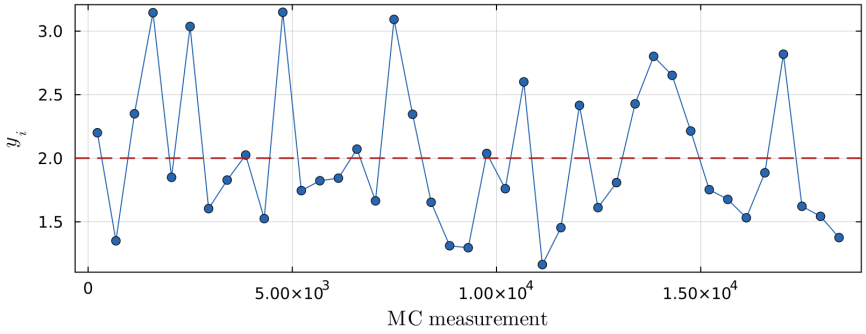



Figure A.4: The bin averages $y_i$ of $\langle \Delta^{\dagger}\Delta \rangle$ at the bin size $M_b = 10\tau$ used in this work, plotted at their bin centres across the full run ($N_b = 41$ bins). The dashed line is the global mean. Unlike the raw stream of Fig. A.1, the bin averages fluctuate randomly about the mean with no slow drift, indicating that successive bins are effectively independent.

# Appendix B

# Current–current correlation function

This appendix collects the technical material behind the current–current correlator and the transport observables of Chs. 5–7: the Kubo derivation, the gauge-invariance identity, the London/Meissner connection to $\rho_s$, the explicit BdG forms of $\langle K_x \rangle$ and $\Lambda_{xx}(q = 0, \omega)$, and the $\delta \to 0$ extrapolation we use to access the regular conductivity. The operator definitions used here ($K_x$, $j_x$, $\Lambda_{xx}$, $\rho_s$, $D$, $\sigma_{xx}^{\mathrm{reg}}$) are stated in §4.5.

## B.1 Kubo response and the Drude–regular decomposition

We introduce a vector potential $A_x(l)$ into the lattice Hamiltonian by Peierls substitution [26], which dresses each $x$–hopping bond with a phase:

$$c^\dagger_{l+\hat{x},\sigma} c_{l,\sigma} \;\longrightarrow\; e^{ieA_x(l)}\, c^\dagger_{l+\hat{x},\sigma} c_{l,\sigma}, \tag{B.1}$$

with $\hbar = c = 1$. Expanding the dressed kinetic energy to second order in $A_x$ and using $A_{-x}(l + \hat{x}) = -A_x(l)$ gives

$$K_{A_x} = K - \sum_i \Big[\, e\, j^p_x(i)\, A_x(i) + \tfrac{e^2}{2}\, k_x(i)\, A^2_x(i) \,\Big], \tag{B.2}$$

where $k_x(i) = -t \sum_\sigma \big(c^\dagger_{i,\sigma} c_{i+\hat{x},\sigma} + \mathrm{h.c.}\big)$ is the bond kinetic energy and $j^p_x(i) = it \sum_\sigma \big(c^\dagger_{i+\hat{x},\sigma} c_{i,\sigma} - c^\dagger_{i,\sigma} c_{i+\hat{x},\sigma}\big)$ is the paramagnetic current. The total current density follows by functional differentiation,

$$j_x(l) = -\frac{\delta K_{A_x}}{\delta A_x(l)} = e\, j^p_x(l) + e^2\, k_x(l)\, A_x(l), \tag{B.3}$$

splitting into a paramagnetic and a diamagnetic part.

In linear response we keep terms to first order in $A_x$ and therefore treat the perturbation $\delta H = -\sum_i [e\, j_x^P(i) A_x(i) + \frac{e^2}{2} k_x(i) A_x^2(i)]$. Applying the Kubo formula to calculate the induced current density gives

$$\langle j_x(l,t)\rangle = ie^2 \sum_j \int_{t_0}^{t} dt' \, \big\langle [j_x^P(l,t),\, j_x^P(j,t')] \big\rangle_0 \, A_x(j,t') + e^2 \langle k_x(l)\rangle_0 \, A_x(l,t), \tag{B.4}$$

where the equilibrium piece $e\langle j_x^P\rangle_0$ has been dropped (zero in the absence of external fields). Fourier transforming and using lattice-translation invariance to collapse the spatial sums, the response becomes

$$\langle j_x(q,\omega)\rangle = -e^2 [\, \langle -K_x\rangle - \Lambda_{xx}(q,\omega)\,]\, A_x(q,\omega), \tag{B.5}$$

coinciding with Eq. (4.14). Here $\Lambda_{xx}(q,\omega)$ is the analytic continuation $i\omega_n \to \omega + i\delta$ of the imaginary-time current–current correlator defined in Eq. (4.15).

For the uniform DC limit we set $q = 0$ and use $A_x(q=0,\omega) = E_x(q=0,\omega)/[i(\omega+i\delta)]$, which converts Eq. (B.5) into a frequency-dependent conductivity,

$$\sigma_{xx}(\omega) = \frac{e^2}{i(\omega+i\delta)} [\, \langle K_x\rangle + \Lambda_{xx}(q=0,\omega)\,]\,. \tag{B.6}$$

The real part of the conductivity decomposes into a regular piece and a Drude delta function,

$$\begin{aligned} &\mathrm{Re}[\sigma_{xx}(\omega)] = \\ &e^2 \Big\{ \frac{1}{\omega} \,\mathrm{Im}\,\Lambda_{xx}(q=0,\omega) \;-\; \pi\,\delta(\omega)[\langle K_x\rangle + \mathrm{Re}\,\Lambda_{xx}(q=0,\omega\to 0)] \Big\}\,. \end{aligned} \tag{B.7}$$

The two pieces are the regular conductivity $\sigma_{xx}^{\mathrm{reg}}(\omega)$ of Eq. (4.20) and the Drude weight $D$ of Eq. (4.19), giving the decomposition $\mathrm{Re}[\sigma_{xx}(\omega)] = D\,\delta(\omega) + \sigma_{xx}^{\mathrm{reg}}(\omega)$ of Eq. (4.18).

## B.2 Gauge invariance and the longitudinal limit

For the static, long-wavelength limits of $\Lambda_{xx}(q, i\omega_n = 0)$,

$$\begin{aligned} \Lambda^L &= \Lambda_{xx}(q_x \to 0,\, q_y = 0,\, i\omega_n = 0), \\ \Lambda^T &= \Lambda_{xx}(q_x = 0,\, q_y \to 0,\, i\omega_n = 0), \end{aligned} \tag{B.8}$$

gauge-invariance leads to $\Lambda^L = -\langle K_x \rangle$. To see this, current conservation together with lattice isotropy forces the static, long-wavelength current response to be purely transverse,

$$\langle j_i(q, \omega = 0) \rangle = f(q) \left( \delta_{ij} - \frac{q_i q_j}{q^2} \right) A_j(q), \tag{B.9}$$

and thus a purely longitudinal vector potential produces no current. Setting $i = j = x$ with $q_x \to 0$, $q_y = 0$ kills the projector ($\delta_{xx} - q_x^2/q_x^2 = 0$), and the response vanishes. Comparing to the Kubo form Eq. (B.5) then forces the bracketed combination to vanish in that limit, $\langle -K_x \rangle - \Lambda^L = 0$, hence

$$\Lambda^L = -\langle K_x \rangle. \tag{B.10}$$

This identity is used as a benchmark for the simulations in Chs. 5–7.

The transverse counterpart $\Lambda^T$ collapses onto $\Lambda^L$ in the normal state and falls below it in the superconducting state, leaving a finite $\rho_s$. The non-commutativity of the $q_x \to 0$ and $q_y \to 0$ limits is the diagnostic for long-range phase coherence. Fig. B.1 shows this directly in the MCMF simulation: it resolves the full $q$-dependence of both cuts and exposes the $q \to 0$ limit by which $\Lambda^L$ and $\Lambda^T$ are extracted. The two limits coincide at high $T$ and split at low $T$.

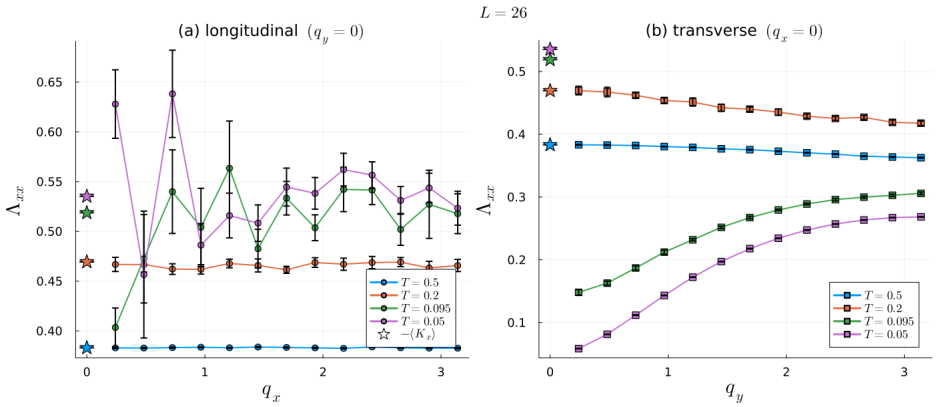



Figure B.1: The two long-wavelength limits of the current–current correlator in the MCMF simulation ($U = 4$, $n = 0.5$, $L = 26$), resolved as a function of $q$ for several temperatures (colors). (a) the longitudinal cut $\Lambda_{xx}(q_x, q_y = 0)$ vs $q_x$ (circles); (b) the transverse cut $\Lambda_{xx}(q_x = 0, q_y)$ vs $q_y$ (squares). In both panels the star at $q = 0$ marks the kinetic energy $-\langle K_x \rangle$ at that temperature.

## B.3 The superfluid stiffness

In a static ($\omega = 0$), long-wavelength ($q_y \to 0$), transverse ($q \cdot A = 0$) vector potential, London's argument identifies the Meissner response as

$$\langle j_x(q_y \to 0)\rangle = -\frac{1}{4\pi\lambda^2}\, A_x(q_y \to 0), \tag{B.11}$$

with $\lambda$ the magnetic-field penetration depth and $1/\lambda^2 = 4\pi n_s e^2/(mc^2)$. Comparing to the transverse limit of the Kubo response (Eq. (B.5)),

$$\langle j_x(q_x = 0, q_y \to 0, \omega = 0)\rangle = -e^2\Big[\langle -K_x\rangle - \Lambda^T\Big] A_x, \tag{B.12}$$

identifies the prefactor with the superfluid density,

$$\frac{D_s}{\pi e^2} \equiv \frac{n_s}{m} = \langle -K_x\rangle - \Lambda^T, \tag{B.13}$$

where $D_s = \pi e^2 n_s/m$ is the superfluid weight. Cooper pairs carry charge $2e$, so the corresponding stiffness is

$$\rho_s = \frac{D_s}{\pi(2e)^2} = \tfrac{1}{4}\Big[\,\langle -K_x\rangle - \Lambda^T\,\Big], \tag{B.14}$$

recovering Eq. (4.17). The link to the penetration depth follows from $1/\lambda^2 \propto n_s$: a finite $\rho_s$ corresponds directly to a finite Meissner kernel and a finite penetration depth.

## B.4 Explicit BdG form of $\langle K_x\rangle$

Starting from $K_x = -(t/N_s)\sum_{r,\sigma}\big(c^\dagger_{r+\hat{x},\sigma}c_{r,\sigma} + \text{h.c.}\big)$, we substitute the Bogoliubov transformation Eq. (4.6) at $\tau = 0$ and use the $\gamma$-basis two-point function Eq. (4.4). For the spin-$\uparrow$ contraction,

$$\langle c^\dagger_{r+\hat{x},\uparrow}c_{r,\uparrow}\rangle = \sum_\lambda U^*_{r+\hat{x},\uparrow,\lambda}\, U_{r,\uparrow,\lambda}\, f(\varepsilon_\lambda). \tag{B.15}$$

The spin-$\downarrow$ contribution maps onto the spin-$\uparrow$ form by the BdG symmetry Eq. (4.8) (which gives $|U_{i\uparrow,\lambda}|^2 = |U_{i\downarrow,\bar{\lambda}}|^2$) together with $f(-\varepsilon_\lambda) = 1 - f(\varepsilon_\lambda)$, collapsing the spin sum into a factor of two. Adding the Hermitian conjugate doubles the result again, giving

$$\langle K_x\rangle = -\frac{4t}{N_s}\sum_\lambda \mathrm{Re}\Big[F^x_{\uparrow\lambda\lambda}(q=0)\Big]\, f(\varepsilon_\lambda) \tag{B.16}$$

with

$$F^{x}_{\uparrow\lambda\lambda}(q=0) = \sum_{i} U^{*}_{i+\hat{x},\uparrow,\lambda}\, U_{i,\uparrow,\lambda}. \tag{B.17}$$

This is the form evaluated at every MC measurement of $\langle K_x \rangle$ throughout the results chapters.

## B.5 Explicit BdG form of $\Lambda_{xx}(q=0,\omega)$

The current–current correlator is defined in imaginary time by

$$\Lambda_{xx}(q, i\omega_n) = \frac{1}{N_s}\int_0^{\beta} d\tau\, e^{i\omega_n \tau}\, \langle j_x(q,\tau)\, j_x(-q,0)\rangle, \tag{B.18}$$

with the current-density operator

$$j_x(q,\tau) = it \sum_{i,\sigma} e^{-iq\cdot r_i}\, e^{H\tau}\Big(c^{\dagger}_{i+\hat{x},\sigma} c_{i,\sigma} - c^{\dagger}_{i,\sigma} c_{i+\hat{x},\sigma}\Big) e^{-H\tau}, \tag{B.19}$$

and $\omega_n = 2\pi n/\beta$ a bosonic Matsubara frequency. To express Eq. (B.18) in the BdG basis we insert the quasi-particle decomposition Eq. (4.6) into $j_x(q,\tau)$. The result expresses the current in the $\gamma$ basis,

$$j_x(\tau,q) = it \sum_{\sigma\lambda\mu} e^{\sigma\tau(\varepsilon_\lambda - \varepsilon_\mu)}\, S^{x}_{\sigma\lambda\mu}(q)\, (\gamma^{\dagger}_{\lambda})^{\sigma}\, (\gamma_{\mu})^{\sigma}, \tag{B.20}$$

with

$$S^{x}_{\sigma\lambda\mu}(q) = F^{x}_{\sigma\lambda\mu}(q) - F^{x*}_{\sigma\mu\lambda}(-q), \qquad F^{x}_{\sigma\lambda\mu}(q) = \sum_{r} e^{-iq\cdot r}\, \big(U^{*}_{r+\hat{x},\sigma,\lambda} U_{r,\sigma,\mu}\big)^{\sigma}, \tag{B.21}$$

where the superscript $\sigma$ on the bracket denotes Hermitian conjugates ($\dagger$) if $\sigma = \downarrow$. The four-fermion product $\langle j_x(\tau,q)\, j_x(0,-q)\rangle$ reduces by Wick's theorem (Eq. (4.7)) to two-point contractions of the $\gamma$'s. Using $\langle \gamma^{\dagger}_{\lambda}\gamma_{\mu}\rangle = \delta_{\lambda\mu} f(\varepsilon_\lambda)$ and combining spin channels through the BdG symmetry Eq. (4.8) gives the closed form

$$\langle j_x(\tau,q)\, j_x(0,-q)\rangle = -t^2\left[\, |g_x(q)|^2 + \sum_{\lambda\mu} e^{\tau(\varepsilon_\lambda - \varepsilon_\mu)}\, |G^{x}_{\lambda\mu}(q)|^2\, f(\varepsilon_\lambda)\, f(-\varepsilon_\mu)\,\right], \tag{B.22}$$

where

$$G^{x}_{\lambda\mu}(q) = S^{x}_{\uparrow\lambda\mu}(q) - S^{x}_{\downarrow\mu\lambda}(q), \qquad g_x(q) = \sum_{\lambda} G^{x}_{\lambda\lambda}(q)\, f(\varepsilon_\lambda). \tag{B.23}$$

The first piece in Eq. (B.22) is the zero-mode contribution; the second carries the full $\tau$ dependence. Performing the $\tau$ integral $\int_0^\beta d\tau\, e^{i\omega_n\tau}(\cdots)$ produces $\beta\,\delta_{n,0}$ from the constant piece and the standard Lindhard denominator from the $\tau$-dependent piece,

$$\Lambda_{xx}(q, i\omega_n) = \frac{t^2}{N_s}\left[\beta\,|g_x(q)|^2\,\delta_{n,0} \;-\; \sum_{\lambda\mu}\frac{f(\varepsilon_\lambda)-f(\varepsilon_\mu)}{\varepsilon_\lambda-\varepsilon_\mu+i\omega_n}\,|G^x_{\lambda\mu}(q)|^2\right]. \tag{B.24}$$

At $q=0$ the zero-mode term contributes only at $i\omega_n=0$. For finite $\omega$ the analytic continuation $i\omega_n\to\omega+i\delta$ leaves the Lehmann form

$$\begin{aligned}
&\Lambda_{xx}(q=0,\omega+i\delta) = \\
&= -\frac{t^2}{N_s}\sum_{\lambda\mu}\frac{f(\varepsilon_\lambda)-f(\varepsilon_\mu)}{\varepsilon_\lambda-\varepsilon_\mu+\omega+i\delta}\,|G^x_{\lambda\mu}(q=0)|^2 \\
&= -\frac{t^2}{N_s}\sum_{\lambda\mu}\left[\frac{\varepsilon_\lambda-\varepsilon_\mu+\omega}{(\varepsilon_\lambda-\varepsilon_\mu+\omega)^2+\delta^2} - i\,\frac{\delta}{(\varepsilon_\lambda-\varepsilon_\mu+\omega)^2+\delta^2}\right] \\
&\quad\times\big(f(\varepsilon_\lambda)-f(\varepsilon_\mu)\big)\,|G^x_{\lambda\mu}(q=0)|^2.
\end{aligned} \tag{B.25}$$

Eq. (B.25) is the form evaluated at every MC measurement. Next, we discuss the way in which the $\delta\to 0$ limit is evaluated.

## B.6 $\delta\to 0$ extrapolation for the regular conductivity

The imaginary part of Eq. (B.25), needed for $\sigma^{\rm reg}_{xx}(\omega)$ (Eq. (4.20)), cannot be evaluated at $\delta=0$ — doing so gives identically zero. We therefore compute $\Lambda_{xx}(q=0,\omega;\delta)$ for a range of broadening $\delta\in[0.5,10]$ MLS (here one MLS is the mean level spacing which we take as $4t/N_s$). A linear regime appears for roughly $1\,\text{MLS}<\delta<8\,\text{MLS}$ (Fig. B.2), which we fit and extrapolate to $\delta\to 0$. This is done independently at each $\omega$, yielding $\Lambda_{xx}(q=0,\omega;\delta\to 0)$. The DC limit of $\sigma^{\rm reg}_{xx}$ is then taken at $\omega<1\,\text{MLS}$, where the result saturates (Fig. B.3).

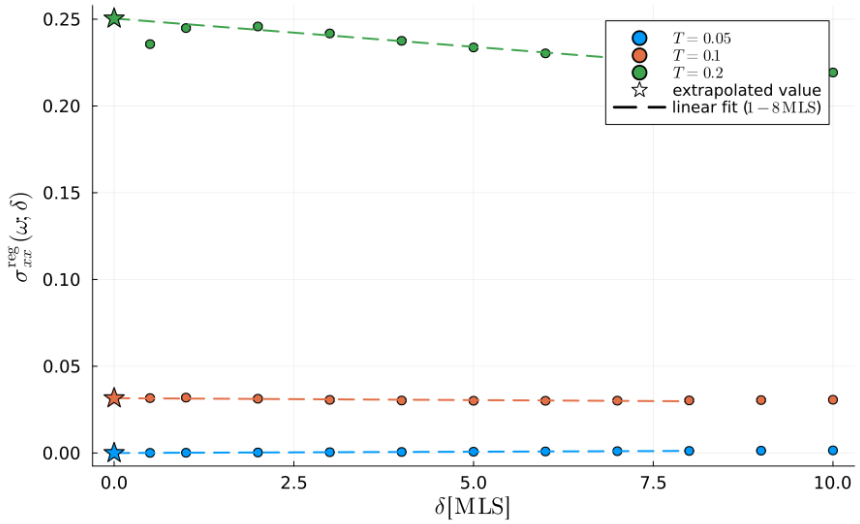



Figure B.2: The $\delta \to 0$ extrapolation of the regular conductivity $\sigma_{xx}^{\mathrm{reg}}(\omega;\delta) = \mathrm{Im}\,\Lambda_{xx}(q = 0, \omega;\delta)/\omega$ as a function of the broadening $\delta$ at a fixed small $\omega \approx 1\,\mathrm{MLS}$, for a few temperatures ($\mathrm{U} = 4$, $n = 0.5$, $L = 20$). For each $T$ the points are fitted by a straight line over the fixed window $\delta \in [1, 8]$ MLS (dashed) and extrapolated to $\delta \to 0$ (star).

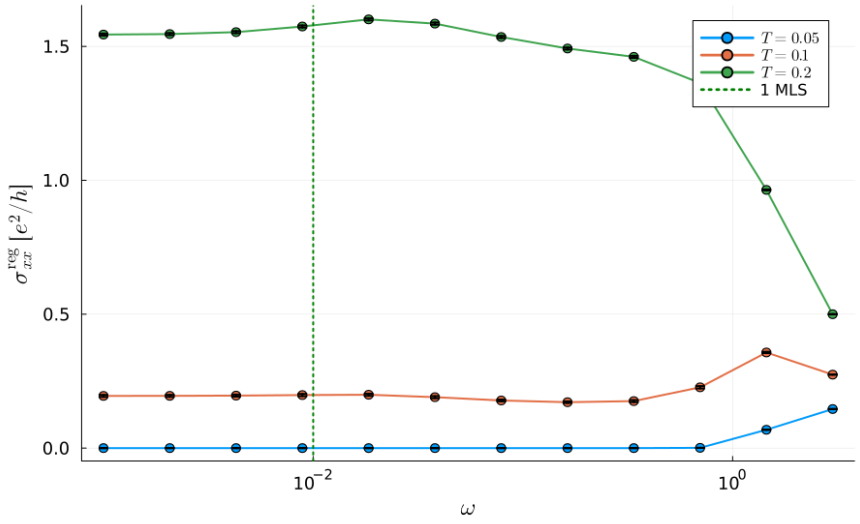



Figure B.3: $\sigma_{xx}^{\mathrm{reg}}(\omega;\delta \to 0)$ as a function of $\omega$ (log scale), after the per-$\omega$ $\delta$-extrapolation, for a few temperatures ($\mathrm{U} = 4$, $n = 0.5$, $L = 20$). The value saturates for $\omega < 1\,\mathrm{MLS}$ (green dotted line), defining the DC limit.

An alternative way to extract the DC limit is to analytically integrate $\sigma_{xx}^{\mathrm{reg}}(\omega;\delta\to 0)$ over a frequency window $\omega\in[0,\Delta\omega]$ and average [38],

$$\sigma_{xx}^{\mathrm{DC}}(\Delta\omega)=\frac{1}{\Delta\omega}\int_0^{\Delta\omega} d\omega\,\sigma_{xx}^{\mathrm{reg}}(\omega;\delta\to 0), \tag{B.26}$$

giving

$$\sigma_{xx}^{\mathrm{DC}}(\Delta\omega)=-\frac{\pi t^2}{N_s\,\Delta\omega}\sum_{\lambda\mu}|G^x_{\lambda\mu}(q=0)|^2\int_0^{\Delta\omega} d\omega\,\frac{f(\varepsilon_\lambda)-f(\varepsilon_\mu)}{\varepsilon_\lambda-\varepsilon_\mu}\,\delta(\varepsilon_\lambda-\varepsilon_\mu+\omega). \tag{B.27}$$

We checked that the two methods agree at about $\Delta\omega = 5\,\mathrm{MLS}$. We do not show results from this approach in the main text: it requires an arbitrary choice of $\Delta\omega$ and is less robust than the $\delta\to 0$ extrapolation, while giving similar values.

# Appendix C

# Explicit BdG forms of the observables

This appendix collects the explicit formulas used to evaluate the observables of Chs. 5–7 from the BdG diagonalized form $\hat{K} = U\,D\,U^{\dagger}$ of §4.4. Each follows mechanically from the $\gamma$-basis two-point function (Eq. (4.4)), the imaginary-time translation (Eq. (4.6)), Wick's theorem (Eq. (4.7)), and the BdG symmetry (Eq. (4.8)).

## C.1 Density (filling)

The filling per site is

$$n = \frac{1}{N_s}\sum_{i,\sigma}\langle c^{\dagger}_{i\sigma}c_{i\sigma}\rangle = \frac{2}{N_s}\sum_{i,\lambda}|U_{i\uparrow,\lambda}|^2\,f(\varepsilon_\lambda), \tag{C.1}$$

where $f(\varepsilon) = 1/(e^{\beta\varepsilon}+1)$ is the Fermi–Dirac function. The factor of 2 reflects the spin sum via the BdG symmetry (Eq. (4.8)).

## C.2 Spectral function, DOS, LDOS

The retarded one-particle Green's function in the BdG basis is

$$G^{R}_{ij\uparrow}(i\omega_n) = -\int_0^{\beta} d\tau e^{-i\omega_n\tau}\langle c_{i\uparrow}(\tau)c^{\dagger}_{j\uparrow}(0)\rangle = \sum_{\lambda}\frac{U_{i\uparrow,\lambda}\,U^{*}_{j\uparrow,\lambda}}{i\omega_n-\varepsilon_\lambda}. \tag{C.2}$$

Substituting into $A_\sigma(k,\omega) = -\frac{1}{\pi N_s}\sum_{ij} e^{-ik\cdot(r_i-r_j)}\,\mathrm{Im}\,G^R_{ij\sigma}(i\omega_n \to \omega + i\delta)$ gives

$$A_\sigma(k,\omega) = \frac{1}{N_s}\sum_\lambda \Big|\sum_i e^{-ik\cdot r_i}\,U_{i\sigma,\lambda}\Big|^2 \delta(\omega-\varepsilon_\lambda). \tag{C.3}$$

The local spectral function follows by setting $i=j$:

$$A_{i\sigma}(\omega) = \sum_\lambda |U_{i\sigma,\lambda}|^2\,\delta(\omega-\varepsilon_\lambda). \tag{C.4}$$

Using $A_{i\uparrow}(\omega) = A_{i\downarrow}(\omega)$ (a consequence of the BdG symmetry, Eq. (4.8)), the LDOS and full DOS are

$$\rho_i(\omega) = 2\sum_\lambda |U_{i\uparrow,\lambda}|^2\,\delta(\omega-\varepsilon_\lambda), \qquad \rho(\omega) = \sum_i \rho_i(\omega). \tag{C.5}$$

Numerically, delta functions are broadened by Lorentzians of width $4t/N_s$ (one mean level spacing).

## C.3 Pair-field structure factor

The expectation value $P_s = \langle \hat\Delta^\dagger\hat\Delta + \hat\Delta\hat\Delta^\dagger\rangle$ (Eq. (4.10)) turns via the commutator $[\hat\Delta,\hat\Delta^\dagger] = 1-\hat n$ into,

$$P_s = 2\langle\hat\Delta^\dagger\hat\Delta\rangle + 1 - n. \tag{C.6}$$

Thus it is sufficient to evaluate the single four-fermion correlator

$$\langle\hat\Delta^\dagger\hat\Delta\rangle = \frac{1}{N_s}\sum_{ij}\langle c^\dagger_{i\uparrow}c^\dagger_{i\downarrow}c_{j\downarrow}c_{j\uparrow}\rangle. \tag{C.7}$$

Substituting Eq. (4.6) at $\tau=0$ converts it into a product of four-$\gamma$ correlator with prefactor $U^*_{i\uparrow,\lambda}U_{i\downarrow,\mu}U^*_{j\downarrow,\nu}U_{j\uparrow,\eta}$. Using Wick's theorem (Eq. (4.7)) on $\langle\gamma^\dagger_\lambda\gamma_\mu\gamma^\dagger_\nu\gamma_\eta\rangle$ and keeping the two non-vanishing contractions (the anomalous $\langle\gamma\gamma\rangle$ and $\langle\gamma^\dagger\gamma^\dagger\rangle$ vanish by Eq. (4.4)) gives

$$\langle\gamma^\dagger_\lambda\gamma_\mu\gamma^\dagger_\nu\gamma_\eta\rangle = f(\varepsilon_\lambda)f(\varepsilon_\nu)\,\delta_{\lambda\mu}\delta_{\nu\eta} + f(\varepsilon_\lambda)\big(1-f(\varepsilon_\mu)\big)\,\delta_{\lambda\eta}\delta_{\mu\nu}. \tag{C.8}$$

Introducing the eigenvector overlap

$$V_{\lambda\mu} = \sum_i U^*_{i\uparrow,\lambda}\,U_{i\downarrow,\mu}, \tag{C.9}$$

allows us to express

$$\langle\hat\Delta^\dagger\hat\Delta\rangle = \frac{1}{N_s}\left[\Big|\sum_\lambda V_{\lambda\lambda}\,f(\varepsilon_\lambda)\Big|^2 + \sum_{\lambda\nu}|V_{\lambda\nu}|^2\,f(\varepsilon_\lambda)\big(1-f(\varepsilon_\nu)\big)\right]. \tag{C.10}$$

The full $P_s$ follows from Eq. (C.6) together with the density Eq. (C.1).

## C.4 Spin susceptibility

The Spin susceptibility is given by

$$\chi_s = 3\int_0^\beta d\tau \langle \hat{S}_z(\tau)\hat{S}_z(0)\rangle, \tag{C.11}$$

where $\hat{S}_z(\tau) = \frac{1}{2}\sum_i (n_{i,\uparrow}(\tau) - n_{i,\downarrow}(\tau))$ is the $z$ component of the spin density operator. Using the BDG diagonalization and the symmetry of the BdG spectrum yields

$$\chi_s = -\frac{3}{2}\sum_{\lambda\mu} \frac{f(\varepsilon_\lambda) - f(\varepsilon_\mu)}{\varepsilon_\lambda - \varepsilon_\mu} \left|L_{\lambda\mu}\right|^2, \tag{C.12}$$

where

$$L_{\lambda\mu} = -\sum_r \left[\left(U^*_{r\uparrow,\lambda} U_{r\uparrow,\mu}\right) + \left(U^*_{r\downarrow,\lambda} U_{r\downarrow,\mu}\right)^*\right], \tag{C.13}$$

is the matrix element of the spin operator between the BdG eigenstates $\lambda$ and $\mu$.

# Appendix D

# Finite-size scaling fit for $T_c^{P_s}$

This appendix documents the two-step fit used in §5.1.2 to extract $T_c^{P_s}$ from the BKT finite-size scaling analysis of the pair-field structure factor, Eq. (5.6). Also, a connection to the finite size scaling of $\Delta_c$ is made, showing that the same exponent and similar $T_c$ are obtained, consistent with the expectation that $\Delta_c$ and $P_s$ are closely related measures of superconducting order.

## D.1 Real-space BKT signature in $P_s(l)$

The finite-size scaling of $P_s$ follows from the spatial decay of the real-space pair correlator $P_s(l)$ (Eq. (4.11)), since $P_s \propto \sum_l P_s(l)$. Fig. D.1 shows that this decay changes character around $T_c$: exponential at high $T$, power-law at low $T$, as expected from a BKT transition.

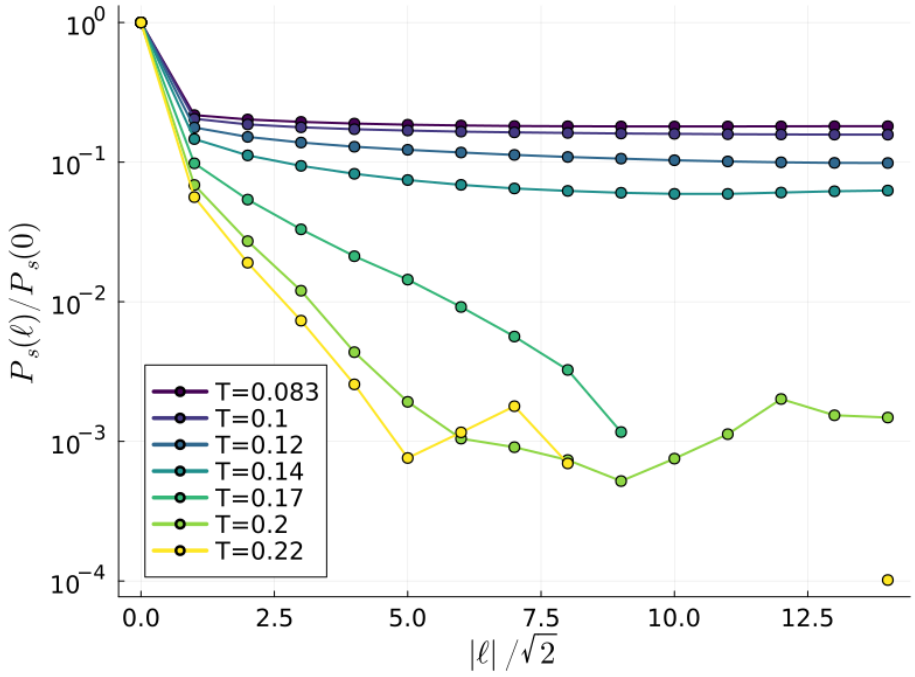


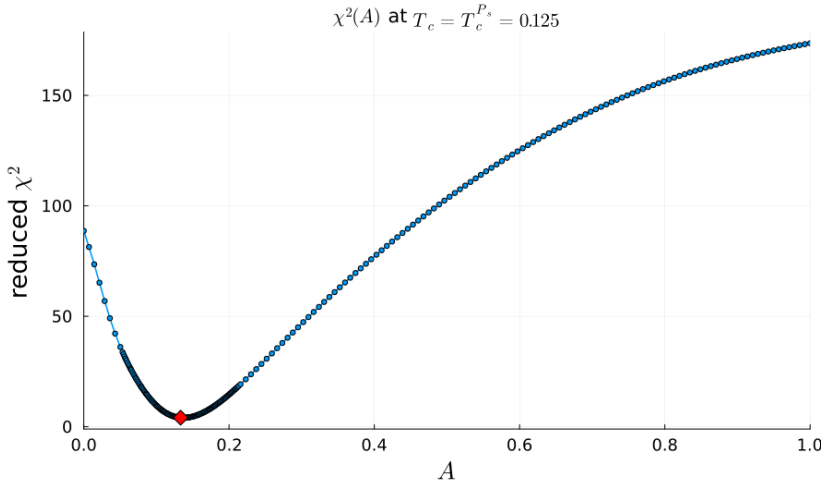

Figure D.1: $P_s(l)$ as defined in Eq. (4.11) for a system of size $L = 28$ with interaction strength $U = 4.0$ and filling $n = 0.875$, for various $T$.The distance $l$ is along the diagonal of the lattice. The $y$-axis is in log scale. The decay crosses from exponential to power-law at $T \approx 0.16$ ($\beta \approx 6$–$7$).

## D.2 Two-step minimization

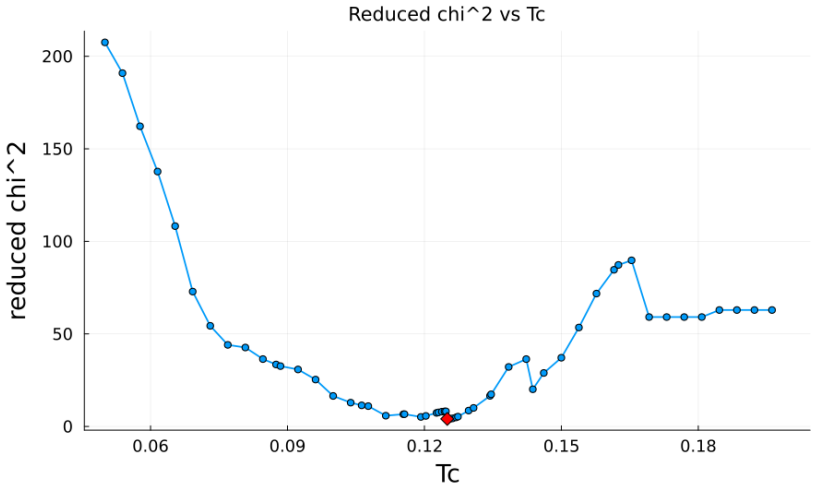
The scaling function $f(x)$ in Eq. (5.6) is unknown. We approximate it by a degree-5 polynomial in the scaling variable $x = L/\xi(T)$, with $\xi(T) = \exp[A/(T - T_c)^{1/2}]$. For each candidate $(A, T_c)$ we compute the scaling variable $x_i = L_i/\xi(T_i)$ for every measurement $(T_i, L_i)$ in the data set, fit the polynomial coefficients $\mathbf{c}^*$ that minimize the residual against $L^{-7/4}P_s$, and record the residual

$$\chi^2(A, T_c) = \sum_i \Big[ L_i^{-7/4} P_s(T_i, L_i) - p(L_i/\xi(T_i); \mathbf{c}^*) \Big]^2 . \tag{D.1}$$

We first minimize over $A$ at fixed $T_c$, giving $A^*(T_c)$ (Fig. D.2a illustrates this inner minimization at $T_c = T_c^{P_s}$), and then over $T_c$ to obtain the global minimum from the $\chi^2(T_c)$ curve (Fig. D.2b).




(a) Inner minimization: $\chi^2(A)$ at fixed $T_c = T_c^{P_s} = 0.125$, for U = 4, $n = 0.5$. The optimal $A^*(T_c)$ is the minimizer (red diamond, here $A^* = 0.133$).




(b) $\chi^2(A^*(T_c), T_c)$ as a function of $T_c$, for U = 4, $n = 0.5$. The global minimum (red diamond) defines $T_c^{P_s}$; the curvature about the minimum sets the statistical (fit) error, which is superseded by the temperature-grid spacing (see text).

Figure D.2: Two-step minimization for the finite-size scaling collapse.

The global minimum of $\chi^2(T_c)$ yields the best-fit values

$$T_c^{P_s} = 0.125 \pm 0.005, \qquad A^* = 0.133 \pm 0.019, \tag{D.2}$$

for U = 4, $n = 0.5$. The error in $A^*$ comes from the curvature of $\chi^2$ about the minimum, which also gives a much smaller statistical error on $T_c$. The

true uncertainty on $T_c$, however, is limited by the finite temperature grid: the two temperatures bracketing $T_c$ are separated by $\Delta T \simeq 0.01$ with no measurements in between, and we take half this spacing as the error. The resulting collapse, with the fitted degree-5 polynomial $p(x; \mathbf{c}^*)$ overlaid, is shown in Fig. D.3. For comparison, the BKT transition temperature extracted from the superfluid stiffness is $T_c^{\rho_s} \approx 0.11 \pm 0.001$. The small offset from $T_c^{P_s}$ is consistent with the coarse temperature grid just above $T_c$ that limits the $P_s$ collapse (§5.1.3).

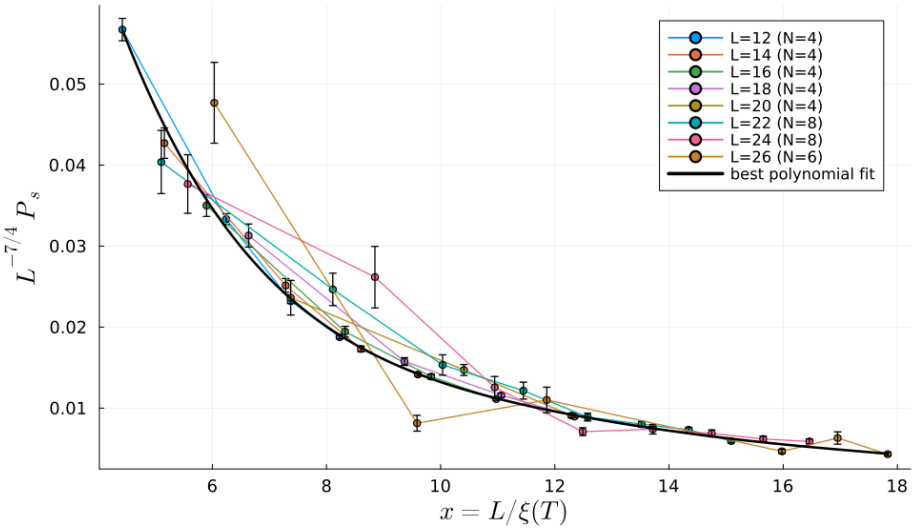



Figure D.3: The finite-size-scaling collapse of $L^{-7/4}P_s$ vs. $x = L/\xi(T)$ at the best-fit $(A^*, T_c^{P_s})$ for $U = 4$, $n = 0.5$, $L = 12$–$26$, with the fitted polynomial $p(x; \mathbf{c}^*)$ overlaid (black curve). This is the same data as Fig. 5.5 with the fit curve drawn.

## D.3 Relation to $\Delta_c$

The observable $\Delta_c$ defined in Eq. (4.9) is a measure of phase coherence from the MC field configurations. we saw that at high temperatures $\Delta_c$ scales as $1/L$ and decreaces to 0, consistent with the absence of long-range order. Conversely, at low temperatures it saturates to a finite value as $L$ increases. Fig. D.4 shows that $\Delta_c$ scales as $1/L$ at high $T$ and changes its scaling drastically at low $T$. $\Delta_c$ measures $\langle |\sum_i \Delta_i| \rangle = \langle \sqrt{\sum_{i,j} \Delta_i \Delta_j^*} \rangle \approx \sqrt{\sum_{ij} \langle \Delta_i \Delta_j^* \rangle}$. The last expression is a very rough approximation, but it shows that $\Delta_c$ is closely related to $P_s$, and we expect them to have similar scaling. At the

critical temperature the scaling of $\Delta_c$ is expected to follow $\Delta_c \sim L^{-\eta(T_c)/2}$, where $\eta(T_c) = 1/4$ is the critical exponent at the BKT transition. Fig. D.4 shows that this is indeed the case, and the critical temperature extracted from this scaling is consistent with the $T_c$ from the $P_s$ collapse.

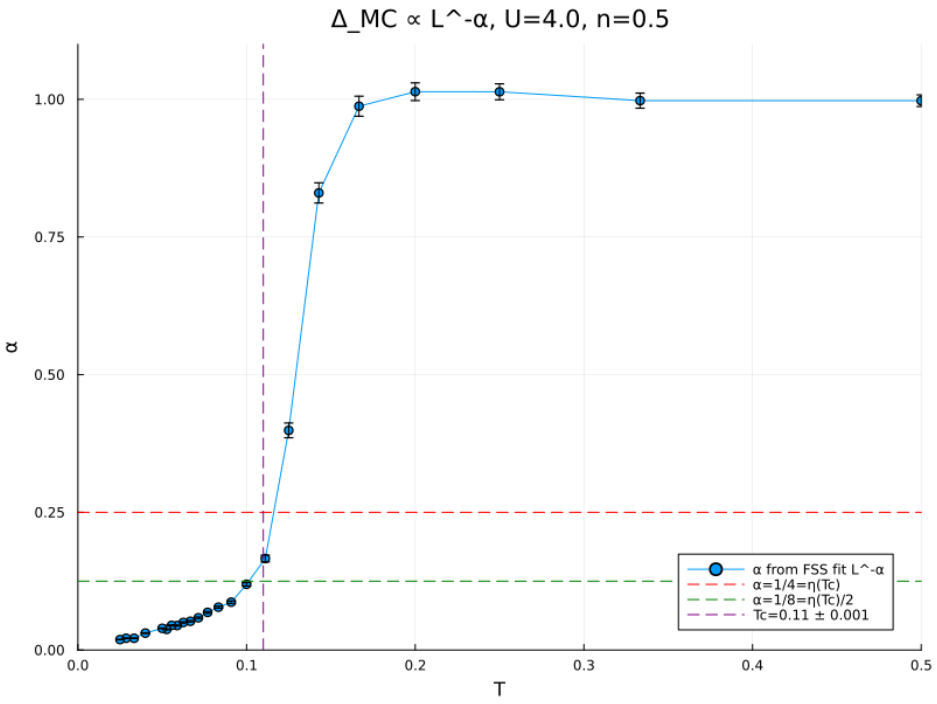



Figure D.4: The exponent $\alpha$ from a power law fit to $\Delta_c(L) = \Delta_0 + L^{-\alpha}$. The vertical line is the critical temperature $T_c$ from the BKT crossing of the superfluid stiffness. The horisontal line is the critical exponent $\eta(T_c) = 1/4$ and half of it.

# Appendix E

# The iso-$T_c = 0.0865$ interaction-disordered family

This appendix presents data for the second iso-$T_c$ family referred to in §6.5. It comprises of systems sharing $T_c \approx 0.0865$, starting from the $U = 10$ uniform system and following decreasing $p$ deeper into the SC-puddle regime. The overall behavior mirrors the $T_c \approx 0.1215$ family of the main text.

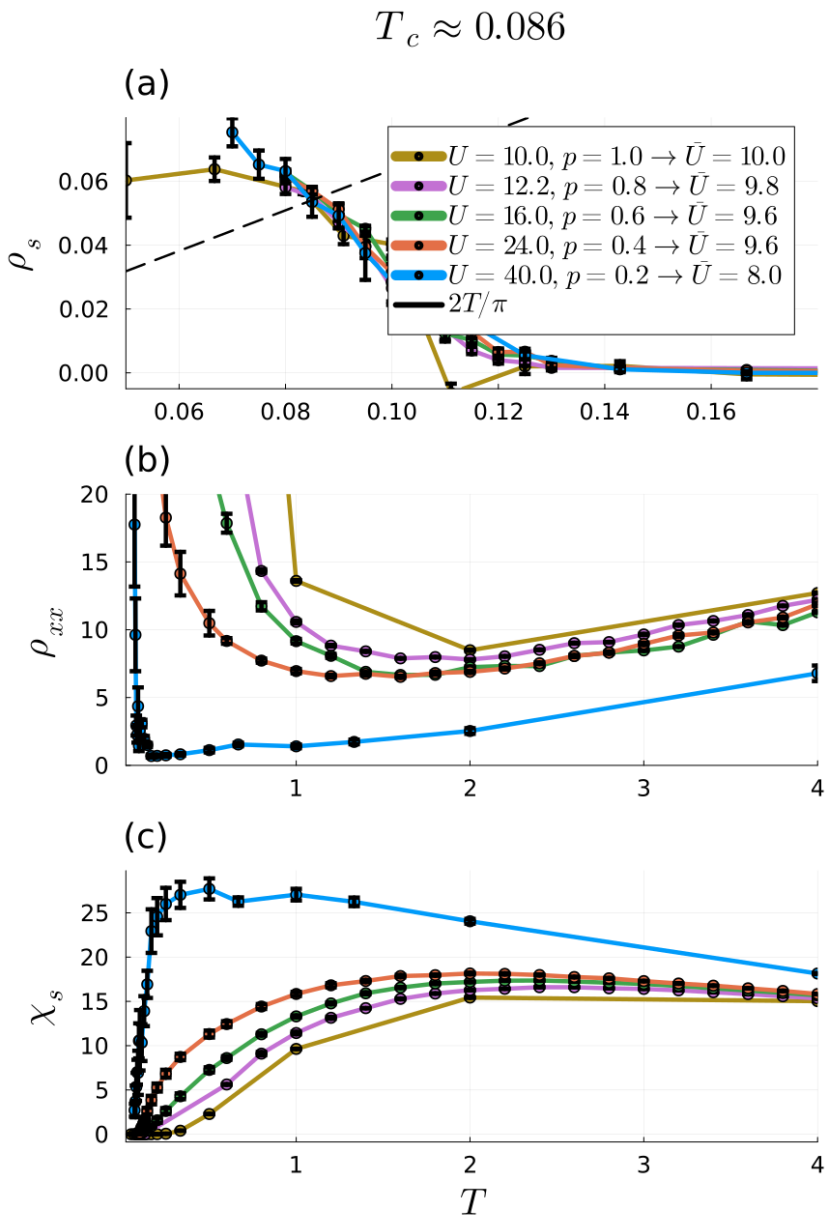



Figure E.1: Same as Fig. 6.5 but for the iso-$T_c$ line of $T_c = 0.0865$.

# Appendix F

# The XY model coupled to a uniform vector potential

In this appendix we show that for an XY model on a torus, the response to a constant uniform vector potential $\mathbf{A}_H$ is similar to the response to a transverse vector potential $\mathbf{A}_T$. For that reason the stiffness can be extracted from the correlation function $\Lambda_{xx}(q = 0, \omega = 0)$ at zero momentum. This is an extention of the proof given in Ref. [36].

We start with the free energy of the XY model on a torus in the presence of a vector potential $\mathbf{A}$:

$$F[\theta, \mathbf{A}] = F_0 + \frac{\rho_s}{2} \int d^2r (\nabla\theta - Q\mathbf{A})^2, \tag{F.1}$$

where $Q = 2e$, the charge of the cooper pairs. The vector potential can be decomposed into four components:

$$\mathbf{A} = \mathbf{A}_0 + \mathbf{A}_L + \mathbf{A}_T + \mathbf{A}_H, \tag{F.2}$$

where $\nabla \cdot \mathbf{A}_T = \nabla \cdot \mathbf{A}_0 = 0$ , $\nabla \times \mathbf{A}_L = 0$ and $\mathbf{A}_H$ is constant. The vector potential $\mathbf{A}_0$ generates a constant uniform magnetic field on the system. We would like to evaluate the response of the system to $\mathbf{A}$. Minimizing the free energy with respect to $\theta$ yields

$$\nabla^2\theta = Q\nabla \cdot \mathbf{A} = Q\nabla \cdot \mathbf{A}_L. \tag{F.3}$$

Thus, it is clear that $\mathbf{A}_L$ is pure gauge because $\nabla\theta = Q\mathbf{A}_L$. On the other hand we cannot gauge away $\mathbf{A}_H$, because $\mathbf{A}_H = \nabla(\mathbf{A}_H \cdot \mathbf{r})$ and the scalar field $\mathbf{A}_H \cdot \mathbf{r}$ does not respect the torus periodicity. Plugging $\nabla\theta = Q\mathbf{A}_L$ back

into the free energy gives us that near its minimum it behaves like

$$\begin{aligned} F_{min}[\mathbf{A}] - F_0 &= \frac{Q^2\rho_s}{2}\int d^2r(\mathbf{A}_0 + \mathbf{A}_T + \mathbf{A}_H)^2 \\ &= \frac{Q^2\rho_s}{2}\int d^2r(\mathbf{A}_0^2 + \mathbf{A}_T^2 + 2\mathbf{A}_0\cdot\mathbf{A}_T + \mathbf{A}_H^2), \end{aligned} \quad \text{(F.4)}$$

where have used the fact the integral over $\mathbf{A_0}$ and $\mathbf{A}_T$ vanishes.

We note that for $\mathbf{A}_0 = 0$ the second derivative $\frac{\delta^2 F}{\delta A_T^2}\Big|_{A_T=0}$ defining the superfluid stiffness $\rho_s$ coincides with $\frac{\delta^2 F}{\delta A_H^2}\Big|_{A_H=0}$. For this reason we claim that the stiffness can be extracted from the value of the current-current correlation function $\Lambda_{xx}(q = 0, \omega = 0)$ strictly at $q = 0$, which corresponds to the response of the system to a constant uniform vector potential. In Fig. 7.5 we give evidence in support of this claim by showing that the $q = 0$ value extrapolates well to the $q_y \to 0$ limit for the systems in the absence of a magnetic field.

For a system in the presence of a magnetic field (non-vanishing $\mathbf{A}_0$), the phase stiffness is defined as the curvature of the free energy with respect to deviation from $\mathbf{A}_0$. Therefore, in this case as well $\rho_s = \frac{\delta^2 F}{\delta A_T^2}\Big|_{A_T=0}$ and the same argument as above can be used to justify its calculation by $\Lambda_{xx}(q = 0, \omega = 0)$. In Fig. 7.5 we show that using this prescription gives sensible results, while calculating $\rho_s$ from the value of the correlation function at the lowest $q_y$ or extrapolating $q_y \to 0$, yields a negative or wildly fluctuating $\rho_s$.

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